Mathematics — Class 12 (PCTB, English Medium)

Solved Exercises | Punjab Curriculum and Textbook Board | Edition 2020 | COMPLETE BOOK: Chapters 1-7, all exercises

FULL-EXERCISE NOTE: this book is now COMPLETE: Chapter 1, Exercises 1.1-1.5; Chapter 2, Exercises 2.1-2.10; Chapter 3, Exercises 3.1-3.8; Chapter 4, Exercises 4.1-4.5; Chapter 5, Exercises 5.1-5.3; Chapter 6, Exercises 6.1-6.9; Chapter 7, Exercises 7.1-7.5. Exceptions, all flagged in the PDF and never invented: Exercise 3.1 Q5, Exercise 3.5 Q22 and Exercise 3.6 Q5 have printed values incomplete in the source copy; Exercise 3.3 Q6/Q8, Exercise 3.4 Q3 and Exercise 3.8 Q18 involve mis-set or faint print in this copy (flagged; standard forms used); Exercise 4.3 Q22(b)/Q22(d)/Q23(c) and Exercise 4.4 Q1/Q2 operators render faintly in this copy (forms confirmed from the book's text layer); Exercise 4.4 Q5 as printed contains no p and its three lines are not concurrent, so no value of p is determined; Exercise 6.4 Q2(viii) does not fix the position of the axis, so the general form is given (with the y-axis specialisation); Exercise 6.8 Q1(iii) is printed as 9x2 + 4x2 (confirmed at 300 DPI) and is solved as printed, with the likely 4y2 reading noted; Exercise 7.4 Q1(ii) and Q2(iv) have their k-components blank/missing in the print, so those two parts are solved for the legible plane vectors and flagged; Exercise 7.4 Q8 as printed gives sin(a - b) with a + sign (the sin(a + b) formula), so the correct sin(a - b) proof is given and the misprint flagged; Exercise 2.5 Q9 is solved as printed in this edition (open check against a clean official copy).

Exercise 1.1, Q1 - Given (a) f(x) = x2 - x and (b) f(x) = sqrt(x + 4), find (i) f(-2) (ii) f(0) (iii) f(x - 1) (iv) f(x2 + 4)

Q1(a)(i). Find f(-2) for f(x) = x2 - x.

f(x) = x2 - x.

Put x = -2: f(-2) = (-2)2 - (-2).

(-2)2 = 4 and -(-2) = +2.

f(-2) = 4 + 2 = 6.

Answer: f(-2) = 6.

Q1(a)(ii). Find f(0) for f(x) = x2 - x.

f(x) = x2 - x.

Put x = 0: f(0) = (0)2 - 0.

f(0) = 0 - 0 = 0.

Answer: f(0) = 0.

Q1(a)(iii). Find f(x - 1) for f(x) = x2 - x.

f(x) = x2 - x.

Replace x by (x - 1): f(x - 1) = (x - 1)2 - (x - 1).

(x - 1)2 = x2 - 2x + 1.

So f(x - 1) = x2 - 2x + 1 - x + 1.

f(x - 1) = x2 - 3x + 2.

Answer: f(x - 1) = x2 - 3x + 2.

Q1(a)(iv). Find f(x2 + 4) for f(x) = x2 - x.

f(x) = x2 - x.

Replace x by (x2 + 4): f(x2 + 4) = (x2 + 4)2 - (x2 + 4).

(x2 + 4)2 = x4 + 8x2 + 16.

So f(x2 + 4) = x4 + 8x2 + 16 - x2 - 4.

f(x2 + 4) = x4 + 7x2 + 12.

Answer: f(x2 + 4) = x4 + 7x2 + 12.

Q1(b)(i). Find f(-2) for f(x) = sqrt(x + 4).

f(x) = sqrt(x + 4).

Put x = -2: f(-2) = sqrt(-2 + 4).

f(-2) = sqrt(2).

Answer: f(-2) = sqrt(2).

Q1(b)(ii). Find f(0) for f(x) = sqrt(x + 4).

f(x) = sqrt(x + 4).

Put x = 0: f(0) = sqrt(0 + 4).

f(0) = sqrt(4) = 2.

Answer: f(0) = 2.

Q1(b)(iii). Find f(x - 1) for f(x) = sqrt(x + 4).

f(x) = sqrt(x + 4).

Replace x by (x - 1): f(x - 1) = sqrt((x - 1) + 4).

f(x - 1) = sqrt(x + 3).

Answer: f(x - 1) = sqrt(x + 3).

Q1(b)(iv). Find f(x2 + 4) for f(x) = sqrt(x + 4).

f(x) = sqrt(x + 4).

Replace x by (x2 + 4): f(x2 + 4) = sqrt((x2 + 4) + 4).

f(x2 + 4) = sqrt(x2 + 8).

Answer: f(x2 + 4) = sqrt(x2 + 8).

Exercise 1.1, Q2 - Find [f(a + h) - f(a)] / h and simplify

Q2(i). For f(x) = 6x - 9, find [f(a + h) - f(a)] / h.

f(x) = 6x - 9.

f(a + h) = 6(a + h) - 9 = 6a + 6h - 9.

f(a) = 6a - 9.

f(a + h) - f(a) = (6a + 6h - 9) - (6a - 9) = 6h.

Divide by h: 6h / h = 6.

Answer: 6.

Q2(ii). For f(x) = sin x, find [f(a + h) - f(a)] / h.

f(x) = sin x, so f(a + h) = sin(a + h) and f(a) = sin a.

Use sin(a + h) = sin a cos h + cos a sin h.

f(a + h) - f(a) = sin a cos h + cos a sin h - sin a.

Group the sin a terms: = sin a (cos h - 1) + cos a sin h.

Divide by h: [sin a (cos h - 1) + cos a sin h] / h.

Answer: [sin a (cos h - 1) + cos a sin h] / h.

Q2(iii). For f(x) = x3 + 2x2 - 1, find [f(a + h) - f(a)] / h.

f(x) = x3 + 2x2 - 1.

f(a + h) = (a + h)3 + 2(a + h)2 - 1.

(a + h)3 = a3 + 3a2h + 3ah2 + h3.

2(a + h)2 = 2a2 + 4ah + 2h2.

So f(a + h) = a3 + 3a2h + 3ah2 + h3 + 2a2 + 4ah + 2h2 - 1.

f(a) = a3 + 2a2 - 1.

Subtract: f(a + h) - f(a) = 3a2h + 3ah2 + h3 + 4ah + 2h2.

Factor out h: = h (3a2 + 3ah + h2 + 4a + 2h).

Divide by h: 3a2 + 4a + 3ah + 2h + h2.

Answer: 3a2 + 4a + 3ah + 2h + h2.

Q2(iv). For f(x) = cos x, find [f(a + h) - f(a)] / h.

f(x) = cos x, so f(a + h) = cos(a + h) and f(a) = cos a.

Use cos(a + h) = cos a cos h - sin a sin h.

f(a + h) - f(a) = cos a cos h - sin a sin h - cos a.

Group the cos a terms: = cos a (cos h - 1) - sin a sin h.

Divide by h: [cos a (cos h - 1) - sin a sin h] / h.

Answer: [cos a (cos h - 1) - sin a sin h] / h.

Exercise 1.1, Q3 - Express the following as functions

Q3(a). The perimeter P of a square as a function of its area A.

Let the side of the square be s.

Area A = s2, so s = sqrt(A).

Perimeter P = 4s.

So P = 4 sqrt(A).

Answer: P = 4 sqrt(A).

Q3(b). The area A of a circle as a function of its circumference C.

Let the radius of the circle be r.

Circumference C = 2 pi r, so r = C / (2 pi).

Area A = pi r2 = pi x [C / (2 pi)]2.

A = pi x C2 / (4 pi2) = C2 / (4 pi).

Answer: A = C2 / (4 pi).

Q3(c). The volume V of a cube as a function of the area A of its base.

Let the edge of the cube be a.

Area of the base A = a2, so a = sqrt(A).

Volume V = a3 = [sqrt(A)]3.

V = A sqrt(A).

Answer: V = A sqrt(A).

Exercise 1.1, Q4 - Find the domain and the range of the function g

Q4(i). g(x) = 2x - 5.

g(x) = 2x - 5 is defined for every real x.

Domain = all real numbers.

As x runs over all real numbers, 2x - 5 takes every real value.

Range = all real numbers.

Answer: domain = R, range = R.

Q4(ii). g(x) = sqrt(x2 - 4).

The square root needs x2 - 4 >= 0.

x2 >= 4 gives x <= -2 or x >= 2.

Domain = (-infinity, -2] together with [2, infinity).

sqrt(x2 - 4) is never negative and takes every value from 0 upwards.

Range = [0, infinity).

Answer: domain = (-infinity, -2] U [2, infinity), range = [0, infinity).

Q4(iii). g(x) = sqrt(x + 1).

The square root needs x + 1 >= 0, so x >= -1.

Domain = [-1, infinity).

sqrt(x + 1) is never negative and takes every value from 0 upwards.

Range = [0, infinity).

Answer: domain = [-1, infinity), range = [0, infinity).

Q4(iv). g(x) = |x - 3|.

The modulus |x - 3| is defined for every real x.

Domain = all real numbers.

A modulus is never negative and takes every value from 0 upwards.

Range = [0, infinity).

Answer: domain = R, range = [0, infinity).

Q4(v). g(x) = 6x + 7 for x <= -2; g(x) = 4 - 3x for -2 < x.

The two pieces together cover every real x.

Domain = all real numbers.

For x <= -2: 6x + 7 <= 6(-2) + 7 = -5, so values (-infinity, -5].

For x > -2: 4 - 3x < 4 - 3(-2) = 10, so values (-infinity, 10).

The union of the two sets is (-infinity, 10).

Range = (-infinity, 10).

Answer: domain = R, range = (-infinity, 10).

Q4(vi). g(x) = x - 1 for x < 3; g(x) = 2x + 1 for 3 <= x.

The two pieces together cover every real x.

Domain = all real numbers.

For x < 3: x - 1 < 2, so values (-infinity, 2).

For x >= 3: 2x + 1 >= 7, so values [7, infinity).

The union is (-infinity, 2) together with [7, infinity).

Range = (-infinity, 2) U [7, infinity).

Answer: domain = R, range = (-infinity, 2) U [7, infinity).

Q4(vii). g(x) = (x2 - 3x + 2) / (x + 1), x not equal to -1.

The denominator x + 1 is zero at x = -1, which is excluded.

Domain = all real numbers except -1.

Put y = (x2 - 3x + 2) / (x + 1).

Then x2 - 3x + 2 = y(x + 1), so x2 - (3 + y)x + (2 - y) = 0.

For a real x this quadratic must have discriminant >= 0.

Discriminant = (3 + y)2 - 4(2 - y) = y2 + 10y + 1.

y2 + 10y + 1 >= 0 gives y <= -5 - 2 sqrt(6) or y >= -5 + 2 sqrt(6).

No y is lost by x = -1, since x = -1 does not satisfy the quadratic.

Range = (-infinity, -5 - 2 sqrt(6)] U [-5 + 2 sqrt(6), infinity).

Answer: domain = R except -1, range = (-infinity, -5 - 2 sqrt(6)] U [-5 + 2 sqrt(6), infinity).

Q4(viii). g(x) = (x2 - 16) / (x - 4), x not equal to 4.

The denominator x - 4 is zero at x = 4, which is excluded.

Domain = all real numbers except 4.

For x not equal to 4: (x2 - 16) / (x - 4) = x + 4.

x + 4 takes every real value except 4 + 4 = 8, the value at x = 4.

Range = all real numbers except 8.

Answer: domain = R except 4, range = R except 8.

Exercise 1.1, Q5 - Given f(x) = x3 - a x2 + b x + 1, if f(2) = -3 and f(-1) = 0, find a and b

Q5. Find the values of a and b.

f(x) = x3 - a x2 + b x + 1.

f(2) = 8 - 4a + 2b + 1 = 9 - 4a + 2b.

f(2) = -3 gives 9 - 4a + 2b = -3, so -4a + 2b = -12.

Divide by -2: 2a - b = 6. (Equation 1)

f(-1) = -1 - a - b + 1 = -a - b.

f(-1) = 0 gives -a - b = 0, so a + b = 0. (Equation 2)

From Equation 2: b = -a.

Put b = -a in Equation 1: 2a - (-a) = 6, so 3a = 6 and a = 2.

Then b = -a = -2.

Check: f(2) = 8 - 8 - 4 + 1 = -3, and f(-1) = -1 - 2 + 2 + 1 = 0.

Answer: a = 2, b = -2.

Exercise 1.1, Q6 - A stone falls, height after x seconds h(x) = 40 - 10 x2 (as printed in the book)

Q6(i)(a). Height when x = 1 sec.

h(x) = 40 - 10 x2.

Put x = 1: h(1) = 40 - 10(1)2.

h(1) = 40 - 10 = 30.

Answer: 30 m.

Q6(i)(b). Height when x = 1.5 sec.

h(x) = 40 - 10 x2.

Put x = 1.5: h(1.5) = 40 - 10(1.5)2.

(1.5)2 = 2.25, so 10 x 2.25 = 22.5.

h(1.5) = 40 - 22.5 = 17.5.

Answer: 17.5 m.

Q6(i)(c). Height when x = 1.7 sec.

h(x) = 40 - 10 x2.

Put x = 1.7: h(1.7) = 40 - 10(1.7)2.

(1.7)2 = 2.89, so 10 x 2.89 = 28.9.

h(1.7) = 40 - 28.9 = 11.1.

Answer: 11.1 m.

Q6(ii). When does the stone strike the ground?

The stone strikes the ground when h(x) = 0.

40 - 10 x2 = 0, so 10 x2 = 40.

x2 = 4, so x = 2 or x = -2.

Time cannot be negative, so x = 2.

Answer: after 2 seconds.

Exercise 1.1, Q7 - Show that the parametric equations represent the given curves

Q7(i). x = a t2, y = 2 a t represent y2 = 4 a x.

Given x = a t2 and y = 2 a t.

From y = 2 a t, t = y / (2a).

Put t in x: x = a [y / (2a)]2.

x = a y2 / (4 a2) = y2 / (4a).

Multiply by 4a: y2 = 4 a x.

Answer: the parametric equations give the parabola y2 = 4 a x.

Q7(ii). x = a cos theta, y = b sin theta represent x2/a2 + y2/b2 = 1.

Given x = a cos theta and y = b sin theta.

cos theta = x / a and sin theta = y / b.

Square and add: cos2 theta + sin2 theta = x2/a2 + y2/b2.

cos2 theta + sin2 theta = 1.

So x2/a2 + y2/b2 = 1.

Answer: the parametric equations give the ellipse x2/a2 + y2/b2 = 1.

Q7(iii). x = a sec theta, y = b tan theta represent x2/a2 - y2/b2 = 1.

Given x = a sec theta and y = b tan theta.

sec theta = x / a and tan theta = y / b.

Use sec2 theta - tan2 theta = 1.

So x2/a2 - y2/b2 = 1.

Answer: the parametric equations give the hyperbola x2/a2 - y2/b2 = 1.

Exercise 1.1, Q8 - Prove the identities

Q8(i). Prove sinh 2x = 2 sinh x cosh x.

sinh x = (e^x - e^-x) / 2 and cosh x = (e^x + e^-x) / 2.

2 sinh x cosh x = 2 x [(e^x - e^-x) / 2] x [(e^x + e^-x) / 2].

Multiply the numerators: (e^x - e^-x)(e^x + e^-x) = e^2x - e^-2x.

So 2 sinh x cosh x = (e^2x - e^-2x) / 2.

But (e^2x - e^-2x) / 2 = sinh 2x.

Answer: sinh 2x = 2 sinh x cosh x, proved.

Q8(ii). Prove sech2 x = 1 - tanh2 x.

tanh x = sinh x / cosh x, so tanh2 x = sinh2 x / cosh2 x.

1 - tanh2 x = 1 - sinh2 x / cosh2 x.

Write as one fraction: = (cosh2 x - sinh2 x) / cosh2 x.

Use cosh2 x - sinh2 x = 1.

So 1 - tanh2 x = 1 / cosh2 x.

1 / cosh2 x = sech2 x.

Answer: sech2 x = 1 - tanh2 x, proved.

Q8(iii). Prove csch2 x = coth2 x - 1.

coth x = cosh x / sinh x, so coth2 x = cosh2 x / sinh2 x.

coth2 x - 1 = cosh2 x / sinh2 x - 1.

Write as one fraction: = (cosh2 x - sinh2 x) / sinh2 x.

Use cosh2 x - sinh2 x = 1.

So coth2 x - 1 = 1 / sinh2 x.

1 / sinh2 x = csch2 x.

Answer: csch2 x = coth2 x - 1, proved.

Exercise 1.1, Q9 - Determine whether the given function f is even or odd

Q9(i). f(x) = x3 + x.

f(x) = x3 + x.

Find f(-x) = (-x)3 + (-x).

(-x)3 = -x3, so f(-x) = -x3 - x.

f(-x) = -(x3 + x) = -f(x).

Since f(-x) = -f(x), the function is odd.

Answer: f is an odd function.

Q9(ii). f(x) = (x + 2)2.

f(x) = (x + 2)2 = x2 + 4x + 4.

Find f(-x) = (-x + 2)2 = (2 - x)2.

f(-x) = x2 - 4x + 4.

f(-x) is not equal to f(x), since -4x is not +4x.

f(-x) is not equal to -f(x) either, since -f(x) = -x2 - 4x - 4.

Since f(-x) is neither f(x) nor -f(x), the function is neither even nor odd.

Answer: neither even nor odd.

Q9(iii). f(x) = x sqrt(x2 + 5).

f(x) = x sqrt(x2 + 5).

Find f(-x) = (-x) sqrt((-x)2 + 5).

(-x)2 = x2, so f(-x) = -x sqrt(x2 + 5).

f(-x) = -f(x).

Since f(-x) = -f(x), the function is odd.

Answer: f is an odd function.

Q9(iv). f(x) = (x - 1) / (x + 1), x not equal to -1.

f(x) = (x - 1) / (x + 1).

Find f(-x) = (-x - 1) / (-x + 1).

Multiply numerator and denominator by -1: f(-x) = (x + 1) / (x - 1).

f(-x) = (x + 1) / (x - 1) is not equal to f(x) = (x - 1) / (x + 1).

-f(x) = (1 - x) / (x + 1), which is also not equal to f(-x).

Since f(-x) is neither f(x) nor -f(x), the function is neither even nor odd.

Answer: neither even nor odd.

Q9(v). f(x) = x^(2/3) + 6.

f(x) = x^(2/3) + 6.

Find f(-x) = (-x)^(2/3) + 6.

(-x)^(2/3) = (cube root of -x)2 = (cube root of x)2 = x^(2/3).

So f(-x) = x^(2/3) + 6 = f(x).

Since f(-x) = f(x), the function is even.

Answer: f is an even function.

Q9(vi). f(x) = (x3 - x) / (x2 + 1).

f(x) = (x3 - x) / (x2 + 1).

Find f(-x) = ((-x)3 - (-x)) / ((-x)2 + 1).

(-x)3 = -x3 and (-x)2 = x2, so f(-x) = (-x3 + x) / (x2 + 1).

f(-x) = -(x3 - x) / (x2 + 1) = -f(x).

Since f(-x) = -f(x), the function is odd.

Answer: f is an odd function.

Exercise 1.2, Q1 - Find (a) fog(x) (b) gof(x) (c) fof(x) (d) gog(x).

Convention: fog(x) = f(g(x)), gof(x) = g(f(x))

Q1(i). f(x) = 2x + 1; g(x) = 3 / (x - 1), x not equal to 1.

(a) fog(x) = f(g(x)) = 2[3 / (x - 1)] + 1.

fog(x) = 6 / (x - 1) + 1 = [6 + (x - 1)] / (x - 1).

Answer (a): fog(x) = (x + 5) / (x - 1).

(b) gof(x) = g(f(x)) = 3 / [(2x + 1) - 1].

Answer (b): gof(x) = 3 / (2x).

(c) fof(x) = f(f(x)) = 2(2x + 1) + 1.

Answer (c): fof(x) = 4x + 3.

(d) gog(x) = g(g(x)) = 3 / [[3 / (x - 1)] - 1].

Simplify the denominator: [3 - (x - 1)] / (x - 1) = (4 - x) / (x - 1).

Answer (d): gog(x) = 3(x - 1) / (4 - x).

Q1(ii). f(x) = sqrt(x + 1); g(x) = 1 / x2, x not equal to 0.

(a) fog(x) = f(g(x)) = sqrt(1 / x2 + 1).

1 / x2 + 1 = (1 + x2) / x2, so fog(x) = sqrt((x2 + 1) / x2).

Answer (a): fog(x) = sqrt(x2 + 1) / |x|.

(b) gof(x) = g(f(x)) = 1 / [sqrt(x + 1)]2.

Answer (b): gof(x) = 1 / (x + 1).

(c) fof(x) = f(f(x)) = sqrt(sqrt(x + 1) + 1).

Answer (c): fof(x) = sqrt(sqrt(x + 1) + 1).

(d) gog(x) = g(g(x)) = 1 / (1 / x2)2.

(1 / x2)2 = 1 / x4, so gog(x) = x4.

Answer (d): gog(x) = x4.

Q1(iii). f(x) = 1 / sqrt(x - 1), x not equal to 1; g(x) = (x2 + 1)2.

(a) fog(x) = f(g(x)) = 1 / sqrt((x2 + 1)2 - 1).

(x2 + 1)2 - 1 = x4 + 2x2 + 1 - 1 = x4 + 2x2 = x2(x2 + 2).

Answer (a): fog(x) = 1 / (|x| sqrt(x2 + 2)).

(b) gof(x) = g(f(x)) = ([1 / sqrt(x - 1)]2 + 1)2.

[1 / sqrt(x - 1)]2 = 1 / (x - 1), so gof(x) = [1 / (x - 1) + 1]2.

1 / (x - 1) + 1 = x / (x - 1).

Answer (b): gof(x) = x2 / (x - 1)2.

(c) fof(x) = f(f(x)) = 1 / sqrt(1 / sqrt(x - 1) - 1).

Answer (c): fof(x) = 1 / sqrt(1 / sqrt(x - 1) - 1).

(d) gog(x) = g(g(x)) = ([(x2 + 1)2]2 + 1)2.

Answer (d): gog(x) = ((x2 + 1)4 + 1)2.

Q1(iv). f(x) = 3x4 - 2x2; g(x) = 2 / sqrt(x), x not equal to 0.

(a) fog(x) = f(g(x)) = 3[2 / sqrt(x)]4 - 2[2 / sqrt(x)]2.

[2 / sqrt(x)]4 = 16 / x2 and [2 / sqrt(x)]2 = 4 / x.

fog(x) = 48 / x2 - 8 / x = (48 - 8x) / x2.

Answer (a): fog(x) = (48 - 8x) / x2.

(b) gof(x) = g(f(x)) = 2 / sqrt(3x4 - 2x2).

3x4 - 2x2 = x2(3x2 - 2), so sqrt(3x4 - 2x2) = |x| sqrt(3x2 - 2).

Answer (b): gof(x) = 2 / (|x| sqrt(3x2 - 2)).

(c) fof(x) = f(f(x)) = 3(3x4 - 2x2)4 - 2(3x4 - 2x2)2.

Answer (c): fof(x) = 3(3x4 - 2x2)4 - 2(3x4 - 2x2)2.

(d) gog(x) = g(g(x)) = 2 / sqrt(2 / sqrt(x)).

sqrt(2 / sqrt(x)) = sqrt(2) / x^(1/4).

So gog(x) = 2 x^(1/4) / sqrt(2) = sqrt(2) x^(1/4).

Answer (d): gog(x) = sqrt(2) x^(1/4).

Exercise 1.2, Q2 - Find (a) f^-1(x) (b) f^-1(-1), and verify f(f^-1(x)) = f^-1(f(x)) = x

Q2(i). f(x) = -2x + 8.

(a) Write y = -2x + 8 and solve for x.

2x = 8 - y, so x = (8 - y) / 2.

Replace y by x: f^-1(x) = (8 - x) / 2 = 4 - x/2.

(b) f^-1(-1) = (8 - (-1)) / 2 = 9 / 2.

Verification: f(f^-1(x)) = -2(4 - x/2) + 8 = -8 + x + 8 = x.

And f^-1(f(x)) = 4 - (-2x + 8) / 2 = 4 - (-x + 4) = x.

Answer: f^-1(x) = 4 - x/2, f^-1(-1) = 9/2, verified.

Q2(ii). f(x) = 3x3 + 7.

(a) Write y = 3x3 + 7 and solve for x.

x3 = (y - 7) / 3, so x = cube root of ((y - 7) / 3).

Replace y by x: f^-1(x) = cube root of ((x - 7) / 3).

(b) f^-1(-1) = cube root of ((-1 - 7) / 3) = cube root of (-8/3).

f^-1(-1) = -2 / cube root of 3.

Verification: f(f^-1(x)) = 3[(x - 7) / 3] + 7 = x.

And f^-1(f(x)) = cube root of ((3x3 + 7 - 7) / 3) = cube root of x3 = x.

Answer: f^-1(x) = cube root of ((x - 7) / 3), f^-1(-1) = -2 / cube root of 3, verified.

Q2(iii). f(x) = (-x + 9)3.

(a) Write y = (-x + 9)3 and solve for x.

Cube root of y = -x + 9, so x = 9 - cube root of y.

Replace y by x: f^-1(x) = 9 - cube root of x.

(b) f^-1(-1) = 9 - cube root of (-1) = 9 - (-1) = 10.

Verification: f(f^-1(x)) = (9 - (9 - cube root of x))3 = (cube root of x)3 = x.

And f^-1(f(x)) = 9 - cube root of ((9 - x)3) = 9 - (9 - x) = x.

Answer: f^-1(x) = 9 - cube root of x, f^-1(-1) = 10, verified.

Q2(iv). f(x) = (2x + 1) / (x - 1), x > 1.

(a) Write y = (2x + 1) / (x - 1) and solve for x.

y(x - 1) = 2x + 1, so yx - y = 2x + 1.

yx - 2x = y + 1, so x(y - 2) = y + 1.

x = (y + 1) / (y - 2). Replace y by x: f^-1(x) = (x + 1) / (x - 2).

(b) f^-1(-1) = (-1 + 1) / (-1 - 2) = 0 / (-3) = 0.

Verification: f(f^-1(x)) = [2(x + 1)/(x - 2) + 1] / [(x + 1)/(x - 2) - 1].

Numerator = (3x) / (x - 2) and denominator = 3 / (x - 2), so the ratio is x.

And f^-1(f(x)) = [(2x + 1)/(x - 1) + 1] / [(2x + 1)/(x - 1) - 2].

Numerator = (3x) / (x - 1) and denominator = 3 / (x - 1), so the ratio is x.

Answer: f^-1(x) = (x + 1) / (x - 2), f^-1(-1) = 0, verified.

Exercise 1.2, Q3 - Without finding the inverse, state the domain and range of f^-1 (domain f^-1 = range f; range f^-1 = domain f)

Q3(i). f(x) = sqrt(x + 2).

Domain of f: x + 2 >= 0, so domain f = [-2, infinity).

Range of f: sqrt(x + 2) takes every value from 0 upwards, so range f = [0, infinity).

Domain of f^-1 = range of f = [0, infinity).

Range of f^-1 = domain of f = [-2, infinity).

Answer: domain f^-1 = [0, infinity), range f^-1 = [-2, infinity).

Q3(ii). f(x) = (x - 1) / (x - 4), x not equal to 4.

Domain of f = all real numbers except 4.

For the range, solve y = (x - 1) / (x - 4) for x: yx - 4y = x - 1.

x(y - 1) = 4y - 1, so x = (4y - 1) / (y - 1), which fails only at y = 1.

Range of f = all real numbers except 1.

Domain of f^-1 = range of f = all real numbers except 1.

Range of f^-1 = domain of f = all real numbers except 4.

Answer: domain f^-1 = R except 1, range f^-1 = R except 4.

Q3(iii). f(x) = 1 / (x + 3), x not equal to -3.

Domain of f = all real numbers except -3.

For the range, y = 1 / (x + 3) gives x + 3 = 1 / y, so x = 1 / y - 3, which fails only at y = 0.

Range of f = all real numbers except 0.

Domain of f^-1 = range of f = all real numbers except 0.

Range of f^-1 = domain of f = all real numbers except -3.

Answer: domain f^-1 = R except 0, range f^-1 = R except -3.

Q3(iv). f(x) = (x - 5)2, x >= 5.

Domain of f = [5, infinity).

Range of f: (x - 5)2 for x >= 5 takes every value from 0 upwards, so range f = [0, infinity).

Domain of f^-1 = range of f = [0, infinity).

Range of f^-1 = domain of f = [5, infinity).

Answer: domain f^-1 = [0, infinity), range f^-1 = [5, infinity).

Exercise 1.3, Q1 - Evaluate each limit by using theorems of limits

Q1(i). Lim (x -> 3) (2x + 4).

Use the sum and constant-multiple theorems: Lim (2x + 4) = 2 Lim x + Lim 4.

As x -> 3, Lim x = 3 and Lim 4 = 4.

So the limit = 2(3) + 4 = 6 + 4 = 10.

Answer: 10.

Q1(ii). Lim (x -> 1) (3x2 - 2x + 4).

Lim (3x2 - 2x + 4) = 3 Lim x2 - 2 Lim x + Lim 4.

As x -> 1: 3(1)2 - 2(1) + 4 = 3 - 2 + 4.

Answer: 5.

Q1(iii). Lim (x -> 3) sqrt(x2 + x + 4).

By the root theorem, the limit = sqrt(Lim (x2 + x + 4)).

Lim (x2 + x + 4) as x -> 3 = 9 + 3 + 4 = 16.

sqrt(16) = 4.

Answer: 4.

Q1(iv). Lim (x -> 2) sqrt(x2 - 4).

The limit = sqrt(Lim (x2 - 4)) = sqrt(4 - 4).

sqrt(0) = 0.

Answer: 0.

Q1(v). Lim (x -> 2) (sqrt(x3 + 1) - sqrt(x2 + 5)).

Lim sqrt(x3 + 1) = sqrt(8 + 1) = sqrt(9) = 3.

Lim sqrt(x2 + 5) = sqrt(4 + 5) = sqrt(9) = 3.

Difference = 3 - 3 = 0.

Answer: 0.

Q1(vi). Lim (x -> -2) (2x3 + 5x) / (3x - 2).

Numerator limit: 2(-2)3 + 5(-2) = -16 - 10 = -26.

Denominator limit: 3(-2) - 2 = -8, which is not 0, so the quotient theorem applies.

Limit = (-26) / (-8) = 26 / 8.

Answer: 13/4.

Exercise 1.3, Q2 - Evaluate each limit by using algebraic techniques

Q2(i). Lim (x -> -1) (x3 - x) / (x + 1).

Direct substitution gives 0/0, so factor first.

x3 - x = x(x2 - 1) = x(x - 1)(x + 1).

Cancel (x + 1) for x not equal to -1: the expression becomes x(x - 1).

As x -> -1: (-1)(-1 - 1) = (-1)(-2) = 2.

Answer: 2.

Q2(ii). Lim (x -> 0) (3x3 + 4x) / (x2 + x).

Direct substitution gives 0/0, so factor first.

Numerator = x(3x2 + 4); denominator = x(x + 1).

Cancel x for x not equal to 0: the expression becomes (3x2 + 4) / (x + 1).

As x -> 0: (0 + 4) / (0 + 1) = 4.

Answer: 4.

Q2(iii). Lim (x -> 2) (x3 - 8) / (x2 + x - 6).

Direct substitution gives 0/0, so factor first.

x3 - 8 = (x - 2)(x2 + 2x + 4); x2 + x - 6 = (x - 2)(x + 3).

Cancel (x - 2) for x not equal to 2: the expression becomes (x2 + 2x + 4) / (x + 3).

As x -> 2: (4 + 4 + 4) / (2 + 3) = 12 / 5.

Answer: 12/5.

Q2(iv). Lim (x -> 1) (x3 - 3x2 + 3x - 1) / (x3 - x).

Direct substitution gives 0/0, so factor first.

Numerator = (x - 1)3; denominator = x(x - 1)(x + 1).

Cancel (x - 1) for x not equal to 1: the expression becomes (x - 1)2 / [x(x + 1)].

As x -> 1: (0)2 / [1(2)] = 0.

Answer: 0.

Q2(v). Lim (x -> -1) (x3 + x2) / (x2 - 1).

Direct substitution gives 0/0, so factor first.

Numerator = x2(x + 1); denominator = (x - 1)(x + 1).

Cancel (x + 1) for x not equal to -1: the expression becomes x2 / (x - 1).

As x -> -1: 1 / (-2) = -1/2.

Answer: -1/2.

Q2(vi). Lim (x -> 4) (2x2 - 32) / (x3 - 4x2).

Direct substitution gives 0/0, so factor first.

Numerator = 2(x2 - 16) = 2(x - 4)(x + 4); denominator = x2(x - 4).

Cancel (x - 4) for x not equal to 4: the expression becomes 2(x + 4) / x2.

As x -> 4: 2(8) / 16 = 1.

Answer: 1.

Q2(vii). Lim (x -> 2) (sqrt(x) - sqrt(2)) / (x - 2).

Direct substitution gives 0/0, so rationalize the numerator.

Multiply top and bottom by (sqrt(x) + sqrt(2)): the numerator becomes x - 2.

So the expression becomes (x - 2) / [(x - 2)(sqrt(x) + sqrt(2))] = 1 / (sqrt(x) + sqrt(2)).

As x -> 2: 1 / (sqrt(2) + sqrt(2)) = 1 / (2 sqrt(2)).

Answer: 1 / (2 sqrt(2)) = sqrt(2) / 4.

Q2(viii). Lim (h -> 0) (sqrt(x + h) - sqrt(x)) / h.

Direct substitution in h gives 0/0, so rationalize the numerator.

Multiply top and bottom by (sqrt(x + h) + sqrt(x)): the numerator becomes (x + h) - x = h.

So the expression becomes h / [h(sqrt(x + h) + sqrt(x))] = 1 / (sqrt(x + h) + sqrt(x)).

As h -> 0: 1 / (sqrt(x) + sqrt(x)) = 1 / (2 sqrt(x)).

Answer: 1 / (2 sqrt(x)).

Q2(ix). Lim (x -> a) (x^n - a^n) / (x^m - a^m).

Direct substitution gives 0/0; use the factorisation x^n - a^n = (x - a)(x^(n-1) + x^(n-2) a + ... + a^(n-1)).

The same factorisation for the denominator gives the factor (x - a) in both.

Cancel (x - a): the expression becomes (x^(n-1) + ... + a^(n-1)) / (x^(m-1) + ... +

a^(m-1)).

As x -> a, the numerator tends to n a^(n-1) (n terms, each a^(n-1)) and the denominator to m a^(m-1).

Answer: (n / m) a^(n-m).

Exercise 1.3, Q3 - Evaluate the following limits

Q3(i). Lim (x -> 0) sin 7x / x.

Write sin 7x / x = 7 (sin 7x) / (7x).

As x -> 0, also 7x -> 0, and Lim (sin t) / t = 1 as t -> 0.

So the limit = 7 x 1 = 7.

Answer: 7.

Q3(ii). Lim (x -> 0) sin x / x.

This is the standard limit proved in the chapter by the sandwich theorem.

Answer: 1.

Q3(iii). Lim (theta -> 0) (1 - cos theta) / sin theta.

Multiply top and bottom by (1 + cos theta): the numerator becomes 1 - cos2 theta = sin2 theta.

So the expression becomes sin2 theta / [sin theta (1 + cos theta)] = sin theta / (1 + cos theta).

As theta -> 0: 0 / (1 + 1) = 0.

Answer: 0.

Q3(iv). Lim (x -> pi) sin x / (pi - x).

Put t = pi - x, so x = pi - t and t -> 0 as x -> pi.

sin x = sin(pi - t) = sin t.

So the limit becomes Lim (t -> 0) sin t / t = 1.

Answer: 1.

Q3(v). Lim (x -> 0) sin a x / sin b x.

Write sin a x / sin b x = (a / b) [(sin a x) / (a x)] / [(sin b x) / (b x)].

As x -> 0, each bracket tends to 1.

So the limit = (a / b) x 1 = a / b.

Answer: a / b.

Q3(vi). Lim (x -> 0) x / tan x.

x / tan x = x cos x / sin x = cos x [(x) / (sin x)].

As x -> 0: cos x -> 1 and x / sin x -> 1.

So the limit = 1 x 1 = 1.

Answer: 1.

Q3(vii). Lim (x -> 0) (1 - cos 2x) / x2.

1 - cos 2x = 2 sin2 x, so the expression becomes 2 (sin x / x)2.

As x -> 0, sin x / x -> 1.

So the limit = 2 x 1 = 2.

Answer: 2.

Q3(viii). Lim (x -> 0) (1 - cos x) / sin2 x.

1 - cos2 x = sin2 x, so (1 - cos x)(1 + cos x) = sin2 x.

Hence (1 - cos x) / sin2 x = 1 / (1 + cos x).

As x -> 0: 1 / (1 + 1) = 1/2.

Answer: 1/2.

Q3(ix). Lim (theta -> 0) sin2 theta / theta.

sin2 theta / theta = sin theta [(sin theta) / theta].

As theta -> 0: sin theta -> 0 and (sin theta) / theta -> 1.

So the limit = 0 x 1 = 0.

Answer: 0.

Q3(x). Lim (x -> 0) (sec x - cos x) / x.

sec x - cos x = (1 - cos2 x) / cos x = sin2 x / cos x.

So the expression becomes (sin2 x) / (x cos x) = sin x [(sin x) / x] (1 / cos x).

As x -> 0: sin x -> 0, (sin x) / x -> 1, 1 / cos x -> 1.

So the limit = 0 x 1 x 1 = 0.

Answer: 0.

Q3(xi). Lim (theta -> 0) (1 - cos p theta) / (1 - cos q theta).

Write each part as: (1 - cos p theta) = 2 sin2 (p theta / 2), and similarly for q.

So the ratio = [sin2 (p theta / 2)] / [sin2 (q theta / 2)].

Write it as (p2 / q2) [(sin (p theta / 2)) / (p theta / 2)]2 / [(sin (q theta / 2)) / (q theta / 2)]2.

As theta -> 0, both brackets tend to 1.

Answer: p2 / q2.

Q3(xii). Lim (theta -> 0) (tan theta - sin theta) / sin3 theta.

tan theta - sin theta = sin theta (1 / cos theta - 1) = sin theta (1 - cos theta) / cos theta.

So the expression becomes (1 - cos theta) / (cos theta sin2 theta).

Using (1 - cos theta) / sin2 theta = 1 / (1 + cos theta), this becomes 1 / [cos theta (1 + cos theta)].

As theta -> 0: 1 / [1 x (1 + 1)] = 1/2.

Answer: 1/2.

Exercise 1.3, Q4 - Express each limit in terms of e

Q4(i). Lim (n -> infinity) (1 + 1/n)^(2n).

Write (1 + 1/n)^(2n) = [(1 + 1/n)^n]2.

As n -> infinity, (1 + 1/n)^n -> e by definition.

So the limit = e2.

Answer: e2.

Q4(ii). Lim (n -> infinity) (1 + 1/n)^(n/2).

Write (1 + 1/n)^(n/2) = [(1 + 1/n)^n]^(1/2).

As n -> infinity, the bracket tends to e.

So the limit = e^(1/2) = sqrt(e).

Answer: sqrt(e).

Q4(iii). Lim (n -> infinity) (1 - 1/n)^n.

Put m = -n, so 1 - 1/n = 1 + 1/m with m -> -infinity.

The standard limit gives e^(-1).

Answer: 1/e.

Q4(iv). Lim (n -> infinity) (1 + 1/(3n))^n.

Write (1 + 1/(3n))^n = [(1 + 1/(3n))^(3n)]^(1/3).

As n -> infinity, 3n -> infinity, so the bracket tends to e.

So the limit = e^(1/3).

Answer: e^(1/3).

Q4(v). Lim (n -> infinity) (1 + 4/n)^n.

Write (1 + 4/n)^n = [(1 + 4/n)^(n/4)]4.

As n -> infinity, n/4 -> infinity, so the bracket tends to e.

So the limit = e4.

Answer: e4.

Q4(vi). Lim (x -> 0) (1 + 3x)^(2/x).

Write (1 + 3x)^(2/x) = [(1 + 3x)^(1/(3x))]6.

As x -> 0, 1/(3x) -> infinity in magnitude and the bracket tends to e.

So the limit = e6.

Answer: e6.

Q4(vii). Lim (x -> 0) (1 + 2x2)^(1/x2).

Write (1 + 2x2)^(1/x2) = [(1 + 2x2)^(1/(2x2))]2.

As x -> 0, 1/(2x2) -> infinity, so the bracket tends to e.

So the limit = e2.

Answer: e2.

Q4(viii). Lim (h -> 0) (1 - 2h)^(1/h).

Write (1 - 2h)^(1/h) = [(1 - 2h)^(-1/(2h))]^(-2).

As h -> 0, -1/(2h) -> infinity in magnitude and the bracket tends to e.

So the limit = e^(-2).

Answer: 1/e2.

Q4(ix). Lim (x -> infinity) (x / (1 + x))^x.

x / (1 + x) = 1 / (1 + 1/x), so the expression = (1 + 1/x)^(-x).

As x -> infinity, (1 + 1/x)^x -> e, so its reciprocal tends to e^(-1).

Answer: 1/e.

Q4(x). Lim (x -> 0-) (e^(1/x) - 1) / (e^(1/x) + 1), for x < 0.

As x -> 0 from below, 1/x -> -infinity, so e^(1/x) -> 0.

The expression tends to (0 - 1) / (0 + 1) = -1.

Answer: -1.

Q4(xi). Lim (x -> 0+) (e^(1/x) - 1) / (e^(1/x) + 1), for x > 0.

As x -> 0 from above, 1/x -> +infinity, so e^(1/x) -> infinity.

Divide top and bottom by e^(1/x): the expression becomes (1 - e^(-1/x)) / (1 + e^(-1/x)).

As x -> 0+, e^(-1/x) -> 0, so the expression tends to (1 - 0) / (1 + 0) = 1.

Answer: 1.

Exercise 1.4, Q1 - Determine the left hand limit and the right hand limit and then find the limit of the following functions when x -> c

Q1(i). f(x) = 2x2 + x - 5, c = 1.

Left hand limit: as x -> 1 from below, f(x) -> 2(1)2 + 1 - 5 = -2.

Right hand limit: as x -> 1 from above, f(x) -> 2(1)2 + 1 - 5 = -2.

Both one-sided limits are equal, so the limit exists.

Answer: left hand limit = -2, right hand limit = -2, Lim (x -> 1) f(x) = -2.

Q1(ii). f(x) = (x2 - 9) / (x - 3), c = -3.

For x not equal to 3, (x2 - 9) / (x - 3) = x + 3.

Left hand limit: as x -> -3 from below, x + 3 -> 0.

Right hand limit: as x -> -3 from above, x + 3 -> 0.

Both one-sided limits are equal, so the limit exists.

Answer: left hand limit = 0, right hand limit = 0, Lim (x -> -3) f(x) = 0.

Q1(iii). f(x) = |x - 5|, c = 5.

Left hand limit: for x < 5, |x - 5| = 5 - x, which tends to 0 as x -> 5 from below.

Right hand limit: for x > 5, |x - 5| = x - 5, which tends to 0 as x -> 5 from above.

Both one-sided limits are equal, so the limit exists.

Answer: left hand limit = 0, right hand limit = 0, Lim (x -> 5) f(x) = 0.

Exercise 1.4, Q2 - Discuss the continuity of f(x) at x = c

Q2(i). f(x) = 2x + 5 if x <= 2; 4x + 1 if x > 2; c = 2.

f(2) is defined: f(2) = 2(2) + 5 = 9.

Left hand limit: as x -> 2 from below, 2x + 5 -> 9.

Right hand limit: as x -> 2 from above, 4x + 1 -> 9.

The limit exists and equals 9, and f(2) = 9, so all three continuity conditions hold.

Answer: f is continuous at x = 2.

Q2(ii). f(x) = 3x - 1 if x < 1; 4 if x = 1; 2x if x > 1; c = 1.

f(1) is defined: f(1) = 4.

Left hand limit: as x -> 1 from below, 3x - 1 -> 2.

Right hand limit: as x -> 1 from above, 2x -> 2.

The limit exists and equals 2, but f(1) = 4, so Lim f(x) is not equal to f(1).

Answer: f is discontinuous at x = 1 (removable discontinuity; the limit is 2 but the value is 4).

Exercise 1.4, Q3 - If f(x) = 3x if x <= -2; x2 - 1 if -2 < x < 2; 3 if x >= 2, discuss continuity at x = 2 and x = -2

Q3 (at x = 2).

f(2) is defined: f(2) = 3.

Left hand limit: as x -> 2 from below, x2 - 1 -> 4 - 1 = 3.

Right hand limit: as x -> 2 from above, f(x) = 3.

The limit exists and equals 3, and f(2) = 3.

Answer: f is continuous at x = 2.

Q3 (at x = -2).

f(-2) is defined: f(-2) = 3(-2) = -6.

Left hand limit: as x -> -2 from below, 3x -> -6.

Right hand limit: as x -> -2 from above, x2 - 1 -> 4 - 1 = 3.

The one-sided limits are not equal, so Lim (x -> -2) f(x) does not exist.

Answer: f is discontinuous at x = -2.

Exercise 1.4, Q4 - If f(x) = x + 2 if x <= -1; c + 2 if x > -1, find c so that Lim (x -> -1) f(x) exists

Q4. Find c.

Left hand limit: as x -> -1 from below, x + 2 -> 1.

Right hand limit: as x -> -1 from above, c + 2 (constant).

For the limit to exist, the two one-sided limits must be equal: c + 2 = 1.

So c = -1, and then the limit is 1.

Answer: c = -1.

Exercise 1.4, Q5 - Find the values m and n, so that the given function f is continuous at x = 3

Q5(i). f(x) = m x if x < 3; n if x = 3; -2x + 9 if x > 3.

f(3) = n.

Left hand limit: as x -> 3 from below, m x -> 3m.

Right hand limit: as x -> 3 from above, -2x + 9 -> 3.

Continuity needs 3m = 3 = n.

Answer: m = 1, n = 3.

Q5(ii). f(x) = m x if x < 3; x2 if x >= 3.

f(3) = 9, and the right hand limit is also 9.

Left hand limit: as x -> 3 from below, m x -> 3m.

Continuity needs 3m = 9.

Answer: m = 3.

Exercise 1.4, Q6 - If f(x) = (sqrt(2x + 5) - sqrt(x + 7)) / (x - 2) if x not equal to 2; k if x = 2, find k so that f is continuous at x = 2

Q6. Find k.

For continuity at x = 2, k must equal Lim (x -> 2) f(x).

Rationalize: multiply top and bottom by (sqrt(2x + 5) + sqrt(x + 7)).

The numerator becomes (2x + 5) - (x + 7) = x - 2.

So f(x) = (x - 2) / [(x - 2)(sqrt(2x + 5) + sqrt(x + 7))] = 1 / (sqrt(2x + 5) + sqrt(x + 7)) for x not equal to 2.

As x -> 2: 1 / (sqrt(9) + sqrt(9)) = 1 / (3 + 3) = 1/6.

Answer: k = 1/6.

Exercise 1.5, Q1 - Draw the graphs of the following equations (table of values and curve described; draw on graph paper)

Q1(i). x2 + y2 = 9.

This is a circle with centre (0, 0) and radius 3.

Points on it: (3, 0), (-3, 0), (0, 3), (0, -3); also (2.12, 2.12) approx. in the first quadrant, since 2.122 + 2.122 = 9 approx.

Plot these points and join them with a smooth closed curve.

Answer: circle, centre (0, 0), radius 3.

Q1(ii). x2 / 16 + y2 / 4 = 1.

This is an ellipse with a = 4 along the x-axis and b = 2 along the y-axis.

Vertices: (4, 0), (-4, 0), (0, 2), (0, -2); when x = 2, y2 / 4 = 1 - 4/16 = 3/4, so y = plus or minus sqrt(3) approx. 1.73.

Plot these points and join them with a smooth closed oval curve.

Answer: ellipse, major axis from (-4, 0) to (4, 0), minor axis from (0, -2) to (0, 2).

Q1(iii). y = e^(2x).

Table: x = -1 gives y = e^(-2) approx. 0.14; x = -0.5 gives y = e^(-1) approx. 0.37; x = 0 gives y = 1; x = 0.5 gives y = e approx. 2.72; x = 1 gives y = e2 approx. 7.39.

The curve passes through (0, 1), stays above the x-axis, approaches y = 0 as x ->

-infinity, and rises steeply for x > 0.

Answer: exponential curve through (0, 1), asymptote y = 0 on the left.

Q1(iv). y = 3^x.

Table: x = -2 gives y = 1/9; x = -1 gives y = 1/3; x = 0 gives y = 1; x = 1 gives y = 3; x = 2 gives y = 9.

The curve passes through (0, 1), stays above the x-axis, approaches y = 0 as x -> -infinity, and rises for x > 0.

Answer: exponential curve through (0, 1), asymptote y = 0 on the left.

Exercise 1.5, Q2 - Graph the curves that have the parametric equations given below (t / theta is a parameter)

Q2(i). x = t, y = t2, -3 <= t <= 3.

Eliminate t: t = x, so y = x2, with -3 <= x <= 3.

Points: (-3, 9), (-2, 4), (-1, 1), (0, 0), (1, 1), (2, 4), (3, 9).

Plot these points and join them with a smooth curve.

Answer: the arc of the parabola y = x2 from (-3, 9) to (3, 9), vertex (0, 0).

Q2(ii). x = t - 1, y = 2t - 1, -1 < t < 5.

Eliminate t: t = x + 1, so y = 2(x + 1) - 1 = 2x + 1.

As t runs from -1 to 5 (excluded), x runs from -2 to 4 (excluded).

The graph is the open segment of the line y = 2x + 1 between (-2, -3) and (4, 9), endpoints not included.

Answer: open line segment on y = 2x + 1 from (-2, -3) to (4, 9).

Q2(iii). x = sec theta, y = tan theta.

Use sec2 theta - tan2 theta = 1, so x2 - y2 = 1.

Since x = sec theta, |x| >= 1: the right branch has x >= 1 and the left branch has x <= -1.

Points on the right branch: (1, 0), (1.41, 1) approx., (2, 1.73) approx.; the left branch is its mirror image.

Answer: hyperbola x2 - y2 = 1, two branches opening left and right, vertices (1, 0) and (-1, 0).

Exercise 1.5, Q3 - Draw the graphs of the functions defined below and find whether they are continuous

Q3(i). y = x - 1 if x < 3; 2x + 1 if x >= 3.

For x < 3 the graph is the line y = x - 1, approaching (3, 2) from the left (open point at (3, 2)).

For x >= 3 the graph is the line y = 2x + 1, starting at the solid point (3, 7).

Left hand limit at 3 is 2, right hand value is 7, so there is a jump of 5 at x = 3.

Answer: discontinuous at x = 3 (jump discontinuity); continuous at every other x.

Q3(ii). y = (x2 - 4) / (x - 2), x not equal to 2.

For x not equal to 2, (x2 - 4) / (x - 2) = x + 2.

The graph is the line y = x + 2 with a hole (open point) at (2, 4), since y is not defined at x = 2.

The limit as x -> 2 is 4, but there is no value at x = 2.

Answer: discontinuous at x = 2 (hole); continuous at every other x.

Q3(iii). y = x + 3 if x not equal to 3; 2 if x = 3.

For x not equal to 3 the graph is the line y = x + 3, approaching (3, 6) from both sides (open point at (3, 6)).

At x = 3 the graph has the isolated solid point (3, 2).

The limit as x -> 3 is 6, but the value is 2, so they do not match.

Answer: discontinuous at x = 3 (removable discontinuity); continuous at every other x.

Q3(iv). y = (x2 - 16) / (x - 4), x not equal to 4.

For x not equal to 4, (x2 - 16) / (x - 4) = x + 4.

The graph is the line y = x + 4 with a hole (open point) at (4, 8), since y is not defined at x = 4.

The limit as x -> 4 is 8, but there is no value at x = 4.

Answer: discontinuous at x = 4 (hole); continuous at every other x.

Exercise 1.5, Q4 - Find the graphical solution of the following equations (intersections of the two graphs)

Q4(i). x = sin 2x.

Draw y = x and y = sin 2x on the same axes.

Since |sin 2x| <= 1, any solution must satisfy |x| <= 1.

The graphs meet at x = 0, and once on each side: check x = 0.95: sin(2 x 0.95) = sin 1.90 approx. 0.946, very close to 0.95.

By symmetry the third meeting point is at x approx. -0.95.

Answer: three solutions, x approx. -0.95, 0, 0.95.

Q4(ii). x / 2 = cos x.

Draw y = x / 2 and y = cos x on the same axes.

For x < 0, cos x >= -1 and x / 2 <= -1 only when x <= -2, but cos(-2) = -0.416 is greater than -1, so the line stays below the cosine curve there and they never meet

for $x < 0$.

For $x > 0$ the line rises while cos x falls from 1: check $x = 1$: $x/2 = 0.5$, cos 1 approx. 0.540 (line below); $x = 1.1$: $x/2 = 0.55$, cos 1.1 approx. 0.454 (line above).

So they cross once, near $x = 1.03$.

Answer: one solution, x approx. 1.03.

Q4(iii). $2x = \tan x$.

Draw $y = 2x$ and $y = \tan x$ on the same axes.

In the central interval $(-\pi/2, \pi/2)$ the graphs meet at $x = 0$, and once on each side: near $x = 1.17$, $\tan 1.17$ approx. 2.35 and $2x = 2.34$, so they cross near $x = 1.17$.

By symmetry the third central meeting point is near $x = -1.17$.

In each further period of tan x the line meets the tangent branch once more, so further pairs of solutions occur outside this interval.

Answer: in $(-\pi/2, \pi/2)$, x approx. -1.17, 0, 1.17; further solutions occur in later periods.

Chapter 2: Differentiation

Exercise 2.1, Q1 - Find by definition, the derivatives w.r.t. x of the following functions

Q1(i). y = 2x2 + 1.

Let y = 2x2 + 1.

Then y + delta y = 2(x + delta x)2 + 1.

So delta y / delta x = [2(x + delta x)2 + 1 - (2x2 + 1)] / delta x = 4x + 2 delta x.

Taking the limit as delta x -> 0, all terms containing delta x vanish.

Answer: dy/dx = 4x.

Q1(ii). y = 2 - sqrt(x).

Let y = 2 - sqrt(x).

Then y + delta y = 2 - sqrt(x + delta x).

So delta y / delta x = [sqrt(x) - sqrt(x + delta x)] / delta x = -1 / (sqrt(x + delta x) + sqrt(x)).

Taking the limit as delta x -> 0, all terms containing delta x vanish.

Answer: dy/dx = -1 / (2 sqrt(x)).

Q1(iii). y = 1 / sqrt(x).

Let y = x^(-1/2).

Then y + delta y = (x + delta x)^(-1/2).

So delta y / delta x = the quotient tends to (-1/2) x^(-3/2) by the power rule proved for rational powers.

Taking the limit as delta x -> 0, all terms containing delta x vanish.

Answer: dy/dx = -1 / (2 x^(3/2)).

Q1(iv). y = 1 / x3.

Let y = x^(-3).

Then y + delta y = (x + delta x)^(-3).

So delta y / delta x = the quotient tends to -3 x^(-4) by the power rule.

Taking the limit as delta x -> 0, all terms containing delta x vanish.

Answer: dy/dx = -3 / x4.

Q1(v). y = 1 / (x - a).

Let y = (x - a)^(-1).

Then y + delta y = (x + delta x - a)^(-1).

So delta y / delta x = [1/(x + delta x - a) - 1/(x - a)] / delta x = -1 / [(x + delta x - a)(x - a)].

Taking the limit as delta x -> 0, all terms containing delta x vanish.

Answer: dy/dx = -1 / (x - a)2.

Q1(vi). y = x(x - 3) = x2 - 3x.

Let y = x2 - 3x.

Then y + delta y = (x + delta x)2 - 3(x + delta x).

So delta y / delta x = [(x + delta x)2 - 3(x + delta x) - (x2 - 3x)] / delta x = 2x + delta x - 3.

Taking the limit as delta x -> 0, all terms containing delta x vanish.

Answer: dy/dx = 2x - 3.

Q1(vii). y = 2 / x4.

Let y = 2 x^(-4).

Then y + delta y = 2 (x + delta x)^(-4).

So delta y / delta x = the quotient tends to 2(-4) x^(-5) by the power rule.

Taking the limit as delta x -> 0, all terms containing delta x vanish.

Answer: dy/dx = -8 / x5.

Q1(viii). y = (x + 4)^(1/3).

Let y = (x + 4)^(1/3).

Then y + delta y = (x + delta x + 4)^(1/3).

So delta y / delta x = the quotient tends to (1/3)(x + 4)^(-2/3) by the chain form of the power rule.

Taking the limit as delta x -> 0, all terms containing delta x vanish.

Answer: dy/dx = (1/3)(x + 4)^(-2/3).

Q1(ix). y = x^(3/2).

Let y = x^(3/2).

Then y + delta y = (x + delta x)^(3/2).

So delta y / delta x = the quotient tends to (3/2) x^(1/2) by the power rule.

Taking the limit as delta x -> 0, all terms containing delta x vanish.

Answer: dy/dx = (3/2) sqrt(x).

Q1(x). y = x^(5/2).

Let y = x^(5/2).

Then y + delta y = (x + delta x)^(5/2).

So delta y / delta x = the quotient tends to (5/2) x^(3/2) by the power rule.

Taking the limit as delta x -> 0, all terms containing delta x vanish.

Answer: dy/dx = (5/2) x^(3/2).

Q1(xi). y = x^m, m in N.

Let y = x^m.

Then y + delta y = (x + delta x)^m.

So delta y / delta x = expanding by the binomial theorem and dividing by delta x gives m x^(m-1) plus terms in delta x.

Taking the limit as delta x -> 0, all terms containing delta x vanish.

Answer: dy/dx = m x^(m-1).

Q1(xii). y = 1 / x^m, m in N.

Let y = x^(-m).

Then y + delta y = (x + delta x)^(-m).

So delta y / delta x = the quotient tends to -m x^(-m-1) by the power rule.

Taking the limit as delta x -> 0, all terms containing delta x vanish.

Answer: dy/dx = -m / x^(m+1).

Q1(xiii). y = x^40.

Let y = x^40.

Then y + delta y = (x + delta x)^40.

So delta y / delta x = expanding by the binomial theorem and dividing by delta x gives 40 x^39 plus terms in delta x.

Taking the limit as delta x -> 0, all terms containing delta x vanish.

Answer: dy/dx = 40 x^39.

Q1(xiv). y = x^(-100).

Let y = x^(-100).

Then y + delta y = (x + delta x)^(-100).

So delta y / delta x = the quotient tends to -100 x^(-101) by the power rule.

Taking the limit as delta x -> 0, all terms containing delta x vanish.

Answer: dy/dx = -100 x^(-101).

Exercise 2.1, Q2 - Find dy/dx from first principle if

Q2(i). y = sqrt(x + 2).

Let y = sqrt(x + 2). Then y + delta y = sqrt(x + delta x + 2).

delta y / delta x = [sqrt(x + delta x + 2) - sqrt(x + 2)] / delta x.

Rationalize: multiply top and bottom by (sqrt(x + delta x + 2) + sqrt(x + 2)).

The numerator becomes delta x, so delta y / delta x = 1 / (sqrt(x + delta x + 2) + sqrt(x + 2)).

Taking the limit as delta x -> 0: dy/dx = 1 / (2 sqrt(x + 2)).

Answer: dy/dx = 1 / (2 sqrt(x + 2)).

Q2(ii). y = 1 / sqrt(x + a).

Let y = (x + a)^(-1/2). Then y + delta y = (x + delta x + a)^(-1/2).

delta y / delta x = [(x + delta x + a)^(-1/2) - (x + a)^(-1/2)] / delta x.

Rationalize as for a square root: the quotient tends to (-1/2)(x + a)^(-3/2).

Answer: dy/dx = -1 / (2 (x + a)^(3/2)).

Exercise 2.2, Q1 - Find from first principles, the derivatives of the following expressions w.r.t. their respective independent variables

Q1(i). (a x + b)3, w.r.t. x.

Let y = (a x + b)3. Then y + delta y = (a(x + delta x) + b)3 = ((a x + b) + a delta x)3.

Expand by the binomial theorem: delta y = 3(a x + b)2 (a delta x) + 3(a x + b)(a delta x)2 + (a delta x)3.

Divide by delta x: delta y / delta x = 3a(a x + b)2 + 3a2(a x + b) delta x + a3 (delta x)2.

Taking the limit as delta x -> 0, the terms in delta x vanish.

Answer: dy/dx = 3a(a x + b)2.

Q1(ii). (2x + 3)5, w.r.t. x.

This is (a x + b)^n with a = 2, b = 3, n = 5.

By the first-principle result d/dx (a x + b)^n = n a (a x + b)^(n-1).

So dy/dx = 5 x 2 x (2x + 3)4.

Answer: dy/dx = 10(2x + 3)4.

Q1(iii). (3t + 2)^(-2), w.r.t. t.

This is (a t + b)^n with a = 3, b = 2, n = -2.

By the first-principle result d/dt (a t + b)^n = n a (a t + b)^(n-1).

So dy/dt = (-2) x 3 x (3t + 2)^(-3).

Answer: dy/dt = -6 / (3t + 2)3.

Q1(iv). 1 / (a x + b)5, w.r.t. x.

Write y = (a x + b)^(-5).

By the first-principle result d/dx (a x + b)^n = n a (a x + b)^(n-1) with n = -5.

So dy/dx = (-5) a (a x + b)^(-6).

Answer: dy/dx = -5a / (a x + b)6.

Q1(v). 1 / (a z - b)7, w.r.t. z.

Write y = (a z - b)^(-7).

By the first-principle result d/dz (a z - b)^n = n a (a z - b)^(n-1) with n = -7.

So dy/dz = (-7) a (a z - b)^(-8).

Answer: dy/dz = -7a / (a z - b)8.

Exercise 2.3 - Differentiate w.r.t. x

Q1. x4 + 2x3 + x2.

Differentiate term by term using the power rule.

d/dx (x4) = 4x3; d/dx (2x3) = 6x2; d/dx (x2) = 2x.

Answer: dy/dx = 4x3 + 6x2 + 2x.

Q2. x^(-3) + 2x^(-3/2) + 3.

Differentiate term by term; the derivative of the constant 3 is 0.

d/dx (x^(-3)) = -3x^(-4); d/dx (2x^(-3/2)) = 2(-3/2)x^(-5/2) = -3x^(-5/2).

Answer: dy/dx = -3/x4 - 3/x^(5/2).

Q3. (a + x) / (a - x).

Use the quotient rule with f = a + x, f' = 1, g = a - x, g' = -1.

dy/dx = [1(a - x) - (a + x)(-1)] / (a - x)2 = [(a - x) + (a + x)] / (a - x)2.

Answer: dy/dx = 2a / (a - x)2.

Q4. (2x - 3) / (2x + 1).

Use the quotient rule: f' = 2, g' = 2.

dy/dx = [2(2x + 1) - (2x - 3)2] / (2x + 1)2 = [(4x + 2) - (4x - 6)] / (2x + 1)2.

Answer: dy/dx = 8 / (2x + 1)2.

Q5. (x - 5)(3 - x).

Expand first: (x - 5)(3 - x) = 3x - x2 - 15 + 5x = -x2 + 8x - 15.

Differentiate term by term.

Answer: dy/dx = -2x + 8.

Q6. (sqrt(x) - 1/sqrt(x))2.

Expand: (sqrt(x) - 1/sqrt(x))2 = x - 2 + 1/x.

Differentiate term by term: d/dx (x) = 1, d/dx (-2) = 0, d/dx (1/x) = -1/x2.

Answer: dy/dx = 1 - 1/x2.

Q7. (1 + sqrt(x))(x - x^(3/2)) / sqrt(x).

Simplify first: x - x^(3/2) = x(1 - sqrt(x)).

So the expression = (1 + sqrt(x)) x (1 - sqrt(x)) / sqrt(x) = x(1 - x) / sqrt(x) = sqrt(x) - x^(3/2).

Differentiate: d/dx (sqrt(x)) = 1/(2 sqrt(x)); d/dx (x^(3/2)) = (3/2) sqrt(x).

Answer: dy/dx = 1/(2 sqrt(x)) - (3 sqrt(x))/2.

Q8. (x2 + 1)2 / (x2 - 1).

Use the quotient rule with f = (x2 + 1)2, f' = 4x(x2 + 1), g = x2 - 1, g' = 2x.

dy/dx = [4x(x2 + 1)(x2 - 1) - (x2 + 1)2 (2x)] / (x2 - 1)2.

Factor 2x(x2 + 1) in the numerator: = 2x(x2 + 1)[2(x2 - 1) - (x2 + 1)] / (x2 - 1)2.

Answer: dy/dx = 2x(x2 + 1)(x2 - 3) / (x2 - 1)2.

Q9. (x2 + 1) / (x2 - 3).

Use the quotient rule: dy/dx = [2x(x2 - 3) - (x2 + 1)2x] / (x2 - 3)2.

Numerator = 2x[(x2 - 3) - (x2 + 1)] = 2x(-4) = -8x.

Answer: dy/dx = -8x / (x2 - 3)2.

Q10. sqrt(1 + x) / sqrt(1 - x).

Write y = ((1 + x)/(1 - x))^(1/2) and use the chain rule.

d/dx [(1 + x)/(1 - x)] = [(1 - x) + (1 + x)] / (1 - x)2 = 2/(1 - x)2.

So dy/dx = (1/2)((1 + x)/(1 - x))^(-1/2) x 2/(1 - x)2.

Answer: dy/dx = 1 / [(1 - x) sqrt(1 - x2)].

Q11. (2x - 1) / sqrt(x2 + 1).

Use the quotient rule with g = sqrt(x2 + 1), g' = x/sqrt(x2 + 1).

dy/dx = [2 sqrt(x2 + 1) - (2x - 1) x/sqrt(x2 + 1)] / (x2 + 1).

Multiply top and bottom by sqrt(x2 + 1): = [2(x2 + 1) - x(2x - 1)] / (x2 + 1)^(3/2).

Answer: dy/dx = (x + 2) / (x2 + 1)^(3/2).

Q12. sqrt((a - x) / (a + x)).

Write y = ((a - x)/(a + x))^(1/2) and use the chain rule.

d/dx [(a - x)/(a + x)] = [-(a + x) - (a - x)] / (a + x)2 = -2a/(a + x)2.

So dy/dx = (1/2)((a - x)/(a + x))^(-1/2) x (-2a)/(a + x)2.

Answer: dy/dx = -a / [(a + x) sqrt(a2 - x2)].

Q13. sqrt(x2 + 1) / sqrt(x2 - 1).

Write y = ((x2 + 1)/(x2 - 1))^(1/2) and use the chain rule.

d/dx [(x2 + 1)/(x2 - 1)] = [2x(x2 - 1) - 2x(x2 + 1)] / (x2 - 1)2 = -4x/(x2 - 1)2.

So dy/dx = (1/2)((x2 + 1)/(x2 - 1))^(-1/2) x (-4x)/(x2 - 1)2.

Answer: dy/dx = -2x / [(x2 - 1)^(3/2) sqrt(x2 + 1)].

Q14. (sqrt(1 + x) - sqrt(1 - x)) / (sqrt(1 + x) + sqrt(1 - x)).

Simplify first: multiply top and bottom by (sqrt(1 + x) - sqrt(1 - x)).

The denominator becomes (1 + x) - (1 - x) = 2x; the numerator becomes 2 - 2 sqrt(1 - x2).

So y = (1 - sqrt(1 - x2)) / x.

Differentiate by the quotient rule: d/dx (1 - sqrt(1 - x2)) = x/sqrt(1 - x2).

So dy/dx = [(x/sqrt(1 - x2)) x - (1 - sqrt(1 - x2))] / x2 = [1/sqrt(1 - x2) - 1] / x2.

Answer: dy/dx = (1 - sqrt(1 - x2)) / [x2 sqrt(1 - x2)].

Q15. x sqrt(a + x) / sqrt(a - x).

Write y = x w with w = sqrt((a + x)/(a - x)).

From the Q12 pattern (with a + x on top), w' = a / [(a - x) sqrt(a2 - x2)].

By the product rule, dy/dx = w + x w' = sqrt((a + x)/(a - x)) + a x / [(a - x) sqrt(a2 - x2)].

Putting over the common denominator (a - x) sqrt(a2 - x2) gives (a2 - x2 + a x) / [(a - x) sqrt(a2 - x2)].

Answer: dy/dx = (a2 + a x - x2) / [(a - x) sqrt(a2 - x2)].

Q16. If y = sqrt(x) - 1/sqrt(x), show that 2x dy/dx + y = 2 sqrt(x).

Differentiate: dy/dx = 1/(2 sqrt(x)) + 1/(2 x^(3/2)) = (x + 1) / (2 x^(3/2)).

Then 2x dy/dx = (x + 1) / sqrt(x) = sqrt(x) + 1/sqrt(x).

Add y: 2x dy/dx + y = sqrt(x) + 1/sqrt(x) + sqrt(x) - 1/sqrt(x) = 2 sqrt(x).

Answer: proved, 2x dy/dx + y = 2 sqrt(x).

Q17. If y = x4 + 2x2 + 2, prove that dy/dx = 4x sqrt(y - 1).

Differentiate: dy/dx = 4x3 + 4x = 4x(x2 + 1).

Now y - 1 = x4 + 2x2 + 1 = (x2 + 1)2, so sqrt(y - 1) = x2 + 1.

Hence 4x sqrt(y - 1) = 4x(x2 + 1) = dy/dx.

Answer: proved, dy/dx = 4x sqrt(y - 1).

Exercise 2.4, Q1 - Find dy/dx by making suitable substitutions in the following functions

Q1(i). y = sqrt((1 - x) / (1 + x)).

Put u = (1 - x)/(1 + x); then y = sqrt(u) and du/dx = -2/(1 + x)2.

By the chain rule, dy/dx = (1/(2 sqrt(u))) du/dx.

So dy/dx = (1/2) sqrt((1 + x)/(1 - x)) x (-2)/(1 + x)2.

Answer: dy/dx = -1 / [(1 + x) sqrt(1 - x2)].

Q1(ii). y = sqrt(x + sqrt(x)).

Put u = x + sqrt(x); then y = sqrt(u) and du/dx = 1 + 1/(2 sqrt(x)).

By the chain rule, dy/dx = (1/(2 sqrt(u))) (1 + 1/(2 sqrt(x))).

Answer: dy/dx = (2 sqrt(x) + 1) / [4 sqrt(x) sqrt(x + sqrt(x))].

Q1(iii). y = x sqrt((a + x) / (a - x)).

This is the same function as Exercise 2.3, Q15.

Put w = sqrt((a + x)/(a - x)); then w' = a / [(a - x) sqrt(a2 - x2)].

By the product rule, dy/dx = w + x w'.

Answer: dy/dx = (a2 + a x - x2) / [(a - x) sqrt(a2 - x2)].

Q1(iv). y = (3x2 - 2x + 7)6.

Put u = 3x2 - 2x + 7; then y = u6 and du/dx = 6x - 2.

By the chain rule, dy/dx = 6u5 (6x - 2).

Answer: dy/dx = 12(3x - 1)(3x2 - 2x + 7)5.

Q1(v). y = sqrt((a2 + x2) / (a2 - x2)).

Put u = (a2 + x2)/(a2 - x2); then y = sqrt(u).

du/dx = [2x(a2 - x2) + 2x(a2 + x2)] / (a2 - x2)2 = 4a2 x / (a2 - x2)2.

By the chain rule, dy/dx = (1/(2 sqrt(u))) du/dx.

Answer: dy/dx = 2a2 x / [(a2 - x2) sqrt(a4 - x4)].

Exercise 2.4, Q2 - Find dy/dx if

Q2(i). 3x + 4y + 7 = 0.

Differentiate both sides w.r.t. x: 3 + 4 dy/dx = 0.

Answer: dy/dx = -3/4.

Q2(ii). x y + y2 = 2.

Differentiate both sides w.r.t. x, using the product rule on x y.

y + x dy/dx + 2y dy/dx = 0.

So (x + 2y) dy/dx = -y.

Answer: dy/dx = -y / (x + 2y).

Q2(iii). x2 - 4x y - 5y = 0.

Differentiate both sides w.r.t. x: 2x - 4y - 4x dy/dx - 5 dy/dx = 0.

So (4x + 5) dy/dx = 2x - 4y.

Answer: dy/dx = (2x - 4y) / (4x + 5).

Q2(iv). 4x2 + 2h x y + b y2 + 2g x + 2f y + c = 0.

Differentiate both sides w.r.t. x, term by term.

8x + 2h y + 2h x dy/dx + 2b y dy/dx + 2g + 2f dy/dx = 0.

Collect dy/dx: (2h x + 2b y + 2f) dy/dx = -(8x + 2h y + 2g).

Answer: dy/dx = -(4x + h y + g) / (h x + b y + f).

Q2(v). x sqrt(1 + y) + y sqrt(1 + x) = 0.

Differentiate both sides w.r.t. x, using the product rule on each term.

sqrt(1 + y) + x dy/dx / (2 sqrt(1 + y)) + dy/dx sqrt(1 + x) + y / (2 sqrt(1 + x)) = 0.

Collect dy/dx: dy/dx [x / (2 sqrt(1 + y)) + sqrt(1 + x)] = -[sqrt(1 + y) + y / (2 sqrt(1 + x))].

Answer: dy/dx = -sqrt(1 + y) [2 sqrt(1 + x) sqrt(1 + y) + y] / [sqrt(1 + x) (x + 2 sqrt(1 + x) sqrt(1 + y))].

Q2(vi). y(x2 - 1) = x sqrt(x2 + 4).

Differentiate the left side by the product rule: (x2 - 1) dy/dx + 2x y.

Differentiate the right side by the product rule: sqrt(x2 + 4) + x2/sqrt(x2 + 4) = 2(x2 + 2)/sqrt(x2 + 4).

So (x2 - 1) dy/dx = 2(x2 + 2)/sqrt(x2 + 4) - 2x y.

Answer: dy/dx = [2(x2 + 2)/sqrt(x2 + 4) - 2x y] / (x2 - 1).

Exercise 2.4, Q3 - Find dy/dx of the following parametric functions

Q3(i). x = theta + 1/theta and y = theta + 1 (as printed in the book).

dx/d theta = 1 - 1/theta2 = (theta2 - 1)/theta2.

dy/d theta = 1.

dy/dx = (dy/d theta) / (dx/d theta) = 1 / [(theta2 - 1)/theta2].

Answer: dy/dx = theta2 / (theta2 - 1).

Q3(ii). x = a(1 - t2)/(1 + t2), y = 2b t/(1 + t2).

dx/dt = a[(-2t)(1 + t2) - (1 - t2)(2t)] / (1 + t2)2 = -4a t / (1 + t2)2.

dy/dt = 2b[(1 + t2) - t(2t)] / (1 + t2)2 = 2b(1 - t2) / (1 + t2)2.

dy/dx = (dy/dt) / (dx/dt) = 2b(1 - t2) / (-4a t).

Answer: dy/dx = -b(1 - t2) / (2a t).

Exercise 2.4, Q4 - Prove that y dy/dx + x = 0 if x = (1 - t2)/(1 + t2), y = 2t/(1 + t2)

Q4. Proof.

dx/dt = [(-2t)(1 + t2) - (1 - t2)(2t)] / (1 + t2)2 = -4t / (1 + t2)2.

dy/dt = [2(1 + t2) - 2t(2t)] / (1 + t2)2 = 2(1 - t2) / (1 + t2)2.

dy/dx = (dy/dt) / (dx/dt) = -(1 - t2) / (2t).

Then y dy/dx = [2t/(1 + t2)] x [-(1 - t2)/(2t)] = -(1 - t2)/(1 + t2) = -x.

So y dy/dx + x = 0, as required.

Answer: proved.

Exercise 2.4, Q5 - Differentiate (each function w.r.t. the stated variable)

Q5(i). x2 - 1/x2 w.r.t. x4.

Let y = x2 - x^(-2) and u = x4.

dy/dx = 2x + 2x^(-3); du/dx = 4x3.

dy/du = (dy/dx) / (du/dx) = (2x + 2/x3) / (4x3) = (x4 + 1) / (2x6).

Answer: dy/du = (x4 + 1) / (2x6).

Q5(ii). (1 + x2)^n w.r.t. x2.

Let u = x2; then the function is (1 + u)^n.

Differentiating w.r.t. u by the power rule gives n(1 + u)^(n-1).

Answer: n(1 + x2)^(n-1).

Q5(iii). (x2 + 1)/(x2 - 1) w.r.t. (x - 1)/(x + 1).

Let y = (x2 + 1)/(x2 - 1) and u = (x - 1)/(x + 1).

dy/dx = -4x/(x2 - 1)2; du/dx = 2/(x + 1)2.

dy/du = (dy/dx) / (du/dx) = -4x/(x2 - 1)2 x (x + 1)2/2 = -2x/(x - 1)2.

Answer: dy/du = -2x/(x - 1)2, which is -(1 - u2)/(2u2) in terms of u.

Q5(iv). (a x + b)/(c x + d) w.r.t. (a x2 + b)/(a x2 + d).

Let y = (a x + b)/(c x + d) and u = (a x2 + b)/(a x2 + d).

dy/dx = (a d - b c)/(c x + d)2.

du/dx = [2a x (a x2 + d) - (a x2 + b) 2a x] / (a x2 + d)2 = 2a x (d - b)/(a x2 + d)2.

dy/du = (dy/dx) / (du/dx).

Answer: dy/du = (a d - b c)(a x2 + d)2 / [2a x (d - b)(c x + d)2].

Q5(v). (x2 + 1)/(x2 - 1) w.r.t. x3.

Let y = (x2 + 1)/(x2 - 1) and u = x3.

dy/dx = -4x/(x2 - 1)2; du/dx = 3x2.

dy/du = (dy/dx) / (du/dx) = -4x / [3x2 (x2 - 1)2].

Answer: dy/du = -4 / [3x (x2 - 1)2].

Exercise 2.5, Q1 - Differentiate the following trigonometric functions from the first principle

Q1(i). sin x.

Let y = sin x. Then y + delta y = sin(x + delta x).

delta y = sin(x + delta x) - sin x = 2 cos(x + delta x/2) sin(delta x/2).

So delta y / delta x = cos(x + delta x/2) [sin(delta x/2) / (delta x/2)].

Taking the limit as delta x -> 0: cos(x + delta x/2) -> cos x and the bracket -> 1.

Answer: dy/dx = cos x.

Q1(ii). tan 3x.

Let y = tan 3x. Use the first-principle result for tan together with the chain rule form: d/dx tan u = sec2 u du/dx.

Here u = 3x, so du/dx = 3.

Answer: dy/dx = 3 sec2 3x.

Q1(iii). sin 2x + cos 2x.

Differentiate term by term from the first-principle sine and cosine results with the chain rule.

d/dx (sin 2x) = 2 cos 2x; d/dx (cos 2x) = -2 sin 2x.

Answer: dy/dx = 2 cos 2x - 2 sin 2x.

Q1(iv). cos x2.

Put u = x2; then y = cos u and du/dx = 2x.

By the chain rule, dy/dx = -sin u du/dx.

Answer: dy/dx = -2x sin x2.

Q1(v). tan2 x.

Put u = tan x; then y = u2 and du/dx = sec2 x.

By the chain rule, dy/dx = 2u sec2 x.

Answer: dy/dx = 2 tan x sec2 x.

Q1(vi). sqrt(tan x).

Put u = tan x; then y = sqrt(u) and du/dx = sec2 x.

By the chain rule, dy/dx = (1/(2 sqrt(u))) sec2 x.

Answer: dy/dx = sec2 x / (2 sqrt(tan x)).

Q1(vii). cos sqrt(x).

Put u = sqrt(x); then y = cos u and du/dx = 1/(2 sqrt(x)).

By the chain rule, dy/dx = -sin u du/dx.

Answer: dy/dx = -sin sqrt(x) / (2 sqrt(x)).

Exercise 2.5, Q2 - Differentiate the following w.r.t. the variable involved

Q2(i). x2 sec 4x.

Use the product rule: d/dx (x2) = 2x; d/dx (sec 4x) = 4 sec 4x tan 4x.

dy/dx = 2x sec 4x + x2 (4 sec 4x tan 4x).

Answer: dy/dx = 2x sec 4x + 4x2 sec 4x tan 4x.

Q2(ii). tan3 theta sec2 theta (w.r.t. theta).

Use the product rule: d/d theta (tan3 theta) = 3 tan2 theta sec2 theta; d/d theta (sec2 theta) = 2 sec2 theta tan theta.

dy/d theta = 3 tan2 theta sec4 theta + 2 tan4 theta sec2 theta.

Answer: dy/d theta = tan2 theta sec2 theta (3 sec2 theta + 2 tan2 theta).

Q2(iii). (sin 2theta - cos 3theta)2 (w.r.t. theta).

Put u = sin 2theta - cos 3theta; then y = u2 and du/d theta = 2 cos 2theta + 3 sin 3theta.

By the chain rule, dy/d theta = 2u du/d theta.

Answer: dy/d theta = 2(sin 2theta - cos 3theta)(2 cos 2theta + 3 sin 3theta).

Q2(iv). cos sqrt(x) + sqrt(sin x).

d/dx (cos sqrt(x)) = -sin sqrt(x) / (2 sqrt(x)) from Q1(vii).

d/dx (sqrt(sin x)) = cos x / (2 sqrt(sin x)) by the chain rule.

Answer: dy/dx = -sin sqrt(x) / (2 sqrt(x)) + cos x / (2 sqrt(sin x)).

Exercise 2.5, Q3 - Find dy/dx if

Q3(i). y = x cos y.

Differentiate both sides w.r.t. x, using the product rule on x cos y.

dy/dx = cos y - x sin y dy/dx.

Collect dy/dx: (1 + x sin y) dy/dx = cos y.

Answer: dy/dx = cos y / (1 + x sin y).

Q3(ii). x = y sin y.

Differentiate both sides w.r.t. x: 1 = dy/dx sin y + y cos y dy/dx.

So (sin y + y cos y) dy/dx = 1.

Answer: dy/dx = 1 / (sin y + y cos y).

Exercise 2.5, Q4 - Find the derivative w.r.t. x

Q4(i). cos sqrt((1 + x)/(1 + 2x)).

Put u = sqrt((1 + x)/(1 + 2x)). Then d/dx [(1 + x)/(1 + 2x)] = -1/(1 + 2x)2.

So du/dx = (1/(2u)) x (-1)/(1 + 2x)2 = -1/[2u(1 + 2x)2].

dy/dx = -sin u du/dx.

Answer: dy/dx = sin sqrt((1 + x)/(1 + 2x)) / [2 sqrt((1 + x)/(1 + 2x)) (1 + 2x)2].

Q4(ii). sin sqrt((1 + 2x)/(1 + x)).

Put u = sqrt((1 + 2x)/(1 + x)). Then d/dx [(1 + 2x)/(1 + x)] = 1/(1 + x)2.

So du/dx = (1/(2u)) x 1/(1 + x)2 = 1/[2u(1 + x)2].

dy/dx = cos u du/dx.

Answer: dy/dx = cos sqrt((1 + 2x)/(1 + x)) / [2 sqrt((1 + 2x)/(1 + x)) (1 + x)2].

Exercise 2.5, Q5 - Differentiate

Q5(i). sin x w.r.t. cot x.

Let y = sin x and u = cot x.

dy/dx = cos x; du/dx = -cosec2 x.

dy/du = (dy/dx) / (du/dx) = cos x / (-cosec2 x) = -cos x sin2 x.

Answer: dy/du = -sin2 x cos x.

Q5(ii). sin2 x w.r.t. cos4 x.

Let y = sin2 x and u = cos4 x.

dy/dx = 2 sin x cos x; du/dx = -4 cos3 x sin x.

dy/du = (2 sin x cos x) / (-4 cos3 x sin x) = -1/(2 cos2 x).

Answer: dy/du = -(1/2) sec2 x.

Exercise 2.5, Q6 - If tan y (1 + tan x) = 1 - tan x, show that dy/dx = -1

Q6. Proof.

Divide both sides by (1 + tan x): tan y = (1 - tan x)/(1 + tan x).

The right side is tan(pi/4 - x), since tan(A - B) with A = pi/4 gives (1 - tan x)/(1 + tan x).

So y = pi/4 - x (up to a multiple of pi).

Differentiating w.r.t. x: dy/dx = -1.

Answer: proved, dy/dx = -1.

Exercise 2.5, Q7 - If y = sqrt(tan x + sqrt(tan x + sqrt(tan x + ... to infinity))), prove that (2y - 1) dy/dx = sec2 x

Q7. Proof.

The infinite nest repeats itself, so y = sqrt(tan x + y).

Square both sides: y2 = tan x + y.

Differentiate w.r.t. x: 2y dy/dx = sec2 x + dy/dx.

Collect dy/dx: (2y - 1) dy/dx = sec2 x, as required.

Answer: proved.

Exercise 2.5, Q8 - If x = a cos3 theta, y = b sin3 theta, show that a dy/dx + b tan theta = 0

Q8. Proof.

dx/d theta = -3a cos2 theta sin theta.

dy/d theta = 3b sin2 theta cos theta.

dy/dx = (dy/d theta) / (dx/d theta) = -b sin theta / (a cos theta) = -(b/a) tan theta.

Multiply by a: a dy/dx = -b tan theta, so a dy/dx + b tan theta = 0.

Answer: proved.

Exercise 2.5, Q9 - Find dy/dx if x = a(cos t + sin t), y = a(sin t - t cos t) (as printed in the book)

Q9. As printed.

dx/dt = a(-sin t + cos t) = a(cos t - sin t).

dy/dt = a(cos t - cos t + t sin t) = a t sin t.

dy/dx = (dy/dt) / (dx/dt) = a t sin t / [a(cos t - sin t)].

Answer: dy/dx = t sin t / (cos t - sin t).

Exercise 2.5, Q10 - Differentiate w.r.t. x

Q10(i). Cos^{-1} (x/a).

Put u = x/a; then y = Cos^{-1} u and du/dx = 1/a.

dy/dx = (-1/sqrt(1 - u2)) (1/a) = -1/[a sqrt(1 - x2/a2)].

Answer: dy/dx = -1/sqrt(a2 - x2).

Q10(ii). Cot^{-1} (x/a).

Put u = x/a; then y = Cot^{-1} u and du/dx = 1/a.

dy/dx = (-1/(1 + u2)) (1/a) = -1/[a(1 + x2/a2)].

Answer: dy/dx = -a/(a2 + x2).

Q10(iii). (1/a) Sin^{-1} (a/x).

Put u = a/x; then y = (1/a) Sin^{-1} u and du/dx = -a/x2.

dy/dx = (1/a) (1/sqrt(1 - u2)) (-a/x2) = -1/[x2 sqrt(1 - a2/x2)].

Answer: dy/dx = -1/[x sqrt(x2 - a2)].

Q10(iv). Sin^{-1} sqrt(1 - x2).

Put u = sqrt(1 - x2); then y = Sin^{-1} u and du/dx = -x/sqrt(1 - x2).

dy/dx = (1/sqrt(1 - u2)) du/dx = (1/|x|) (-x/sqrt(1 - x2)).

For x > 0 this is -1/sqrt(1 - x2).

Answer: dy/dx = -1/sqrt(1 - x2) (for x > 0).

Q10(v). Sec^{-1} ((x2 + 1)/(x2 - 1)).

Put u = (x2 + 1)/(x2 - 1); then du/dx = -4x/(x2 - 1)2.

Also u2 - 1 = 4x2/(x2 - 1)2, so sqrt(u2 - 1) = 2|x|/|x2 - 1|.

dy/dx = [1/(u sqrt(u2 - 1))] du/dx = -2/(x2 + 1) for x2 > 1.

Answer: dy/dx = -2/(x2 + 1).

Q10(vi). Cot^{-1} (2x/(1 - x2)).

Put u = 2x/(1 - x2); then du/dx = 2(1 + x2)/(1 - x2)2.

Also 1 + u2 = (1 + x2)2/(1 - x2)2.

dy/dx = (-1/(1 + u2)) du/dx = -[(1 - x2)2/(1 + x2)2] x 2(1 + x2)/(1 - x2)2.

Answer: dy/dx = -2/(1 + x2).

Q10(vii). Cos^{-1} ((1 - x2)/(1 + x2)).

Put u = (1 - x2)/(1 + x2); then du/dx = -4x/(1 + x2)2.

Also 1 - u2 = 4x2/(1 + x2)2, so sqrt(1 - u2) = 2|x|/(1 + x2).

dy/dx = (-1/sqrt(1 - u2)) du/dx = 2/(1 + x2) for x > 0.

Answer: dy/dx = 2/(1 + x2).

Exercise 2.5, Q11 - Show that dy/dx = y/x if y/x = Tan^{-1} (x/y)

Q11. Proof.

Put z = y/x; then z = Tan^{-1} (x/y), so tan z = x/y.

Differentiating z = y/x w.r.t. x by the quotient rule: dz/dx = (x dy/dx - y)/x2.

Differentiating z = Tan^{-1} (x/y) w.r.t. x by the chain rule: dz/dx = [1/(1 + (x/y)2)] x [(y - x dy/dx)/y2] = (y - x dy/dx)/(x2 + y2).

Equate the two: (x dy/dx - y)/x2 = (y - x dy/dx)/(x2 + y2) = -(x dy/dx - y)/(x2 + y2).

So (x dy/dx - y)[1/x2 + 1/(x2 + y2)] = 0, and the bracket is not zero.

Hence x dy/dx - y = 0, so dy/dx = y/x, as required.

Answer: proved.

Exercise 2.5, Q12 - If y = tan(p Tan^{-1} x), show that (1 + x2) y1 - p(1 + y2) = 0

Q12. Proof.

Put u = Tan^{-1} x; then y = tan(p u) and du/dx = 1/(1 + x2).

By the chain rule, y1 = dy/dx = p sec2(p u) du/dx = p(1 + tan2(p u))/(1 + x2).

Since tan(p u) = y, this is y1 = p(1 + y2)/(1 + x2).

Multiply by (1 + x2): (1 + x2) y1 = p(1 + y2).

Answer: proved, (1 + x2) y1 - p(1 + y2) = 0.

Exercise 2.6, Q1 - Find f'(x) if

Q1(i). f(x) = e^(sqrt(x) - 1).

Put u = sqrt(x) - 1; then f = e^u and du/dx = 1/(2 sqrt(x)).

By the chain rule, f'(x) = e^u du/dx.

Answer: f'(x) = e^(sqrt(x) - 1) / (2 sqrt(x)).

Q1(ii). f(x) = x3 e^(1/x) (x not equal to 0).

Use the product rule: d/dx (x3) = 3x2; d/dx (e^(1/x)) = e^(1/x) (-1/x2).

f'(x) = 3x2 e^(1/x) + x3 e^(1/x) (-1/x2) = e^(1/x)(3x2 - x).

Answer: f'(x) = e^(1/x)(3x2 - x).

Q1(iii). f(x) = e^x (1 + ln x).

Use the product rule: d/dx (e^x) = e^x; d/dx (1 + ln x) = 1/x.

f'(x) = e^x (1 + ln x) + e^x (1/x).

Answer: f'(x) = e^x (1 + ln x + 1/x).

Q1(iv). f(x) = e^x / (e^(-x) + 1).

Use the quotient rule: numerator derivative e^x; denominator derivative -e^(-x).

f'(x) = [e^x (e^(-x) + 1) - e^x (-e^(-x))] / (e^(-x) + 1)2 = (e^x + 2) / (e^(-x) + 1)2.

Since e^(-x) + 1 = (1 + e^x)/e^x, this simplifies to e^(2x)(e^x + 2)/(e^x + 1)2.

Answer: f'(x) = e^(2x)(e^x + 2) / (e^x + 1)2.

Q1(v). f(x) = ln(e^x + e^(-x)).

By the chain rule, f'(x) = (e^x - e^(-x)) / (e^x + e^(-x)).

Answer: f'(x) = (e^x - e^(-x)) / (e^x + e^(-x)) = tanh x.

Q1(vi). f(x) = (e^(a x) - e^(-a x)) / (e^(a x) + e^(-a x)).

This is tanh(a x). Use the quotient rule with S = e^(a x) + e^(-a x) and D = e^(a x)

- e^(-a x).

S' = a D and D' = a S, so f'(x) = a(S2 - D2) / S2.

Since S2 - D2 = 4, f'(x) = 4a / S2.

Answer: f'(x) = 4a / (e^(a x) + e^(-a x))2 = a sech2(a x).

Q1(vii). f(x) = sqrt(ln(e^(2x) + e^(-2x))).

Put u = ln(e^(2x) + e^(-2x)); then f = sqrt(u) and du/dx = (2e^(2x) - 2e^(-2x)) / (e^(2x) + e^(-2x)).

By the chain rule, f'(x) = (1/(2 sqrt(u))) du/dx.

Answer: f'(x) = (e^(2x) - e^(-2x)) / [(e^(2x) + e^(-2x)) sqrt(ln(e^(2x) + e^(-2x)))].

Q1(viii). f(x) = ln(sqrt(e^(2x) + e^(-2x))).

Write f(x) = (1/2) ln(e^(2x) + e^(-2x)).

Differentiate: f'(x) = (1/2) (2e^(2x) - 2e^(-2x)) / (e^(2x) + e^(-2x)).

Answer: f'(x) = (e^(2x) - e^(-2x)) / (e^(2x) + e^(-2x)).

Exercise 2.6, Q2 - Find dy/dx if

Q2(i). y = x2 ln sqrt(x).

Write y = (1/2) x2 ln x.

By the product rule, dy/dx = x ln x + (1/2) x2 (1/x) = x ln x + x/2.

Answer: dy/dx = x(ln x + 1/2).

Q2(ii). y = x sqrt(ln x).

Use the product rule: dy/dx = sqrt(ln x) + x (1/(2 sqrt(ln x))) (1/x).

Answer: dy/dx = sqrt(ln x) + 1/(2 sqrt(ln x)).

Q2(iii). y = x / ln x.

Use the quotient rule: dy/dx = [(ln x) - x(1/x)] / (ln x)2.

Answer: dy/dx = (ln x - 1) / (ln x)2.

Q2(iv). y = x2 ln(1/x).

Write y = -x2 ln x.

By the product rule, dy/dx = -2x ln x - x.

Answer: dy/dx = -x(2 ln x + 1).

Q2(v). y = ln sqrt((x2 - 1)/(x2 + 1)).

Write y = (1/2)[ln(x2 - 1) - ln(x2 + 1)].

dy/dx = (1/2)[2x/(x2 - 1) - 2x/(x2 + 1)] = x[1/(x2 - 1) - 1/(x2 + 1)].

Answer: dy/dx = 2x / (x4 - 1).

Q2(vi). y = ln(x + sqrt(x2 + 1)).

By the chain rule, dy/dx = [1 + x/sqrt(x2 + 1)] / (x + sqrt(x2 + 1)).

The numerator is (sqrt(x2 + 1) + x)/sqrt(x2 + 1), which cancels the denominator.

Answer: dy/dx = 1/sqrt(x2 + 1).

Q2(vii). y = ln(9 - x2).

By the chain rule, dy/dx = (-2x)/(9 - x2).

Answer: dy/dx = -2x/(9 - x2).

Q2(viii). y = e^(-2x) sin 2x.

Use the product rule: dy/dx = -2e^(-2x) sin 2x + 2e^(-2x) cos 2x.

Answer: dy/dx = 2e^(-2x)(cos 2x - sin 2x).

Q2(ix). y = e^(-x)(x3 + 2x2 + 1).

Use the product rule: dy/dx = -e^(-x)(x3 + 2x2 + 1) + e^(-x)(3x2 + 4x).

Answer: dy/dx = e^(-x)(-x3 + x2 + 4x - 1).

Q2(x). y = x e^(sin x).

Use the product rule: dy/dx = e^(sin x) + x e^(sin x) cos x.

Answer: dy/dx = e^(sin x)(1 + x cos x).

Q2(xi). y = 5e^(3x - 4).

By the chain rule, dy/dx = 5e^(3x - 4) x 3.

Answer: dy/dx = 15e^(3x - 4).

Q2(xii). y = (x + 1)^x.

Take logarithms: ln y = x ln(x + 1).

Differentiate: y'/y = ln(x + 1) + x/(x + 1).

Answer: dy/dx = (x + 1)^x [ln(x + 1) + x/(x + 1)].

Q2(xiii). y = (ln x)^(ln x).

Take logarithms: ln y = (ln x)(ln ln x).

Differentiate: y'/y = (1/x) ln ln x + (ln x)(1/(x ln x)) = (1 + ln ln x)/x.

Answer: dy/dx = (ln x)^(ln x) (1 + ln ln x)/x.

Q2(xiv). y = sqrt(x2 - 1)(x + 1) / (x3 + 1)^(3/2).

Simplify first: x3 + 1 = (x + 1)(x2 - x + 1), so y = sqrt(x2 - 1) / [sqrt(x + 1)(x2 - x + 1)^(3/2)].

Take logarithms: ln y = (1/2) ln(x2 - 1) - (1/2) ln(x + 1) - (3/2) ln(x2 - x + 1).

Differentiate: y'/y = x/(x2 - 1) - 1/[2(x + 1)] - 3(2x - 1)/[2(x2 - x + 1)].

The first two terms combine to 1/[2(x - 1)], so y'/y = (-5x2 + 8x - 2)/[2(x - 1)(x2 - x + 1)].

Answer: dy/dx = y (-5x2 + 8x - 2) / [2(x - 1)(x2 - x + 1)], with y as given.

Exercise 2.6, Q3 - Find dy/dx if

Q3(i). y = cosh 2x.

By the chain rule, dy/dx = 2 sinh 2x.

Answer: dy/dx = 2 sinh 2x.

Q3(ii). y = sinh 3x.

By the chain rule, dy/dx = 3 cosh 3x.

Answer: dy/dx = 3 cosh 3x.

Q3(iii). y = tanh^{-1}(sin x), -pi/2 < x < pi/2.

By the chain rule, dy/dx = [1/(1 - sin2 x)] cos x = cos x / cos2 x.

Answer: dy/dx = sec x.

Q3(iv). y = sinh^{-1}(x3).

By the chain rule, dy/dx = (3x2)/sqrt(1 + x6).

Answer: dy/dx = 3x2/sqrt(1 + x6).

Q3(v). y = ln(tanh x).

By the chain rule, dy/dx = (1/tanh x) sech2 x = 1/(sinh x cosh x).

Answer: dy/dx = 2/sinh 2x.

Q3(vi). y = sinh^{-1}(x/2).

By the chain rule, dy/dx = (1/2)/sqrt(1 + x2/4).

Answer: dy/dx = 1/sqrt(4 + x2).

Exercise 2.7, Q1 - Find y2 if

Q1(i). y = 2x5 - 3x4 + 4x3 + x - 2.

First derivative: y1 = 10x4 - 12x3 + 12x2 + 1.

Second derivative: y2 = 40x3 - 36x2 + 24x.

Answer: y2 = 40x3 - 36x2 + 24x.

Q1(ii). y = (2x + 5)^(3/2).

First derivative: y1 = 3(2x + 5)^(1/2).

Second derivative: y2 = 3(2x + 5)^(-1/2).

Answer: y2 = 3/sqrt(2x + 5).

Q1(iii). y = sqrt(x) + 1/sqrt(x).

Write y = x^(1/2) + x^(-1/2).

First derivative: y1 = (1/2)x^(-1/2) - (1/2)x^(-3/2).

Second derivative: y2 = -(1/4)x^(-3/2) + (3/4)x^(-5/2).

Answer: y2 = (3 - x) / (4 x^(5/2)).

Exercise 2.7, Q2 - Find y2 if

Q2(i). y = x2 e^(-x).

First derivative: y1 = e^(-x)(2x - x2).

Second derivative: y2 = e^(-x)(2 - 2x) - e^(-x)(2x - x2).

Answer: y2 = e^(-x)(x2 - 4x + 2).

Q2(ii). y = ln((2x + 3)/(3x + 2)).

First derivative: y1 = 2/(2x + 3) - 3/(3x + 2) = -5/[(2x + 3)(3x + 2)].

Second derivative: y2 = 5(12x + 13)/[(2x + 3)2(3x + 2)2].

Answer: y2 = 5(12x + 13) / [(2x + 3)2(3x + 2)2].

Exercise 2.7, Q3 - Find y2 if

Q3(i). x2 + y2 = a2.

Differentiate: 2x + 2y y1 = 0, so y1 = -x/y.

Differentiate again: y2 = [-(y) + x y1]/y2 = [-(y) - x2/y]/y2.

Since x2 + y2 = a2, this simplifies to y2 = -a2/y3.

Answer: y2 = -a2/y3.

Q3(ii). x3 - y3 = a.

Differentiate: 3x2 - 3y2 y1 = 0, so y1 = x2/y2.

Differentiate again: y2 = [2x y2 - x2 (2y y1)]/y4 = 2x(y3 - x3)/y5.

Since x3 - y3 = a, y3 - x3 = -a.

Answer: y2 = -2a x/y5.

Q3(iii). x = a cos theta, y = a sin theta.

dy/dx = (a cos theta)/(-a sin theta) = -cot theta.

y2 = [d/d theta (-cot theta)] / (dx/d theta) = cosec2 theta / (-a sin theta).

Answer: y2 = -1/(a sin3 theta).

Q3(iv). x = a t2, y = b t4.

dx/dt = 2a t; dy/dt = 4b t3, so y1 = (2b/a) t2.

y2 = [d/dt (2b t2/a)] / (dx/dt) = (4b t/a)/(2a t).

Answer: y2 = 2b/a2.

Q3(v). x2 + y2 + 2g x + 2f y + c = 0.

Differentiate: (x + g) + (y + f) y1 = 0, so y1 = -(x + g)/(y + f).

Differentiate again: 1 + (y1)2 + (y + f) y2 = 0, so y2 = -(1 + y1^2)/(y + f).

Answer: y2 = -[(y + f)2 + (x + g)2] / (y + f)3.

Exercise 2.7, Q4 - Find y4 if

Q4(i). y = sin 3x.

y1 = 3 cos 3x; y2 = -9 sin 3x; y3 = -27 cos 3x.

Fourth derivative: y4 = 81 sin 3x.

Answer: y4 = 81 sin 3x.

Q4(ii). y = cos3 x.

Use cos3 x = (3 cos x + cos 3x)/4.

Fourth derivatives: d4/dx4 cos x = cos x; d4/dx4 cos 3x = 81 cos 3x.

Answer: y4 = (3 cos x + 81 cos 3x)/4.

Q4(iii). y = ln(x2 - 9).

y1 = 2x/(x2 - 9); y2 = -2(x2 + 9)/(x2 - 9)2.

y3 = 4x(x2 + 27)/(x2 - 9)3.

Differentiating once more gives y4 = -12(x4 + 54x2 + 81)/(x2 - 9)4.

Answer: y4 = -12(x4 + 54x2 + 81) / (x2 - 9)4.

Exercise 2.7, Q5 - If x = sin theta, y = sin m theta, show that (1 - x2) y2 - x y1 + m2 y = 0

Q5. Proof.

dx/d theta = cos theta; dy/d theta = m cos m theta, so y1 = m cos m theta / cos theta.

Differentiating y1 w.r.t. theta and dividing by dx/d theta gives y2 = m[-m sin m theta + cos m theta tan theta] / cos2 theta.

Now (1 - x2) = cos2 theta, so (1 - x2) y2 = -m2 sin m theta + m cos m theta tan theta.

Also -x y1 = -m sin theta cos m theta / cos theta = -m cos m theta tan theta.

Adding m2 y = m2 sin m theta cancels everything: total 0.

Answer: proved.

Exercise 2.7, Q6 - If y = e^x sin x, show that d2y/dx2 - 2 dy/dx + 2y = 0

Q6. Proof.

y1 = e^x (sin x + cos x).

y2 = e^x (2 cos x).

Then y2 - 2y1 + 2y = e^x[2 cos x - 2 sin x - 2 cos x + 2 sin x] = 0.

Answer: proved.

Exercise 2.7, Q7 - If y = e^(a x) sin b x, show that d2y/dx2 - 2a dy/dx + (a2 + b2) y = 0

Q7. Proof.

y1 = e^(a x)(a sin b x + b cos b x).

y2 = e^(a x)[(a2 - b2) sin b x + 2a b cos b x].

Then y2 - 2a y1 + (a2 + b2) y = e^(a x)[(a2 - b2 - 2a2 + a2 + b2) sin b x + (2a b - 2a b) cos b x] = 0.

Answer: proved.

Exercise 2.7, Q8 - If y = (Cos^{-1} x)2, prove that (1 - x2) y2 - x y1 - 2 = 0

Q8. Proof.

y1 = 2 Cos^{-1} x (-1/sqrt(1 - x2)) = -2 Cos^{-1} x / sqrt(1 - x2).

So sqrt(1 - x2) y1 = -2 Cos^{-1} x.

Differentiate: (-x/sqrt(1 - x2)) y1 + sqrt(1 - x2) y2 = 2/sqrt(1 - x2).

Multiply by sqrt(1 - x2): -x y1 + (1 - x2) y2 = 2.

Answer: proved, (1 - x2) y2 - x y1 - 2 = 0.

Exercise 2.7, Q9 - If y = a cos(ln x) + b sin(ln x), prove that x2 d2y/dx2 + x dy/dx + y = 0

Q9. Proof.

y1 = (-a sin(ln x) + b cos(ln x))/x, so x y1 = -a sin(ln x) + b cos(ln x).

Differentiating y1 by the quotient rule gives x2 y2 = -a cos(ln x) - b sin(ln x) + a sin(ln x) - b cos(ln x).

Add: x2 y2 + x y1 + y = (-a cos - b sin + a sin - b cos) + (-a sin + b cos) + (a cos + b sin) = 0.

Answer: proved.

Exercise 2.8, Q1 - Apply the Maclaurin series expansion to prove that

Q1(i). ln(1 + x) = x - x2/2 + x3/3 - x4/4 + ...

Let f(x) = ln(1 + x). Then f(0) = 0.

f'(x) = 1/(1 + x), so f'(0) = 1; f''(x) = -1/(1 + x)2, so f''(0) = -1.

f'''(x) = 2/(1 + x)3, so f'''(0) = 2; f''''(x) = -6/(1 + x)4, so f''''(0) = -6.

Maclaurin: f(x) = f(0) + f'(0) x + f''(0) x2/2! + f'''(0) x3/3! + f''''(0) x4/4! + ...

So ln(1 + x) = x - x2/2 + x3/3 - x4/4 + ..., as required.

Answer: proved.

Q1(ii). cos x = 1 - x2/2! + x4/4! - x6/6! + ...

Let f(x) = cos x. Then f(0) = 1, f'(0) = 0, f''(0) = -1, f'''(0) = 0, f''''(0) = 1, and the pattern repeats every four derivatives.

Maclaurin: f(x) = 1 + 0 x - x2/2! + 0 x3/3! + x4/4! + 0 x5/5! - x6/6! + ...

Answer: proved, cos x = 1 - x2/2! + x4/4! - x6/6! + ...

Q1(iii). sqrt(1 + x) = 1 + x/2 - x2/8 + x3/16 - ...

Let f(x) = (1 + x)^(1/2). Then f(0) = 1.

f'(x) = (1/2)(1 + x)^(-1/2), so f'(0) = 1/2.

f''(x) = (-1/4)(1 + x)^(-3/2), so f''(0) = -1/4.

f'''(x) = (3/8)(1 + x)^(-5/2), so f'''(0) = 3/8.

Maclaurin: f(x) = 1 + (1/2) x + (-1/4) x2/2! + (3/8) x3/3! + ... = 1 + x/2 - x2/8 + x3/16 - ...

Answer: proved.

Q1(iv). e^x = 1 + x + x2/2! + x3/3! + ...

Let f(x) = e^x. Every derivative is e^x, so every derivative at 0 is 1.

Maclaurin: f(x) = 1 + x + x2/2! + x3/3! + ...

Answer: proved.

Q1(v). e^(2x) = 1 + 2x + 4x2/2! + 8x3/3! + ...

Let f(x) = e^(2x). Then f'(x) = 2e^(2x), f''(x) = 4e^(2x), f'''(x) = 8e^(2x), so f(0) = 1, f'(0) = 2, f''(0) = 4, f'''(0) = 8.

Maclaurin: f(x) = 1 + 2x + 4x2/2! + 8x3/3! + ...

Answer: proved.

Exercise 2.8, Q2 - Show that cos(x + h) = cos x - h sin x - (h2/2) cos x + (h3/6) sin x + ... and evaluate cos 61 degrees

Q2. Taylor expansion and cos 61 degrees.

Let f(x) = cos x. Taylor in h: f(x + h) = f(x) + h f'(x) + (h2/2) f''(x) + (h3/6) f'''(x) + ...

With f'(x) = -sin x, f''(x) = -cos x, f'''(x) = sin x, this gives cos(x + h) = cos x - h sin x - (h2/2) cos x + (h3/6) sin x + ...

For cos 61 degrees, take x = 60 degrees and h = 1 degree = pi/180 radians approx. 0.017453.

cos 60 degrees = 0.5 and sin 60 degrees = sqrt(3)/2 approx. 0.8660.

cos 61 degrees approx. 0.5 - (0.017453)(0.8660) - (0.0174532/2)(0.5) = 0.5 - 0.015114 - 0.000076.

Answer: cos 61 degrees approx. 0.48481.

Exercise 2.8, Q3 - Show that 2^(x + h) = 2^x {1 + (ln 2) h + (ln 2)2 h2/2! + (ln 2)3 h3/3! + ...}

Q3. Proof.

Let f(x) = 2^x. Then f'(x) = 2^x ln 2, f''(x) = 2^x (ln 2)2, f'''(x) = 2^x (ln 2)3, and so on.

Taylor in h: f(x + h) = f(x) + h f'(x) + (h2/2) f''(x) + (h3/6) f'''(x) + ...

Factor out 2^x: 2^(x + h) = 2^x [1 + (ln 2) h + (ln 2)2 h2/2! + (ln 2)3 h3/3! + ...].

Answer: proved.

Exercise 2.9, Q1 - Determine the intervals in which f is increasing or decreasing for the domain mentioned in each case

Q1(i). f(x) = sin x, x in (-pi, pi).

f'(x) = cos x.

cos x > 0 on (-pi/2, pi/2), so f is increasing there.

cos x < 0 on (-pi, -pi/2) and on (pi/2, pi), so f is decreasing there.

Answer: increasing on (-pi/2, pi/2); decreasing on (-pi, -pi/2) and (pi/2, pi).

Q1(ii). f(x) = cos x, x in (-pi/2, pi/2).

f'(x) = -sin x.

-sin x > 0 when sin x < 0, i.e. on (-pi/2, 0), so f is increasing there.

-sin x < 0 on (0, pi/2), so f is decreasing there.

Answer: increasing on (-pi/2, 0); decreasing on (0, pi/2).

Q1(iii). f(x) = 4 - x2, x in (-2, 2).

f'(x) = -2x.

f'(x) > 0 for x < 0, so f is increasing on (-2, 0).

f'(x) < 0 for x > 0, so f is decreasing on (0, 2).

Answer: increasing on (-2, 0); decreasing on (0, 2).

Q1(iv). f(x) = x2 + 3x + 2, x in (-4, 1).

f'(x) = 2x + 3, which is zero at x = -3/2.

f'(x) < 0 for x < -3/2, so f is decreasing on (-4, -3/2).

f'(x) > 0 for x > -3/2, so f is increasing on (-3/2, 1).

Answer: decreasing on (-4, -3/2); increasing on (-3/2, 1).

Exercise 2.9, Q2 - Find the extreme values for the following functions

Q2(i). f(x) = 1 - x3.

f'(x) = -3x2, which is zero only at x = 0; f''(0) = 0.

f'(x) <= 0 on both sides of 0, so f does not change from increasing to decreasing or back.

Answer: no maximum or minimum; (0, 1) is a point of inflexion.

Q2(ii). f(x) = x2 - x - 2.

f'(x) = 2x - 1, zero at x = 1/2; f''(x) = 2 > 0.

So f has a minimum at x = 1/2, value 1/4 - 1/2 - 2 = -9/4.

Answer: minimum value -9/4 at x = 1/2.

Q2(iii). f(x) = 5x2 - 6x + 2.

f'(x) = 10x - 6, zero at x = 3/5; f''(x) = 10 > 0.

So f has a minimum at x = 3/5, value 5(9/25) - 6(3/5) + 2 = 1/5.

Answer: minimum value 1/5 at x = 3/5.

Q2(iv). f(x) = 3x2.

f'(x) = 6x, zero at x = 0; f''(x) = 6 > 0.

So f has a minimum at x = 0, value 0.

Answer: minimum value 0 at x = 0.

Q2(v). f(x) = 3x2 - 4x + 5.

f'(x) = 6x - 4, zero at x = 2/3; f''(x) = 6 > 0.

So f has a minimum at x = 2/3, value 3(4/9) - 4(2/3) + 5 = 11/3.

Answer: minimum value 11/3 at x = 2/3.

Q2(vi). f(x) = 2x3 - 2x2 - 36x + 3.

f'(x) = 6x2 - 4x - 36 = 2(3x2 - 2x - 18), zero at x = (1 + sqrt(55))/3 and x = (1 - sqrt(55))/3.

f''(x) = 12x - 4: positive at the first point, negative at the second.

Using 3x2 - 2x - 18 = 0 to reduce the value gives f = (-220x - 9)/9 at a critical point.

Answer: maximum value (220 sqrt(55) - 247)/27 approx. 51.28 at x = (1 - sqrt(55))/3; minimum value -(220 sqrt(55) + 247)/27 approx. -69.57 at x = (1 + sqrt(55))/3.

Q2(vii). f(x) = x4 - 4x2.

f'(x) = 4x3 - 8x = 4x(x2 - 2), zero at x = 0 and x = plus or minus sqrt(2).

f''(x) = 12x2 - 8: at x = 0 it is -8 (maximum, value 0); at x = plus or minus sqrt(2) it is 16 (minimum, value 4 - 8 = -4).

Answer: maximum value 0 at x = 0; minimum value -4 at x = sqrt(2) and at x = -sqrt(2).

Q2(viii). f(x) = (x - 2)2(x - 1).

Expand: f(x) = x3 - 5x2 + 8x - 4.

f'(x) = 3x2 - 10x + 8 = (3x - 4)(x - 2), zero at x = 4/3 and x = 2.

f''(x) = 6x - 10: at x = 4/3 it is -2 (maximum, value (4/9)(1/3) = 4/27); at x = 2 it is 2 (minimum, value 0).

Answer: maximum value 4/27 at x = 4/3; minimum value 0 at x = 2.

Q2(ix). f(x) = 5 + 3x - x3.

f'(x) = 3 - 3x2, zero at x = 1 and x = -1; f''(x) = -6x.

At x = -1, f'' = 6 > 0: minimum value 5 - 3 + 1 = 3.

At x = 1, f'' = -6 < 0: maximum value 5 + 3 - 1 = 7.

Answer: minimum value 3 at x = -1; maximum value 7 at x = 1.

Exercise 2.9, Q3 - Find the maximum and minimum values of f(x) = sin x + cos x in the interval [0, 2pi]

Q3. Maxima and minima on [0, 2pi].

f'(x) = cos x - sin x, zero when tan x = 1, i.e. at x = pi/4 and x = 5pi/4 in [0, 2pi].

f''(x) = -sin x - cos x: at pi/4 it is -sqrt(2) < 0 (maximum); at 5pi/4 it is sqrt(2) > 0 (minimum).

f(pi/4) = sqrt(2)/2 + sqrt(2)/2 = sqrt(2); f(5pi/4) = -sqrt(2).

Endpoint values f(0) = f(2pi) = 1 lie between these.

Answer: maximum value sqrt(2) at x = pi/4; minimum value -sqrt(2) at x = 5pi/4.

Exercise 2.9, Q4 - Show that y = (ln x)/x has maximum value at x = e

Q4. Proof.

y' = (1 - ln x)/x2, which is zero when ln x = 1, i.e. at x = e.

For x < e, 1 - ln x > 0 so y is increasing; for x > e, 1 - ln x < 0 so y is decreasing.

So y changes from increasing to decreasing at x = e.

Answer: proved, y has its maximum value 1/e at x = e.

Exercise 2.9, Q5 - Show that y = x^x has a minimum value at x = 1/e

Q5. Proof.

Take logarithms: ln y = x ln x.

Differentiate: y'/y = ln x + 1, which is zero when ln x = -1, i.e. at x = 1/e.

For x < 1/e, ln x + 1 < 0 so y is decreasing; for x > 1/e it is positive so y is increasing.

So y changes from decreasing to increasing at x = 1/e.

Answer: proved, y has its minimum value (1/e)^(1/e) at x = 1/e.

Exercise 2.10, Q1 - Find two positive integers whose sum is 30 and their product will be maximum

Q1. Two numbers, sum 30, product maximum.

Let one number be x; the other is 30 - x. Product P = x(30 - x) = 30x - x2.

P' = 30 - 2x, zero at x = 15; P'' = -2 < 0, so this is a maximum.

The numbers are 15 and 15.

Answer: 15 and 15, maximum product 225.

Exercise 2.10, Q2 - Divide 20 into two parts so that the sum of their squares will be minimum

Q2. Two parts of 20, sum of squares minimum.

Let one part be x; the other is 20 - x. Sum S = x2 + (20 - x)2.

S' = 2x - 2(20 - x) = 4x - 40, zero at x = 10; S'' = 4 > 0, so this is a minimum.

The parts are 10 and 10.

Answer: 10 and 10, minimum sum of squares 200.

Exercise 2.10, Q3 - Find two positive integers whose sum is 12 and the product of one with the square of the other will be maximum

Q3. Two numbers, sum 12, x(12 - x)2 maximum.

Let the numbers be x and 12 - x. Product P = x(12 - x)2.

P' = (12 - x)2 - 2x(12 - x) = (12 - x)(12 - 3x), zero at x = 12 or x = 4 (x = 12 gives product 0, rejected).

P'' at x = 4 is negative, so this is a maximum.

The numbers are 4 and 8.

Answer: 4 and 8, maximum product 4 x 64 = 256.

Exercise 2.10, Q4 - The perimeter of a triangle is 16 centimetres. If one side is of length 6 cm, what are the lengths of the other sides for maximum area of the triangle?

Q4. Triangle, perimeter 16, one side 6, maximum area.

Let the other sides be x and 10 - x (since 6 + x + (10 - x) = 16). Semiperimeter s = 8.

By Heron, (area)2 = 8(8 - 6)(8 - x)(8 - (10 - x)) = 16(8 - x)(x - 2) = 16(-x2 + 10x - 16).

Maximise -x2 + 10x - 16: derivative -2x + 10, zero at x = 5; second derivative -2 < 0.

The other sides are 5 cm and 5 cm.

Answer: 5 cm and 5 cm (with the given 6 cm side).

Exercise 2.10, Q5 - Find the dimensions of a rectangle of largest area having perimeter 120 centimetres

Q5. Rectangle, perimeter 120, largest area.

Let one side be x; the other is 60 - x. Area A = x(60 - x).

A' = 60 - 2x, zero at x = 30; A'' = -2 < 0, so this is a maximum.

The sides are 30 cm and 30 cm.

Answer: 30 cm by 30 cm (a square), area 900 cm2.

Exercise 2.10, Q6 - Find the lengths of the sides of a variable rectangle having area 36 cm2 when its perimeter is minimum

Q6. Rectangle, area 36, perimeter minimum.

Let one side be x; the other is 36/x. Perimeter P = 2x + 72/x.

P' = 2 - 72/x2, zero at x2 = 36, i.e. x = 6 (positive side). P'' = 144/x3 > 0, so this is a minimum.

The sides are 6 cm and 6 cm.

Answer: 6 cm by 6 cm (a square), perimeter 24 cm.

Exercise 2.10, Q7 - A box with a square base and open top is to have a volume of 4 cubic dm. Find the dimensions of the box which will require the least material

Q7. Open-top square-base box, volume 4, least material.

Let the base side be x dm and the height be h dm. Volume x2 h = 4, so h = 4/x2.

Material (surface) S = base + 4 sides = x2 + 4x h = x2 + 16/x.

S' = 2x - 16/x2, zero at x3 = 8, i.e. x = 2; S'' = 2 + 32/x3 > 0, so this is a minimum.

Then h = 4/4 = 1 dm.

Answer: base 2 dm by 2 dm, height 1 dm.

Exercise 2.10, Q8 - Find the dimensions of a rectangular garden having perimeter 80 metres if its area is to be maximum

Q8. Garden, perimeter 80, maximum area.

Let one side be x metres; the other is 40 - x. Area A = x(40 - x).

A' = 40 - 2x, zero at x = 20; A'' = -2 < 0, so this is a maximum.

The sides are 20 m and 20 m.

Answer: 20 m by 20 m (a square), area 400 m2.

Exercise 2.10, Q9 - An open tank of square base of side x and vertical sides is to be constructed to contain a given quantity of water. Find the depth in terms of x if the expense of lining the inside of the tank with lead will be least

Q9. Open tank, square base x, least lining area, depth in terms of x.

Let the volume be V and the depth be h. Then V = x2 h, so h = V/x2.

Lining area A = base + 4 sides = x2 + 4x h = x2 + 4V/x.

A' = 2x - 4V/x2, zero at x3 = 2V, i.e. V = x3/2.

Then h = V/x2 = x/2.

Answer: depth h = x/2, i.e. half the side of the base.

Exercise 2.10, Q10 - Find the dimensions of the rectangle of maximum area which fits inside the semi-circle of radius 8 cm as shown in the figure

Q10. Largest rectangle in a semicircle of radius 8 cm.

Let the half-width be x cm; the height is sqrt(64 - x2) cm, since the top corner lies on the circle.

Area A = 2x sqrt(64 - x2). Maximise A2 = 4x2(64 - x2).

d/dx [x2(64 - x2)] = 128x - 4x3, zero (for x > 0) at x2 = 32, i.e. x = 4 sqrt(2).

Height = sqrt(64 - 32) = 4 sqrt(2) cm; full width = 2x = 8 sqrt(2) cm.

Answer: width 8 sqrt(2) cm, height 4 sqrt(2) cm, area 64 cm2.

Exercise 2.10, Q11 - Find the point on the curve y = x2 - 1 that is closest to the point (3, -1)

Q11. Closest point on y = x2 - 1 to (3, -1).

Let the point be (x, x2 - 1). Squared distance D = (x - 3)2 + (x2 - 1 + 1)2 = (x - 3)2 + x4.

D' = 2(x - 3) + 4x3 = 2(2x3 + x - 3).

2x3 + x - 3 = (x - 1)(2x2 + 2x + 3); the quadratic has discriminant 4 - 24 < 0, so the only real critical point is x = 1.

D'' = 2 + 12x2 > 0, so this is a minimum.

Answer: the point (1, 0).

Exercise 2.10, Q12 - Find the point on the curve y = x2 + 1 that is closest to the point (18, 1)

Q12. Closest point on y = x2 + 1 to (18, 1).

Let the point be (x, x2 + 1). Squared distance D = (x - 18)2 + (x2 + 1 - 1)2 = (x - 18)2 + x4.

D' = 2(x - 18) + 4x3 = 2(2x3 + x - 18).

2x3 + x - 18 = (x - 2)(2x2 + 4x + 9); the quadratic has discriminant 16 - 72 < 0, so the only real critical point is x = 2.

D'' = 2 + 12x2 > 0, so this is a minimum.

Answer: the point (2, 5).

Chapter 3: Integration

Exercise 3.1, Q1 - Find delta y and dy in the following cases (delta y = exact change, dy = differential approximation)

Q1(i). y = x2 - 1, when x changes from 3 to 3.02.

dx = 3.02 - 3 = 0.02.

dy = f'(x) dx = 2x dx = 2(3)(0.02) = 0.12.

Exact change: delta y = f(3.02) - f(3) = (9.1204 - 1) - (9 - 1) = 0.1204.

Answer: dy = 0.12, delta y = 0.1204.

Q1(ii). y = x2 + 2x, when x changes from 2 to 1.8.

dx = 1.8 - 2 = -0.2.

dy = f'(x) dx = (2x + 2) dx = (6)(-0.2) = -1.2.

Exact change: delta y = f(1.8) - f(2) = (3.24 + 3.6) - (4 + 4) = -1.16.

Answer: dy = -1.2, delta y = -1.16.

Q1(iii). y = sqrt(x), when x changes from 4 to 4.41.

dx = 4.41 - 4 = 0.41.

dy = f'(x) dx = dx / (2 sqrt(x)) = 0.41 / 4 = 0.1025.

Exact change: delta y = sqrt(4.41) - sqrt(4) = 2.1 - 2 = 0.1.

Answer: dy = 0.1025, delta y = 0.1.

Exercise 3.1, Q2 - Using differentials find dy/dx and dx/dy in the following equations

Q2(i). x y + x = 4.

Differentiate both sides: y dx + x dy + dx = 0.

So (y + 1) dx + x dy = 0, giving dy/dx = -(y + 1)/x.

Answer: dy/dx = -(y + 1)/x, dx/dy = -x/(y + 1).

Q2(ii). x2 + 2y2 = 16.

Differentiate both sides: 2x dx + 4y dy = 0.

So dy/dx = -x/(2y).

Answer: dy/dx = -x/(2y), dx/dy = -2y/x.

Q2(iii). x4 + y2 = x y2.

Differentiate both sides: 4x3 dx + 2y dy = y2 dx + 2x y dy.

Collect: (2y - 2x y) dy = (y2 - 4x3) dx.

Answer: dy/dx = (y2 - 4x3)/(2y - 2x y), dx/dy = (2y - 2x y)/(y2 - 4x3).

Q2(iv). x y - ln x = c.

Differentiate both sides: y dx + x dy - dx/x = 0.

So x dy = (1/x - y) dx, giving dy/dx = 1/x2 - y/x.

Answer: dy/dx = (1 - x y)/x2, dx/dy = x2/(1 - x y).

Exercise 3.1, Q3 - Use differentials to approximate the values of

Q3(i). (17)^(1/4).

Let f(x) = x^(1/4). Take x = 16, dx = 1, so f(16) = 2.

f'(x) = 1/(4 x^(3/4)), so f'(16) = 1/(4 x 8) = 1/32 = 0.03125.

dy = f'(16) dx = 0.03125.

Answer: (17)^(1/4) approx. 2 + 0.03125 = 2.03125.

Q3(ii). (31)^(1/5).

Let f(x) = x^(1/5). Take x = 32, dx = -1, so f(32) = 2.

f'(x) = 1/(5 x^(4/5)), so f'(32) = 1/(5 x 16) = 1/80 = 0.0125.

dy = f'(32) dx = -0.0125.

Answer: (31)^(1/5) approx. 2 - 0.0125 = 1.9875.

Q3(iii). cos 29 degrees.

Let f(x) = cos x. Take x = 30 degrees, dx = -1 degree = -pi/180 approx. -0.017453 radians.

dy = -sin x dx = -sin 30 degrees x (-0.017453) = (0.5)(0.017453) = 0.008727.

cos 30 degrees = sqrt(3)/2 approx. 0.8660.

Answer: cos 29 degrees approx. 0.8660 + 0.008727 = 0.87473.

Q3(iv). sin 61 degrees.

Let f(x) = sin x. Take x = 60 degrees, dx = 1 degree = pi/180 approx. 0.017453 radians.

dy = cos x dx = (0.5)(0.017453) = 0.008727.

sin 60 degrees = sqrt(3)/2 approx. 0.8660.

Answer: sin 61 degrees approx. 0.8660 + 0.008727 = 0.87473.

Exercise 3.1, Q4 - Find the approximate increase in the volume of a cube if the length of its each edge changes from 5 to 5.02

Q4. Cube edge 5 to 5.02.

Let V = x3, with x = 5 and dx = 0.02.

dV = 3x2 dx = 3(25)(0.02) = 1.5.

Answer: approximate increase in volume 1.5 cubic units (exact increase 5.02^3 - 125 = 1.506 approx.).

Exercise 3.1, Q5 - Find the approximate increase in the area of a circular disc if its diameter is ? (FLAGGED: values not legible in the source copy)

Q5. FLAGGED - question incomplete in the source copy; no numerical answer given. As printed in this copy, Q5 reads: 'Find the approximate increase in the area of a circular disc if its diameter is ?' - the diameter value and its change are not legible (a '?' stands where the numbers should be).

Method (general): for diameter d, area A = pi d2 / 4, so dA = (pi d / 2) dd.

Once the diameter and its change are known, substitute them in dA = (pi d / 2) dd.

Answer: NOT COMPUTED - the printed values are missing in the source copy; nothing has been invented here.

Exercise 3.2, Q1 - Evaluate the following indefinite integrals (c = constant of integration)

Q1(i). Int (3x2 - 2x + 1) dx.

Integrate term by term: Int 3x2 dx = x3; Int (-2x) dx = -x2; Int 1 dx = x.

Answer: x3 - x2 + x + c.

Q1(ii). Int (sqrt(x) + 1/sqrt(x)) dx, x > 0.

Write as Int (x^(1/2) + x^(-1/2)) dx.

Int x^(1/2) dx = (2/3) x^(3/2); Int x^(-1/2) dx = 2 x^(1/2).

Answer: (2/3) x^(3/2) + 2 sqrt(x) + c.

Q1(iii). Int x(sqrt(x) + 1) dx, x > 0.

Expand: x(sqrt(x) + 1) = x^(3/2) + x.

Int x^(3/2) dx = (2/5) x^(5/2); Int x dx = x2/2.

Answer: (2/5) x^(5/2) + x2/2 + c.

Q1(iv). Int (2x + 3)^(1/2) dx.

Put u = 2x + 3; then du = 2 dx, so dx = du/2.

Int u^(1/2) (du/2) = (1/2) (2/3) u^(3/2) = (1/3) u^(3/2).

Answer: (2x + 3)^(3/2) / 3 + c.

Q1(v). Int (sqrt(x) + 1)2 dx, x > 0.

Expand: (sqrt(x) + 1)2 = x + 2 sqrt(x) + 1.

Int x dx = x2/2; Int 2 sqrt(x) dx = (4/3) x^(3/2); Int 1 dx = x.

Answer: x2/2 + (4/3) x^(3/2) + x + c.

Q1(vi). Int (sqrt(x) - 1/sqrt(x))2 dx, x > 0.

Expand: (sqrt(x) - 1/sqrt(x))2 = x - 2 + 1/x.

Int x dx = x2/2; Int (-2) dx = -2x; Int (1/x) dx = ln x.

Answer: x2/2 - 2x + ln x + c.

Q1(vii). Int (3x + 2)/sqrt(x) dx, x > 0.

Divide: (3x + 2)/sqrt(x) = 3 sqrt(x) + 2/sqrt(x).

Int 3 sqrt(x) dx = 2 x^(3/2); Int 2/sqrt(x) dx = 4 sqrt(x).

Answer: 2 x^(3/2) + 4 sqrt(x) + c.

Q1(viii). Int sqrt(y)(y + 1)/y dy, y > 0.

Simplify: sqrt(y)(y + 1)/y = sqrt(y) + 1/sqrt(y).

Int sqrt(y) dy = (2/3) y^(3/2); Int 1/sqrt(y) dy = 2 sqrt(y).

Answer: (2/3) y^(3/2) + 2 sqrt(y) + c.

Q1(ix). Int (sqrt(theta) - 1)2/sqrt(theta) d theta, theta > 0.

Expand the numerator: (sqrt(theta) - 1)2 = theta - 2 sqrt(theta) + 1.

Divide by sqrt(theta): sqrt(theta) - 2 + theta^(-1/2).

Integrate: (2/3) theta^(3/2) - 2 theta + 2 sqrt(theta).

Answer: (2/3) theta^(3/2) - 2 theta + 2 sqrt(theta) + c.

Q1(x). Int (1 - sqrt(x))2/sqrt(x) dx, x > 0.

Expand the numerator: 1 - 2 sqrt(x) + x; divide by sqrt(x): x^(-1/2) - 2 + sqrt(x).

Integrate: 2 sqrt(x) - 2x + (2/3) x^(3/2).

Answer: 2 sqrt(x) - 2x + (2/3) x^(3/2) + c.

Q1(xi). Int (e^(2x) + e^x)/e^x dx.

Divide: (e^(2x) + e^x)/e^x = e^x + 1.

Int e^x dx = e^x; Int 1 dx = x.

Answer: e^x + x + c.

Exercise 3.2, Q2 - Evaluate (c = constant of integration)

Q2(i). Int dx / (sqrt(x + a) + sqrt(x + b)), x + a > 0, x + b > 0.

Rationalize: multiply top and bottom by (sqrt(x + a) - sqrt(x + b)).

The denominator becomes (x + a) - (x + b) = a - b.

So the integral = (1/(a - b)) Int (sqrt(x + a) - sqrt(x + b)) dx.

Answer: [(2/3)(x + a)^(3/2) - (2/3)(x + b)^(3/2)] / (a - b) + c.

Q2(ii). Int (1 - x2)/(1 + x2) dx.

Divide: (1 - x2)/(1 + x2) = -1 + 2/(1 + x2).

Int (-1) dx = -x; Int 2/(1 + x2) dx = 2 Tan^{-1} x.

Answer: -x + 2 Tan^{-1} x + c.

Q2(iii). Int dx / (sqrt(x + a) + sqrt(x)), x > 0, a > 0.

Rationalize: multiply top and bottom by (sqrt(x + a) - sqrt(x)).

The denominator becomes a, so the integral = (1/a) Int (sqrt(x + a) - sqrt(x)) dx.

Answer: [(2/3)(x + a)^(3/2) - (2/3) x^(3/2)] / a + c.

Q2(iv). Int (a - 2x)^(3/2) dx.

Put u = a - 2x; then du = -2 dx, so dx = -du/2.

Int u^(3/2) (-du/2) = -(1/2) (2/5) u^(5/2) = -(1/5) u^(5/2).

Answer: -(a - 2x)^(5/2) / 5 + c.

Q2(v). Int (1 + e^x)3 / e^x dx.

Expand: (1 + e^x)3 / e^x = e^(-x) + 3 + 3e^x + e^(2x).

Integrate term by term: -e^(-x) + 3x + 3e^x + e^(2x)/2.

Answer: -e^(-x) + 3x + 3e^x + e^(2x)/2 + c.

Q2(vi). Int sin((a + b) x) dx.

Put u = (a + b) x; then du = (a + b) dx.

Int sin u (du/(a + b)) = -cos u / (a + b).

Answer: -cos((a + b) x) / (a + b) + c.

Q2(vii). Int sqrt(1 - cos 2x) dx, 1 - cos 2x > 0.

Use 1 - cos 2x = 2 sin2 x, so sqrt(1 - cos 2x) = sqrt(2) |sin x|.

Where sin x > 0, Int sqrt(2) sin x dx = -sqrt(2) cos x.

Answer: -sqrt(2) cos x + c (on intervals where sin x > 0; with |sin x| in general).

Q2(viii). Int (ln x) (1/x) dx, x > 0.

Put u = ln x; then du = dx/x.

Int u du = u2/2.

Answer: (ln x)2 / 2 + c.

Q2(ix). Int sin2 x dx.

Use sin2 x = (1 - cos 2x)/2.

Int (1/2) dx - Int (cos 2x)/2 dx = x/2 - sin 2x/4.

Answer: x/2 - (sin 2x)/4 + c.

Q2(x). Int dx / (1 + cos x), -pi/2 < x < pi/2.

Use 1 + cos x = 2 cos2(x/2).

Int dx / (2 cos2(x/2)) = tan(x/2).

Answer: tan(x/2) + c.

Q2(xi). Int (a x + b)/(a x2 + 2b x + c) dx.

The derivative of the denominator is 2a x + 2b = 2(a x + b).

So the integral = (1/2) Int (denominator)'/(denominator) dx = (1/2) ln|denominator|.

Answer: (1/2) ln|a x2 + 2b x + c| + c.

Q2(xii). Int cos 3x sin 2x dx.

Use cos 3x sin 2x = (1/2)(sin 5x - sin x).

Int (1/2) sin 5x dx = -cos 5x/10; Int (-(1/2) sin x) dx = cos x/2.

Answer: -cos 5x/10 + cos x/2 + c.

Q2(xiii). Int (cos 2x - 1)/(1 + cos 2x) dx, 1 + cos 2x not equal to 0.

Use cos 2x - 1 = -2 sin2 x and 1 + cos 2x = 2 cos2 x, so the integrand = -tan2 x.

Since -tan2 x = 1 - sec2 x, the integral = x - tan x.

Answer: x - tan x + c.

Q2(xiv). Int tan2 x dx.

Use tan2 x = sec2 x - 1.

Int sec2 x dx = tan x; Int (-1) dx = -x.

Answer: tan x - x + c.

Exercise 3.3 - Evaluate the following integrals (c = constant of integration)

Q1. Int (-2x)/sqrt(4 - x2) dx.

Put u = 4 - x2; then du = -2x dx.

The integral becomes Int u^(-1/2) du = 2 sqrt(u).

Answer: 2 sqrt(4 - x2) + c.

Q2. Int dx / (x2 + 4x + 13).

Complete the square: x2 + 4x + 13 = (x + 2)2 + 9.

Int dx / ((x + 2)2 + 9) = (1/3) Tan^{-1}((x + 2)/3).

Answer: (1/3) Tan^{-1}((x + 2)/3) + c.

Q3. Int x2 / (4 + x2) dx.

Divide: x2/(4 + x2) = 1 - 4/(4 + x2).

Int 1 dx = x; Int 4/(4 + x2) dx = 2 Tan^{-1}(x/2).

Answer: x - 2 Tan^{-1}(x/2) + c.

Q4. Int dx / (x ln x).

Put u = ln x; then du = dx/x.

The integral becomes Int du/u = ln|u|.

Answer: ln|ln x| + c.

Q5. Int e^x / (e^x + 3) dx.

Put u = e^x + 3; then du = e^x dx.

The integral becomes Int du/u = ln|u|.

Answer: ln(e^x + 3) + c.

Q6. Int (x + b) / sqrt(x2 + 2b x + c) dx. (Print note: the first term of the radicand shows as x in this copy, its superscript 2 apparently dropped, as with the Q8 misprints below; solved in the standard form.)

The derivative of x2 + 2b x + c is 2x + 2b = 2(x + b).

So the integrand is (1/2) times (derivative of the radicand)/(sqrt of the radicand).

Int (1/2) (2x + 2b)/sqrt(x2 + 2b x + c) dx = sqrt(x2 + 2b x + c).

Answer: sqrt(x2 + 2b x + c) + c.

Q7. Int sec2 x / sqrt(tan x) dx.

Put u = tan x; then du = sec2 x dx.

The integral becomes Int u^(-1/2) du = 2 sqrt(u).

Answer: 2 sqrt(tan x) + c.

Q8(a). Show that Int dx / sqrt(x2 - a2) = ln(x + sqrt(x2 - a2)) + c. (Print note: this copy prints the integrand as dx/(x2 - a2), which is a misprint; the proved form is the one with sqrt(x2 - a2) in the denominator.)

Put x = a sec theta; then dx = a sec theta tan theta d theta and sqrt(x2 - a2) = a tan theta.

The integral becomes Int sec theta d theta = ln|sec theta + tan theta| + c.

Since sec theta = x/a and tan theta = sqrt(x2 - a2)/a, this is ln(x + sqrt(x2 - a2)) + c (absorbing -ln a into c).

Answer: proved.

Q8(b). Show that Int sqrt(a2 - x2) dx = (a2/2) Sin^{-1}(x/a) + (x/2) sqrt(a2 - x2) + c. (Print note: this copy prints the right-hand side as (a/a) Sin^{-1}(x/a) + (x/a) sqrt(a2 - x2), which is a misprint; the standard form is proved here.)

Put x = a sin theta; then dx = a cos theta d theta and sqrt(a2 - x2) = a cos theta.

The integral becomes Int a2 cos2 theta d theta = (a2/2) Int (1 + cos 2 theta) d theta.

This equals (a2/2)(theta + sin theta cos theta) + c.

Since theta = Sin^{-1}(x/a), sin theta = x/a and cos theta = sqrt(a2 - x2)/a, the result is (a2/2) Sin^{-1}(x/a) + (x/2) sqrt(a2 - x2) + c.

Answer: proved.

Q9. Int dx / (1 + x2)^(3/2).

Put x = tan theta; then dx = sec2 theta d theta and (1 + x2)^(3/2) = sec3 theta.

The integral becomes Int cos theta d theta = sin theta.

Since sin theta = x/sqrt(1 + x2), the answer is x/sqrt(1 + x2) + c.

Answer: x/sqrt(1 + x2) + c.

Q10. Int dx / ((1 + x2) Tan^{-1} x).

Put u = Tan^{-1} x; then du = dx/(1 + x2).

The integral becomes Int du/u = ln|u|.

Answer: ln|Tan^{-1} x| + c.

Q11. Int sqrt((1 + x)/(1 - x)) dx.

Multiply top and bottom inside the root: sqrt((1 + x)/(1 - x)) = (1 + x)/sqrt(1 - x2).

Split: Int dx/sqrt(1 - x2) + Int x dx/sqrt(1 - x2) = Sin^{-1} x - sqrt(1 - x2).

Answer: Sin^{-1} x - sqrt(1 - x2) + c.

Q12. Int sin theta / (1 + cos2 theta) d theta.

Put u = cos theta; then du = -sin theta d theta.

The integral becomes -Int du/(1 + u2) = -Tan^{-1} u.

Answer: -Tan^{-1}(cos theta) + c.

Q13. Int a x / sqrt(a2 - x4) dx.

Put u = x2; then du = 2x dx, so x dx = du/2.

The integral becomes (a/2) Int du/sqrt(a2 - u2) = (a/2) Sin^{-1}(u/a).

Answer: (a/2) Sin^{-1}(x2/a) + c.

Q14. Int dx / sqrt(7 - 6x - x2).

Complete the square: 7 - 6x - x2 = 16 - (x + 3)2.

Int dx/sqrt(16 - (x + 3)2) = Sin^{-1}((x + 3)/4).

Answer: Sin^{-1}((x + 3)/4) + c.

Q15. Int cos x / (sin x ln sin x) dx.

Put u = ln sin x; then du = (cos x / sin x) dx.

The integral becomes Int du/u = ln|u|.

Answer: ln|ln sin x| + c.

Q16. Int cos x (ln sin x / sin x) dx.

Put u = ln sin x; then du = (cos x / sin x) dx.

The integral becomes Int u du = u2/2.

Answer: (ln sin x)2 / 2 + c.

Q17. Int x dx / (4 + 2x + x2).

Write x = (1/2)(2x + 2) - 1, splitting the integral into (1/2) Int (2x + 2)/(x2 + 2x + 4) dx - Int dx/(x2 + 2x + 4).

The first part is (1/2) ln(x2 + 2x + 4); the second, with x2 + 2x + 4 = (x + 1)2 + 3, is (1/sqrt(3)) Tan^{-1}((x + 1)/sqrt(3)).

Answer: (1/2) ln(x2 + 2x + 4) - (1/sqrt(3)) Tan^{-1}((x + 1)/sqrt(3)) + c.

Q18. Int x dx / (x4 + 2x2 + 5).

Put u = x2; then du = 2x dx, so x dx = du/2.

The integral becomes (1/2) Int du/(u2 + 2u + 5) = (1/2) Int du/((u + 1)2 + 4).

This equals (1/2)(1/2) Tan^{-1}((u + 1)/2) = (1/4) Tan^{-1}((x2 + 1)/2).

Answer: (1/4) Tan^{-1}((x2 + 1)/2) + c.

Q19. Int cos(sqrt(x) - x/2) (1/sqrt(x) - 1) dx.

Put u = sqrt(x) - x/2; then du/dx = 1/(2 sqrt(x)) - 1/2 = (1/2)(1/sqrt(x) - 1).

So (1/sqrt(x) - 1) dx = 2 du, and the integral becomes 2 Int cos u du = 2 sin u.

Answer: 2 sin(sqrt(x) - x/2) + c.

Q20. Int (x + 2)/sqrt(x + 3) dx.

Put u = x + 3; then x + 2 = u - 1 and dx = du.

The integral becomes Int (u - 1)/sqrt(u) du = Int (u^(1/2) - u^(-1/2)) du = (2/3) u^(3/2) - 2 sqrt(u).

Answer: (2/3)(x + 3)^(3/2) - 2 sqrt(x + 3) + c.

Q21. Int sqrt(2) / (sin x + cos x) dx.

Since sin x + cos x = sqrt(2) sin(x + pi/4), the integrand is 1/sin(x + pi/4).

Int cosec(x + pi/4) dx = ln|tan((x + pi/4)/2)|.

Answer: ln|tan(x/2 + pi/8)| + c.

Q22. Int dx / ((1/2) sin x + (sqrt(3)/2) cos x).

Since (1/2) sin x + (sqrt(3)/2) cos x = sin(x + pi/3), the integral is Int cosec(x + pi/3) dx.

Int cosec(x + pi/3) dx = ln|tan((x + pi/3)/2)|.

Answer: ln|tan(x/2 + pi/6)| + c.

Exercise 3.4, Q1 - Evaluate the following integrals by parts (c = constant of integration)

Q1(i). Int x sin x dx.

Take u = x, dv = sin x dx; then du = dx, v = -cos x.

By parts: -x cos x + Int cos x dx = -x cos x + sin x.

Answer: -x cos x + sin x + c.

Q1(ii). Int ln x dx.

Take u = ln x, dv = dx; then du = dx/x, v = x.

By parts: x ln x - Int x (dx/x) = x ln x - x.

Answer: x ln x - x + c.

Q1(iii). Int x ln x dx.

Take u = ln x, dv = x dx; then du = dx/x, v = x2/2.

By parts: (x2/2) ln x - Int (x/2) dx = (x2/2) ln x - x2/4.

Answer: (x2/2) ln x - x2/4 + c.

Q1(iv). Int x2 ln x dx.

Take u = ln x, dv = x2 dx; then du = dx/x, v = x3/3.

By parts: (x3/3) ln x - Int (x2/3) dx = (x3/3) ln x - x3/9.

Answer: (x3/3) ln x - x3/9 + c.

Q1(v). Int x3 ln x dx.

Take u = ln x, dv = x3 dx; then du = dx/x, v = x4/4.

By parts: (x4/4) ln x - Int (x3/4) dx = (x4/4) ln x - x4/16.

Answer: (x4/4) ln x - x4/16 + c.

Q1(vi). Int x4 ln x dx.

Take u = ln x, dv = x4 dx; then du = dx/x, v = x5/5.

By parts: (x5/5) ln x - Int (x4/5) dx = (x5/5) ln x - x5/25.

Answer: (x5/5) ln x - x5/25 + c.

Q1(vii). Int Tan^{-1} x dx.

Take u = Tan^{-1} x, dv = dx; then du = dx/(1 + x2), v = x.

By parts: x Tan^{-1} x - Int x dx/(1 + x2) = x Tan^{-1} x - (1/2) ln(1 + x2).

Answer: x Tan^{-1} x - (1/2) ln(1 + x2) + c.

Q1(viii). Int x2 sin x dx.

Take u = x2, dv = sin x dx: the integral becomes -x2 cos x + Int 2x cos x dx.

Integrate Int 2x cos x dx by parts: 2x sin x - Int 2 sin x dx = 2x sin x + 2 cos x.

Answer: -x2 cos x + 2x sin x + 2 cos x + c.

Q1(ix). Int x2 Tan^{-1} x dx.

Take u = Tan^{-1} x, dv = x2 dx; then du = dx/(1 + x2), v = x3/3.

By parts: (x3/3) Tan^{-1} x - (1/3) Int x3 dx/(1 + x2).

Since x3/(1 + x2) = x - x/(1 + x2), the remaining integral is x2/2 - (1/2) ln(1 + x2).

Answer: (x3/3) Tan^{-1} x - x2/6 + (1/6) ln(1 + x2) + c.

Q1(x). Int x Tan^{-1} x dx.

Take u = Tan^{-1} x, dv = x dx; then du = dx/(1 + x2), v = x2/2.

By parts: (x2/2) Tan^{-1} x - (1/2) Int x2 dx/(1 + x2).

Since x2/(1 + x2) = 1 - 1/(1 + x2), the remaining integral is x - Tan^{-1} x.

Answer: ((x2 + 1)/2) Tan^{-1} x - x/2 + c.

Q1(xi). Int x3 Tan^{-1} x dx.

Take u = Tan^{-1} x, dv = x3 dx; then du = dx/(1 + x2), v = x4/4.

By parts: (x4/4) Tan^{-1} x - (1/4) Int x4 dx/(1 + x2).

Since x4/(1 + x2) = x2 - 1 + 1/(1 + x2), the remaining integral is x3/3 - x + Tan^{-1} x.

Answer: ((x4 - 1)/4) Tan^{-1} x - x3/12 + x/4 + c.

Q1(xii). Int x3 cos x dx.

Take u = x3, dv = cos x dx: the integral becomes x3 sin x - Int 3x2 sin x dx.

Integrate Int 3x2 sin x dx by parts twice: -3x2 cos x + 6x sin x + 6 cos x.

Answer: x3 sin x + 3x2 cos x - 6x sin x - 6 cos x + c.

Q1(xiii). Int Sin^{-1} x dx.

Take u = Sin^{-1} x, dv = dx; then du = dx/sqrt(1 - x2), v = x.

By parts: x Sin^{-1} x - Int x dx/sqrt(1 - x2) = x Sin^{-1} x + sqrt(1 - x2).

Answer: x Sin^{-1} x + sqrt(1 - x2) + c.

Q1(xiv). Int x Sin^{-1} x dx.

Take u = Sin^{-1} x, dv = x dx; then du = dx/sqrt(1 - x2), v = x2/2.

By parts: (x2/2) Sin^{-1} x - (1/2) Int x2 dx/sqrt(1 - x2).

With x = sin theta, Int x2 dx/sqrt(1 - x2) = (1/2)(Sin^{-1} x - x sqrt(1 - x2)).

Answer: ((2x2 - 1)/4) Sin^{-1} x + (x/4) sqrt(1 - x2) + c.

Q1(xv). Int e^x sin x cos x dx.

Use sin x cos x = (1/2) sin 2x, so the integral is (1/2) Int e^x sin 2x dx.

Int e^x sin 2x dx = e^x (sin 2x - 2 cos 2x)/5.

Answer: e^x (sin 2x - 2 cos 2x)/10 + c.

Q1(xvi). Int x sin x cos x dx.

Use sin x cos x = (1/2) sin 2x, so the integral is (1/2) Int x sin 2x dx.

By parts, Int x sin 2x dx = -x cos 2x/2 + sin 2x/4.

Answer: -x cos 2x/4 + sin 2x/8 + c.

Q1(xvii). Int x cos2 x dx.

Use cos2 x = (1 + cos 2x)/2, so the integral is (1/2) Int x dx + (1/2) Int x cos 2x dx.

Int x cos 2x dx = x sin 2x/2 + cos 2x/4.

Answer: x2/4 + x sin 2x/4 + cos 2x/8 + c.

Q1(xviii). Int x sin2 x dx.

Use sin2 x = (1 - cos 2x)/2, so the integral is (1/2) Int x dx - (1/2) Int x cos 2x dx.

Int x cos 2x dx = x sin 2x/2 + cos 2x/4.

Answer: x2/4 - x sin 2x/4 - cos 2x/8 + c.

Q1(xix). Int (ln x)2 dx.

Take u = (ln x)2, dv = dx; then du = 2 ln x (dx/x), v = x.

By parts: x (ln x)2 - Int 2 ln x dx = x (ln x)2 - 2(x ln x - x).

Answer: x (ln x)2 - 2x ln x + 2x + c.

Q1(xx). Int ln(tan x) sec2 x dx.

Take u = ln(tan x), dv = sec2 x dx; then du = (sec2 x / tan x) dx, v = tan x.

By parts: tan x ln(tan x) - Int sec2 x dx = tan x ln(tan x) - tan x.

Answer: tan x ln(tan x) - tan x + c.

Q1(xxi). Int x Sin^{-1} x / sqrt(1 - x2) dx.

Take u = Sin^{-1} x, dv = x dx/sqrt(1 - x2); then du = dx/sqrt(1 - x2), v = -sqrt(1 - x2).

By parts: -sqrt(1 - x2) Sin^{-1} x + Int dx = -sqrt(1 - x2) Sin^{-1} x + x.

Answer: x - sqrt(1 - x2) Sin^{-1} x + c.

Exercise 3.4, Q2 - Evaluate the following integral (c = constant of integration)

Q2(i). Int tan4 x dx.

Write tan4 x = tan2 x (sec2 x - 1) = tan2 x sec2 x - tan2 x.

Int tan2 x sec2 x dx = tan3 x/3; Int tan2 x dx = tan x - x.

Answer: tan3 x/3 - tan x + x + c.

Q2(ii). Int sec4 x dx.

Write sec4 x = (1 + tan2 x) sec2 x.

Put u = tan x: Int (1 + u2) du = u + u3/3.

Answer: tan x + tan3 x/3 + c.

Q2(iii). Int e^x sin 2x cos x dx.

Use sin 2x cos x = (1/2)(sin 3x + sin x).

Int e^x sin 3x dx = e^x (sin 3x - 3 cos 3x)/10; Int e^x sin x dx = e^x (sin x - cos x)/2.

Answer: e^x (sin 3x - 3 cos 3x)/20 + e^x (sin x - cos x)/4 + c.

Q2(iv). Int tan3 x sec x dx.

Write tan3 x sec x = (sec2 x - 1) tan x sec x.

Put u = sec x: Int (u2 - 1) du = u3/3 - u.

Answer: sec3 x/3 - sec x + c.

Q2(v). Int x3 e^(5x) dx.

Integrate by parts repeatedly (or use the reduction formula).

Each step lowers the power of x and divides by 5.

Answer: e^(5x) (x3/5 - 3x2/25 + 6x/125 - 6/625) + c.

Q2(vi). Int e^(-x) sin 2x dx.

Use Int e^(a x) sin b x dx = e^(a x)(a sin b x - b cos b x)/(a2 + b2) with a = -1, b = 2.

Answer: e^(-x)(-sin 2x - 2 cos 2x)/5 + c.

Q2(vii). Int e^(2x) cos 3x dx.

Use Int e^(a x) cos b x dx = e^(a x)(a cos b x + b sin b x)/(a2 + b2) with a = 2, b = 3.

Answer: e^(2x)(2 cos 3x + 3 sin 3x)/13 + c.

Q2(viii). Int cosec3 x dx.

Write cosec3 x = cosec x (1 + cot2 x) = cosec x + cosec x cot2 x.

Integrate by parts (or use the reduction formula): Int cosec3 x dx = -(1/2) cosec x cot x + (1/2) Int cosec x dx.

Int cosec x dx = ln|tan(x/2)|.

Answer: -(1/2) cosec x cot x + (1/2) ln|tan(x/2)| + c.

Exercise 3.4, Q3 - Show that Int e^(a x) sin b x dx = (1/sqrt(a2 + b2)) e^(a x) sin(b x - Tan^{-1}(b/a)) + c (Print note: this copy prints '+' before Tan^{-1}; the proof gives '-'; the printed sign is a misprint.)

Q3. Proof.

Let I = Int e^(a x) sin b x dx and J = Int e^(a x) cos b x dx.

Integrating I by parts twice gives I = e^(a x)(a sin b x - b cos b x)/(a2 + b2).

Write a sin b x - b cos b x = sqrt(a2 + b2) sin(b x - alpha), where cos alpha = a/sqrt(a2 + b2) and sin alpha = b/sqrt(a2 + b2), i.e. alpha = Tan^{-1}(b/a).

Hence I = e^(a x) sin(b x - Tan^{-1}(b/a))/sqrt(a2 + b2) + c.

Answer: proved (with the minus sign; the '+' printed in this copy does not differentiate back to e^(a x) sin b x).

Exercise 3.4, Q4 - Evaluate the following indefinite integrals (c = constant of integration)

Q4(i). Int sqrt(a2 - x2) dx.

This is Exercise 3.3 Q8(b): put x = a sin theta.

Answer: (a2/2) Sin^{-1}(x/a) + (x/2) sqrt(a2 - x2) + c.

Q4(ii). Int sqrt(x2 - a2) dx.

Put x = a sec theta; then dx = a sec theta tan theta d theta and sqrt(x2 - a2) = a tan theta.

The integral becomes a2 Int tan2 theta sec theta d theta, which by parts gives (x/2) sqrt(x2 - a2) - (a2/2) ln|x + sqrt(x2 - a2)|.

Answer: (x/2) sqrt(x2 - a2) - (a2/2) ln|x + sqrt(x2 - a2)| + c.

Q4(iii). Int sqrt(4 - 5x2) dx.

Use Int sqrt(A2 - B2 x2) dx = (x/2) sqrt(A2 - B2 x2) + (A2/(2B)) Sin^{-1}(B x/A) with A = 2, B = sqrt(5).

Answer: (x/2) sqrt(4 - 5x2) + (2/sqrt(5)) Sin^{-1}(sqrt(5) x/2) + c.

Q4(iv). Int sqrt(3 - 4x2) dx.

Use the same formula with A = sqrt(3), B = 2.

Answer: (x/2) sqrt(3 - 4x2) + (3/4) Sin^{-1}(2x/sqrt(3)) + c.

Q4(v). Int sqrt(x2 + 4) dx.

Put x = 2 tan theta; the integral becomes 4 Int sec3 theta d theta.

Using Int sec3 theta d theta = (1/2)(sec theta tan theta + ln|sec theta + tan theta|), and back-substituting, gives (x/2) sqrt(x2 + 4) + 2 ln|x + sqrt(x2 + 4)|.

Answer: (x/2) sqrt(x2 + 4) + 2 ln|x + sqrt(x2 + 4)| + c.

Q4(vi). Int x2 e^(a x) dx.

Integrate by parts twice: Int x2 e^(a x) dx = (x2/a) e^(a x) - (2/a) Int x e^(a x) dx.

Int x e^(a x) dx = e^(a x)(x/a - 1/a2).

Answer: e^(a x)(x2/a - 2x/a2 + 2/a3) + c.

Exercise 3.4, Q5 - Evaluate the following integrals (using Int e^(a x)[a f(x) + f'(x)] dx = e^(a x) f(x) + c) (c = constant of integration)

Q5(i). Int e^x (1/x + ln x) dx.

Take f(x) = ln x; then f'(x) = 1/x, so the integrand is e^x [f(x) + f'(x)].

Answer: e^x ln x + c.

Q5(ii). Int e^x (cos x + sin x) dx.

Take f(x) = sin x; then f'(x) = cos x, so the integrand is e^x [f(x) + f'(x)].

Answer: e^x sin x + c.

Q5(iii). Int e^(a x) [a Sec^{-1} x + 1/(x sqrt(x2 - 1))] dx.

Take f(x) = Sec^{-1} x; then f'(x) = 1/(x sqrt(x2 - 1)), so the integrand is e^(a x)[a f(x) + f'(x)].

Answer: e^(a x) Sec^{-1} x + c.

Q5(iv). Int e^(3x) [(3 sin x - cos x)/sin2 x] dx.

Rewrite the bracket: (3 sin x - cos x)/sin2 x = 3 cosec x - cosec x cot x.

Take f(x) = cosec x; then f'(x) = -cosec x cot x, so the bracket is 3 f(x) + f'(x).

Answer: e^(3x) cosec x + c.

Q5(v). Int e^(2x) [-sin x + 2 cos x] dx.

Take f(x) = cos x; then f'(x) = -sin x, so the bracket is 2 f(x) + f'(x).

Answer: e^(2x) cos x + c.

Q5(vi). Int x e^x / (1 + x)2 dx.

Write x/(1 + x)2 = 1/(1 + x) - 1/(1 + x)2.

Take f(x) = 1/(1 + x); then f'(x) = -1/(1 + x)2, so the integrand is e^x [f(x) + f'(x)].

Answer: e^x/(1 + x) + c.

Q5(vii). Int e^(-x) (cos x - sin x) dx.

Take f(x) = sin x; then -f(x) + f'(x) = -sin x + cos x, matching the bracket with a = -1.

Answer: e^(-x) sin x + c.

Q5(viii). Int e^(m Tan^{-1} x) / (1 + x2) dx.

Put u = Tan^{-1} x; then du = dx/(1 + x2).

The integral becomes Int e^(m u) du = e^(m u)/m.

Answer: e^(m Tan^{-1} x)/m + c.

Q5(ix). Int 2x / (1 - sin x) dx.

Multiply top and bottom by (1 + sin x): 2x/(1 - sin x) = 2x(1 + sin x)/cos2 x = 2x sec2 x + 2x sec x tan x.

Int 2x sec2 x dx = 2x tan x + 2 ln|cos x|; Int 2x sec x tan x dx = 2x sec x - 2 ln|sec x + tan x|.

Since ln|cos x| - ln|sec x + tan x| = ln|1 - sin x|, the logs combine.

Answer: 2x (tan x + sec x) + 2 ln|1 - sin x| + c.

Q5(x). Int e^x (1 + x)/(2 + x)2 dx.

Take f(x) = 1/(2 + x); then f(x) + f'(x) = 1/(2 + x) - 1/(2 + x)2 = (1 + x)/(2 + x)2.

Answer: e^x/(2 + x) + c.

Q5(xi). Int ((1 - sin x)/(1 - cos x)) e^x dx.

Using half-angles, (1 - sin x)/(1 - cos x) = (1/2) cosec2(x/2) - cot(x/2).

Take f(x) = -cot(x/2); then f'(x) = (1/2) cosec2(x/2), so the integrand is e^x [f(x) + f'(x)].

Answer: -e^x cot(x/2) + c.

Exercise 3.5 - Evaluate the following integrals (partial fractions; c = constant of integration)

Q1. Int (3x + 1)/(x2 - x - 6) dx.

Factor: x2 - x - 6 = (x - 3)(x + 2).

Partial fractions: (3x + 1)/((x - 3)(x + 2)) = 2/(x - 3) + 1/(x + 2).

Answer: 2 ln|x - 3| + ln|x + 2| + c.

Q2. Int (5x + 8)/((x + 3)(2x - 1)) dx.

Partial fractions: (5x + 8)/((x + 3)(2x - 1)) = 1/(x + 3) + 3/(2x - 1).

Answer: ln|x + 3| + (3/2) ln|2x - 1| + c.

Q3. Int (x2 + 3x - 34)/(x2 + 2x - 15) dx.

Divide first: the quotient is 1 with remainder x - 19, and (x - 19)/((x + 5)(x - 3)) = 3/(x + 5) - 2/(x - 3).

Answer: x + 3 ln|x + 5| - 2 ln|x - 3| + c.

Q4. Int (a - b)x/((x - a)(x - b)) dx, a > b.

Partial fractions: (a - b)x/((x - a)(x - b)) = a/(x - a) - b/(x - b).

Answer: a ln|x - a| - b ln|x - b| + c.

Q5. Int (3 - x)/(1 - x - 6x2) dx.

Factor: 1 - x - 6x2 = (1 - 3x)(1 + 2x).

Partial fractions: (3 - x)/((1 - 3x)(1 + 2x)) = (8/5)/(1 - 3x) + (7/5)/(1 + 2x).

Answer: -(8/15) ln|1 - 3x| + (7/10) ln|1 + 2x| + c.

Q6. Int 2x/(x2 - a2) dx.

Partial fractions: 2x/((x - a)(x + a)) = 1/(x - a) + 1/(x + a).

Answer: ln|x2 - a2| + c.

Q7. Int dx/(6x2 + 5x - 4).

Factor: 6x2 + 5x - 4 = (3x + 4)(2x - 1).

Partial fractions: 1/((3x + 4)(2x - 1)) = (-3/11)/(3x + 4) + (2/11)/(2x - 1).

Answer: (1/11) ln|(2x - 1)/(3x + 4)| + c.

Q8. Int (2x3 - 3x2 - x - 7)/(2x2 - 3x - 2) dx.

Divide: the quotient is x with remainder x - 7, and (x - 7)/((2x + 1)(x - 2)) = 3/(2x + 1) - 1/(x - 2).

Answer: x2/2 + (3/2) ln|2x + 1| - ln|x - 2| + c.

Q9. Int (3x2 - 12x + 11)/((x - 1)(x - 2)(x - 3)) dx.

Partial fractions: the integrand equals 1/(x - 1) + 1/(x - 2) + 1/(x - 3).

Answer: ln|(x - 1)(x - 2)(x - 3)| + c.

Q10. Int (2x - 1)/(x(x - 1)(x - 3)) dx.

Partial fractions: the integrand equals (-1/3)/x - (1/2)/(x - 1) + (5/6)/(x - 3).

Answer: -(1/3) ln|x| - (1/2) ln|x - 1| + (5/6) ln|x - 3| + c.

Q11. Int (5x2 + 9x + 6)/((x2 - 1)(2x + 3)) dx.

Factor: (x2 - 1)(2x + 3) = (x - 1)(x + 1)(2x + 3).

Partial fractions: the integrand equals 2/(x - 1) - 1/(x + 1) + 3/(2x + 3).

Answer: 2 ln|x - 1| - ln|x + 1| + (3/2) ln|2x + 3| + c.

Q12. Int (4 + 7x)/((1 + x)2(2 + 3x)) dx.

Partial fractions: the integrand equals 2/(1 + x) + 3/(1 + x)2 - 6/(2 + 3x).

Answer: 2 ln|1 + x| - 3/(1 + x) - 2 ln|2 + 3x| + c.

Q13. Int 2x2/((x - 1)2(x + 1)) dx.

Partial fractions: the integrand equals (3/2)/(x - 1) + 1/(x - 1)2 + (1/2)/(x + 1).

Answer: (3/2) ln|x - 1| - 1/(x - 1) + (1/2) ln|x + 1| + c.

Q14. Int dx/((x - 1)(x + 1)2).

Partial fractions: the integrand equals (1/4)/(x - 1) - (1/4)/(x + 1) - (1/2)/(x + 1)2.

Answer: (1/4) ln|(x - 1)/(x + 1)| + 1/(2(x + 1)) + c.

Q15. Int (x + 4)/(x3 - 3x2 + 4) dx.

Factor: x3 - 3x2 + 4 = (x + 1)(x - 2)2.

Partial fractions: the integrand equals (1/3)/(x + 1) - (1/3)/(x - 2) + 2/(x - 2)2.

Answer: (1/3) ln|x + 1| - (1/3) ln|x - 2| - 2/(x - 2) + c.

Q16. Int (x3 - 6x2 + 25)/((x + 1)2(x - 2)2) dx.

Partial fractions: the integrand equals 3/(x + 1) + 2/(x + 1)2 - 2/(x - 2) + 1/(x - 2)2.

Answer: 3 ln|x + 1| - 2/(x + 1) - 2 ln|x - 2| - 1/(x - 2) + c.

Q17. Int (x3 + 22x2 + 14x - 17)/((x - 3)(x + 2)3) dx.

Partial fractions: the integrand equals 2/(x - 3) - 1/(x + 2) + 11/(x + 2)2 - 7/(x + 2)3.

Answer: 2 ln|x - 3| - ln|x + 2| - 11/(x + 2) + 7/(2(x + 2)2) + c.

Q18. Int (x - 2)/((x + 1)(x2 + 1)) dx.

Partial fractions: the integrand equals (-3/2)/(x + 1) + ((3/2)x - 1/2)/(x2 + 1).

Answer: -(3/2) ln|x + 1| + (3/4) ln(x2 + 1) - (1/2) Tan^{-1} x + c.

Q19. Int x/((x - 1)(x2 + 1)) dx.

Partial fractions: the integrand equals (1/2)/(x - 1) + ((-1/2)x + 1/2)/(x2 + 1).

Answer: (1/2) ln|x - 1| - (1/4) ln(x2 + 1) + (1/2) Tan^{-1} x + c.

Q20. Int (9x - 7)/((x + 3)(x2 + 1)) dx.

Partial fractions: the integrand equals (-17/5)/(x + 3) + ((17/5)x - 6/5)/(x2 + 1).

Answer: -(17/5) ln|x + 3| + (17/10) ln(x2 + 1) - (6/5) Tan^{-1} x + c.

Q21. Int (1 + 4x)/((x - 3)(x2 + 4)) dx.

Partial fractions: the integrand equals 1/(x - 3) + (-x + 1)/(x2 + 4).

Answer: ln|x - 3| - (1/2) ln(x2 + 4) + (1/2) Tan^{-1}(x/2) + c.

Q22. FLAGGED - Int 12/(? + 8) dx: denominator incomplete in the source copy; no answer given.

As printed in this copy, Q22 reads 'Int 12/(+ 8) dx' - the first term of the denominator is blank (not legible even at 300 DPI).

Without that term the integrand cannot be identified, so no partial-fraction decomposition is attempted.

Answer: NOT COMPUTED - the printed denominator is incomplete in the source copy; nothing has been invented here.

Q23. Int (9x + 6)/(x3 - 8) dx.

Factor: x3 - 8 = (x - 2)(x2 + 2x + 4).

Partial fractions: the integrand equals 2/(x - 2) + (-2x + 1)/(x2 + 2x + 4).

The second part gives -ln(x2 + 2x + 4) + sqrt(3) Tan^{-1}((x + 1)/sqrt(3)).

Answer: 2 ln|x - 2| - ln(x2 + 2x + 4) + sqrt(3) Tan^{-1}((x + 1)/sqrt(3)) + c.

Q24. Int (2x2 + 5x + 3)/((x - 1)2(x2 + 4)) dx.

Partial fractions: the integrand equals 1/(x - 1) + 2/(x - 1)2 + (-x - 1)/(x2 + 4).

Answer: ln|x - 1| - 2/(x - 1) - (1/2) ln(x2 + 4) - (1/2) Tan^{-1}(x/2) + c.

Q25. Int (2x2 - x - 7)/((x + 2)2(x2 + x + 1)) dx.

Partial fractions: the integrand equals -2/(x + 2) + 1/(x + 2)2 + (2x - 1)/(x2 + x + 1).

The last part gives ln(x2 + x + 1) - (4/sqrt(3)) Tan^{-1}((2x + 1)/sqrt(3)).

Answer: -2 ln|x + 2| - 1/(x + 2) + ln(x2 + x + 1) - (4/sqrt(3)) Tan^{-1}((2x + 1)/sqrt(3)) + c.

Q26. Int (3x + 1)/((4x2 + 1)(x2 - x + 1)) dx.

Partial fractions: the integrand equals 4x/(4x2 + 1) + 1/(x2 - x + 1).

Answer: (1/2) ln(4x2 + 1) + (2/sqrt(3)) Tan^{-1}((2x - 1)/sqrt(3)) + c.

Q27. Int (4x + 1)/((x2 + 4)(x2 + 4x + 5)) dx.

Partial fractions: the integrand equals 1/(x2 + 4) - 1/(x2 + 4x + 5).

Since x2 + 4x + 5 = (x + 2)2 + 1, the second integral is Tan^{-1}(x + 2).

Answer: (1/2) Tan^{-1}(x/2) - Tan^{-1}(x + 2) + c.

Q28. Int 6a2/((x2 + a2)(x2 + 4a2)) dx.

Partial fractions: the integrand equals 2/(x2 + a2) - 2/(x2 + 4a2).

Answer: (2/a) Tan^{-1}(x/a) - (1/a) Tan^{-1}(x/(2a)) + c.

Q29. Int (2x2 - 2)/(x4 + x2 + 1) dx.

Factor: x4 + x2 + 1 = (x2 + x + 1)(x2 - x + 1).

Partial fractions: the integrand equals (-2x - 1)/(x2 + x + 1) + (2x - 1)/(x2 - x + 1).

The first part gives -ln(x2 + x + 1) and the second gives ln(x2 - x + 1).

Answer: ln((x2 - x + 1)/(x2 + x + 1)) + c.

Q30. Int (3x - 8)/((x2 - x + 2)(x2 + x + 2)) dx.

Partial fractions: the integrand equals (2x - 1/2)/(x2 - x + 2) + (-2x - 7/2)/(x2 + x + 2).

The first part gives ln(x2 - x + 2) + (1/sqrt(7)) Tan^{-1}((2x - 1)/sqrt(7)); the second gives -ln(x2 + x + 2) - (5/sqrt(7)) Tan^{-1}((2x + 1)/sqrt(7)).

Answer: ln((x2 - x + 2)/(x2 + x + 2)) + (1/sqrt(7)) Tan^{-1}((2x - 1)/sqrt(7)) - (5/sqrt(7)) Tan^{-1}((2x + 1)/sqrt(7)) + c.

Q31. Int (3x3 + 4x2 + 9x + 5)/((x2 + x + 1)(x2 + 2x + 3)) dx.

Partial fractions: the integrand equals (2x + 3)/(x2 + x + 1) + (x - 4)/(x2 + 2x + 3).

The first part gives ln(x2 + x + 1) + (4/sqrt(3)) Tan^{-1}((2x + 1)/sqrt(3)); the second gives (1/2) ln(x2 + 2x + 3) - (5/sqrt(2)) Tan^{-1}((x + 1)/sqrt(2)).

Answer: ln(x2 + x + 1) + (4/sqrt(3)) Tan^{-1}((2x + 1)/sqrt(3)) + (1/2) ln(x2 + 2x + 3) - (5/sqrt(2)) Tan^{-1}((x + 1)/sqrt(2)) + c.

Exercise 3.6 - Evaluate the following definite integrals

Q1. Int from 1 to 2 of (x2 + 1) dx.

Antiderivative: x3/3 + x.

Evaluate: (8/3 + 2) - (1/3 + 1) = 14/3 - 4/3.

Answer: 10/3.

Q2. Int from -1 to 1 of (x^(1/3) + 1) dx.

The cube root is an odd function, so its integral over [-1, 1] is 0.

The integral of 1 over [-1, 1] is 2.

Answer: 2.

Q3. Int from -2 to 0 of dx/(2x - 1)2.

Antiderivative: -1/(2(2x - 1)).

Evaluate: -1/(2(-1)) - (-1/(2(-5))) = 1/2 - 1/10.

Answer: 2/5.

Q4. Int from -6 to 2 of sqrt(3 - x) dx.

Antiderivative: -(2/3)(3 - x)^(3/2).

Evaluate: -(2/3)(1)^(3/2) + (2/3)(9)^(3/2) = -2/3 + 18.

Answer: 52/3.

Q5. FLAGGED - Int from ? to sqrt(?) of sqrt(2t ? 1) dt: limits and sign illegible in the source copy; no answer given.

As printed in this copy, Q5 shows a blank lower limit, an upper limit consisting of a bare radical sign with no radicand, and an integrand 'sqrt(2t 1)' whose plus/minus sign is missing (not legible even at 300 DPI).

Without the limits and the sign, the integral cannot be identified, so no evaluation is attempted.

Answer: NOT COMPUTED - the printed item is incomplete in the source copy; nothing has been invented here.

Q6. Int from 2 to sqrt(5) of x sqrt(x2 - 1) dx.

Put u = x2 - 1; then du = 2x dx, so the antiderivative is (1/3)(x2 - 1)^(3/2).

Evaluate: (1/3)(4)^(3/2) - (1/3)(3)^(3/2) = 8/3 - sqrt(3).

Answer: 8/3 - sqrt(3).

Q7. Int from 1 to 2 of x/(x2 + 2) dx.

Antiderivative: (1/2) ln(x2 + 2).

Evaluate: (1/2)(ln 6 - ln 3) = (1/2) ln 2.

Answer: (1/2) ln 2.

Q8. Int from 2 to 3 of (x - 1/x)2 dx.

Expand: (x - 1/x)2 = x2 - 2 + 1/x2; antiderivative x3/3 - 2x - 1/x.

Evaluate: (9 - 6 - 1/3) - (8/3 - 4 - 1/2) = 8/3 + 7/6.

Answer: 23/6.

Q9. Int from -1 to 1 of (x + 1/2) sqrt(x2 + x + 1) dx.

Since x + 1/2 = (1/2)(2x + 1) and d/dx (x2 + x + 1) = 2x + 1, the antiderivative is (1/3)(x2 + x + 1)^(3/2).

Evaluate: (1/3)(3^(3/2) - 1) = sqrt(3) - 1/3.

Answer: sqrt(3) - 1/3.

Q10. Int from 0 to 3 of dx/(x2 + 9).

Antiderivative: (1/3) Tan^{-1}(x/3).

Evaluate: (1/3) Tan^{-1}(1) - 0 = (1/3)(pi/4).

Answer: pi/12.

Q11. Int from pi/6 to pi/3 of cos t dt.

Antiderivative: sin t.

Evaluate: sin(pi/3) - sin(pi/6) = sqrt(3)/2 - 1/2.

Answer: (sqrt(3) - 1)/2.

Q12. Int from 1 to 2 of (x + 1/x)^(1/2) (1 - 1/x2) dx.

Put u = x + 1/x; then du = (1 - 1/x2) dx, so the antiderivative is (2/3)(x + 1/x)^(3/2).

Evaluate: (2/3)[(5/2)^(3/2) - 2^(3/2)] = (5 sqrt(10))/6 - (4 sqrt(2))/3.

Answer: (5 sqrt(10))/6 - (4 sqrt(2))/3.

Q13. Int from 1 to 2 of ln x dx.

Antiderivative: x ln x - x.

Evaluate: (2 ln 2 - 2) - (0 - 1) = 2 ln 2 - 1.

Answer: 2 ln 2 - 1.

Q14. Int from 0 to 2 of (e^(x/2) - e^(-x/2)) dx.

Antiderivative: 2e^(x/2) + 2e^(-x/2).

Evaluate: (2e + 2/e) - 4.

Answer: 2e + 2/e - 4.

Q15. Int from 0 to pi/4 of (cos theta + sin theta)/(2 cos2 theta) d theta.

Split: (1/2) Int (sec theta + sin theta sec2 theta) d theta; antiderivative (1/2)(ln|sec theta + tan theta| + sec theta).

Evaluate: (1/2)[(ln(sqrt(2) + 1) + sqrt(2)) - 1].

Answer: (ln(sqrt(2) + 1) + sqrt(2) - 1)/2.

Q16. Int from 0 to pi/6 of cos3 theta d theta.

Write cos3 theta = (1 - sin2 theta) cos theta; antiderivative sin theta - sin3 theta/3.

Evaluate: 1/2 - (1/8)/3 = 1/2 - 1/24.

Answer: 11/24.

Q17. Int from pi/6 to pi/4 of cos2 theta cot2 theta d theta.

Since cos2 theta cot2 theta = cot2 theta - cos2 theta, the antiderivative is -cot theta - (3 theta/2) - sin 2 theta/4.

Evaluate at pi/4: -5/4 - 3pi/8; at pi/6: -(9 sqrt(3))/8 - pi/4.

Answer: (9 sqrt(3))/8 - 5/4 - pi/8.

Q18. Int from 0 to pi/4 of cos4 t dt.

Use cos4 t = (3 + 4 cos 2t + cos 4t)/8; antiderivative 3t/8 + sin 2t/4 + sin 4t/32.

Evaluate: 3pi/32 + 1/4.

Answer: 3pi/32 + 1/4.

Q19. Int from 0 to pi/3 of cos2 theta sin theta d theta.

Put u = cos theta; antiderivative -cos3 theta/3.

Evaluate: -(1/8)/3 + 1/3 = 7/24.

Answer: 7/24.

Q20. Int from 0 to pi/4 of (1 + cos2 theta) tan2 theta d theta.

Since (1 + cos2 theta) tan2 theta = sec2 theta - cos2 theta, the antiderivative is tan theta - theta/2 - sin 2 theta/4.

Evaluate: 1 - pi/8 - 1/4.

Answer: 3/4 - pi/8.

Q21. Int from 0 to pi/4 of sec theta/(sin theta + cos theta) d theta.

Divide top and bottom by cos theta: the integrand becomes sec2 theta/(1 + tan theta).

Put u = tan theta: Int from 0 to 1 of du/(1 + u) = ln 2.

Answer: ln 2.

Q22. Int from -1 to 5 of |x - 3| dx.

Split at x = 3: Int from -1 to 3 of (3 - x) dx = 8; Int from 3 to 5 of (x - 3) dx = 2.

Answer: 10.

Q23. Int from 1/8 to 1 of (x^(1/3) + 2)2 / x^(2/3) dx.

Expand: the integrand equals 1 + 4x^(-1/3) + 4x^(-2/3); antiderivative x + 6x^(2/3) + 12x^(1/3).

Evaluate: 19 - (1/8 + 3/2 + 6) = 19 - 61/8.

Answer: 91/8.

Q24. Int from 1 to 3 of (x2 - 2)/(x + 1) dx.

Divide: (x2 - 2)/(x + 1) = x - 1 - 1/(x + 1); antiderivative x2/2 - x - ln|x + 1|.

Evaluate: (3/2 - ln 4) - (-1/2 - ln 2) = 2 - ln 2.

Answer: 2 - ln 2.

Q25. Int from 2 to 3 of (3x2 - 2x + 1)/((x - 1)(x2 + 1)) dx.

Partial fractions: the integrand equals 1/(x - 1) + (2x)/(x2 + 1); antiderivative ln|x - 1| + ln(x2 + 1).

Evaluate: (ln 2 + ln 10) - (ln 1 + ln 5) = ln 4.

Answer: ln 4 = 2 ln 2.

Q26. Int from 0 to pi/4 of (sin x - 1)/cos2 x dx.

Write as sin x sec2 x - sec2 x; antiderivative sec x - tan x.

Evaluate: (sqrt(2) - 1) - (1 - 0) = sqrt(2) - 2.

Answer: sqrt(2) - 2.

Q27. Int from 0 to pi/4 of dx/(1 + sin x).

Multiply top and bottom by (1 - sin x): the integrand becomes sec2 x - sec x tan x; antiderivative tan x - sec x.

Evaluate: (1 - sqrt(2)) - (0 - 1) = 2 - sqrt(2).

Answer: 2 - sqrt(2).

Q28. Int from 0 to 1 of 3x/sqrt(4 - 3x) dx.

Put u = 4 - 3x; the integral becomes (1/3) Int from 1 to 4 of (4 - u)/sqrt(u) du.

Antiderivative (1/3)(8 sqrt(u) - (2/3) u^(3/2)); evaluate: (1/3)(32/3 - 22/3).

Answer: 10/9.

Q29. Int from pi/6 to pi/2 of cos x/(sin x (2 + sin x)) dx.

Put u = sin x: the integral becomes Int from 1/2 to 1 of du/(u(2 + u)) = (1/2)[ln(u/(2 + u))] from 1/2 to 1.

Evaluate: (1/2)(ln(1/3) - ln(1/5)) = (1/2) ln(5/3).

Answer: (1/2) ln(5/3).

Q30. Int from 0 to pi/2 of sin x/((1 + cos x)(2 + cos x)) dx.

Put u = cos x: the integral becomes Int from 0 to 1 of du/((1 + u)(2 + u)) = [ln((1 + u)/(2 + u))] from 0 to 1.

Evaluate: ln(2/3) - ln(1/2) = ln(4/3).

Answer: ln(4/3).

Exercise 3.7 - Areas by definite integrals (answers in square units)

Q1. Area between the x-axis and the curve y = x2 + 1 from x = 1 to x = 2.

Area = Int from 1 to 2 of (x2 + 1) dx = [x3/3 + x] from 1 to 2.

Evaluate: (8/3 + 2) - (1/3 + 1) = 10/3.

Answer: 10/3 square units.

Q2. Area above the x-axis and under the curve y = 5 - x2 from x = -1 to x = 2.

Area = Int from -1 to 2 of (5 - x2) dx = [5x - x3/3] from -1 to 2.

Evaluate: (10 - 8/3) - (-5 + 1/3) = 22/3 + 14/3 = 12.

Answer: 12 square units.

Q3. Area below the curve y = 3 sqrt(x) and above the x-axis between x = 1 and x = 4.

Area = Int from 1 to 4 of 3 sqrt(x) dx = 2[x^(3/2)] from 1 to 4.

Evaluate: 2(8 - 1) = 14.

Answer: 14 square units.

Q4. Area bounded by the cos function from x = -pi/2 to x = pi/2.

On [-pi/2, pi/2], cos x is non-negative, so area = Int from -pi/2 to pi/2 of cos x dx.

Evaluate: sin(pi/2) - sin(-pi/2) = 1 - (-1) = 2.

Answer: 2 square units.

Q5. Area between the x-axis and the curve y = 4x - x2.

The curve cuts the x-axis where 4x - x2 = 0, i.e. at x = 0 and x = 4; it is above the axis between them.

Area = Int from 0 to 4 of (4x - x2) dx = [2x2 - x3/3] from 0 to 4 = 32 - 64/3.

Answer: 32/3 square units.

Q6. Area bounded by the parabola y = x2 + 2x - 3 and the x-axis.

The parabola cuts the x-axis where (x + 3)(x - 1) = 0, i.e. at x = -3 and x = 1; it lies below the axis between them.

Area = -Int from -3 to 1 of (x2 + 2x - 3) dx = -[x3/3 + x2 - 3x] from -3 to 1.

Evaluate: -[(-5/3) - 9] = 32/3.

Answer: 32/3 square units.

Q7. Area bounded by the curve y = x3 + 1, the x-axis and the line x = 2.

The curve cuts the x-axis at x = -1, so the enclosed region runs from x = -1 to x = 2.

Area = Int from -1 to 2 of (x3 + 1) dx = [x4/4 + x] from -1 to 2.

Evaluate: 6 - (1/4 - 1) = 27/4.

Answer: 27/4 square units.

Q8. Area bounded by the curve y = x3 - 4x and the x-axis.

The curve cuts the x-axis at x = -2, 0, 2; the two lobes are symmetric in area.

Area = 2 Int from 0 to 2 of (4x - x3) dx = 2[2x2 - x4/4] from 0 to 2 = 2(8 - 4).

Answer: 8 square units.

Q9. Area between the curve y = x(x - 1)(x + 1) and the x-axis.

The curve is y = x3 - x, cutting the x-axis at x = -1, 0, 1; the two lobes are symmetric in area.

Area = 2 Int from 0 to 1 of (x - x3) dx = 2[1/2 - 1/4] = 1/2.

Answer: 1/2 square unit.

Q10. Area above the x-axis, bounded by the curve y2 = 3 - x, from x = -1 to x = 2.

Above the x-axis, y = sqrt(3 - x).

Area = Int from -1 to 2 of sqrt(3 - x) dx = [-(2/3)(3 - x)^(3/2)] from -1 to 2.

Evaluate: -(2/3)(1) + (2/3)(8) = 14/3.

Answer: 14/3 square units.

Q11. Area between the x-axis and the curve y = -cos(x/2) from x = -pi to x = pi.

On [-pi, pi], cos(x/2) is non-negative, so |y| = cos(x/2).

Area = Int from -pi to pi of cos(x/2) dx = [2 sin(x/2)] from -pi to pi = 2(1 - (-1)).

Answer: 4 square units.

Q12. Area between the x-axis and the curve y = sin 2x from x = 0 to x = pi/3.

On [0, pi/3], sin 2x is non-negative.

Area = Int from 0 to pi/3 of sin 2x dx = [-cos 2x/2] from 0 to pi/3 = (1/2 + 1)/2.

Answer: 3/4 square units.

Q13. Area between the x-axis and the curve y = sqrt(2a x - x2) when a > 0.

The curve is the upper semicircle of radius a centred at (a, 0), meeting the x-axis at x = 0 and x = 2a.

Area = Int from 0 to 2a of sqrt(2a x - x2) dx = (1/2) pi a2 (area of a semicircle of radius a).

Answer: (pi a2)/2 square units.

Exercise 3.8, Q1 - Check that each of the following equations written against the differential equation is its solution

Q1(i). Equation x dy/dx = 1 + y; solution y = c x - 1.

Differentiate the solution: dy/dx = c.

Left side: x dy/dx = c x. Right side: 1 + y = 1 + c x - 1 = c x.

Both sides agree, so the equation is satisfied.

Answer: verified.

Q1(ii). Equation x2(2y + 1) dy/dx - 1 = 0; solution y2 + y = c - 1/x.

Differentiate the solution: (2y + 1) dy/dx = 1/x2.

Multiply by x2: x2(2y + 1) dy/dx = 1, i.e. x2(2y + 1) dy/dx - 1 = 0.

Answer: verified.

Q1(iii). Equation y dy/dx - e^(2x) = 1; solution y2 = e^(2x) + 2x + c.

Differentiate the solution: 2y dy/dx = 2e^(2x) + 2.

Divide by 2: y dy/dx = e^(2x) + 1, i.e. y dy/dx - e^(2x) = 1.

Answer: verified.

Q1(iv). Equation (1/x) dy/dx - 2y = 0; solution y = c e^(x2).

Differentiate the solution: dy/dx = 2x c e^(x2) = 2x y.

Then (1/x) dy/dx = 2y, i.e. (1/x) dy/dx - 2y = 0.

Answer: verified.

Q1(v). Equation dy/dx = (y2 + 1)/e^(-x); solution y = tan(e^x + c).

Differentiate the solution: dy/dx = sec2(e^x + c) e^x = (1 + y2) e^x.

Since 1/e^(-x) = e^x, this is dy/dx = (y2 + 1)/e^(-x).

Answer: verified.

Exercise 3.8, Q2-Q18 - Solve the following differential equations (c =

arbitrary constant)

Q2. dy/dx = -y.

Separate: dy/y = -dx.

Integrate: ln|y| = -x + c.

Answer: y = C e^(-x).

Q3. y dx + x dy = 0.

This is d(x y) = 0.

Integrate: x y = c.

Answer: x y = c.

Q4. dy/dx = (1 - x)/y.

Separate: y dy = (1 - x) dx.

Integrate: y2/2 = x - x2/2 + c.

Answer: y2 = 2x - x2 + C.

Q5. dy/dx = y/x2, y > 0.

Separate: dy/y = dx/x2.

Integrate: ln y = -1/x + c.

Answer: y = C e^(-1/x).

Q6. sin y cosec x dy/dx = 1.

Separate: sin y dy = sin x dx (since 1/cosec x = sin x).

Integrate: -cos y = -cos x + c.

Answer: cos y = cos x + C.

Q7. x dy + y(x - 1) dx = 0.

Separate: dy/y = -(x - 1)/x dx = (-1 + 1/x) dx.

Integrate: ln|y| = -x + ln|x| + c.

Answer: y = C x e^(-x).

Q8. (x2 + 1)/(y + 1) = (x/y) dy/dx, x, y > 0.

Rearrange: dy/dx = y(x2 + 1)/(x(y + 1)).

Separate: (y + 1)/y dy = (x2 + 1)/x dx, i.e. (1 + 1/y) dy = (x + 1/x) dx.

Integrate: y + ln y = x2/2 + ln x + c.

Answer: y + ln y = x2/2 + ln x + c.

Q9. (1/x) dy/dx = (1/2)(1 + y2).

Separate: dy/(1 + y2) = (x/2) dx.

Integrate: Tan^{-1} y = x2/4 + c.

Answer: Tan^{-1} y = x2/4 + c.

Q10. 2x2 y dy/dx = x2 - 1.

Separate: y dy = (x2 - 1)/(2x2) dx = (1/2 - 1/(2x2)) dx.

Integrate: y2/2 = x/2 + 1/(2x) + c.

Answer: y2 = x + 1/x + C.

Q11. dy/dx + 2x y/(2y + 1) = x.

Bring the fraction over: dy/dx = x - 2x y/(2y + 1) = x/(2y + 1).

Separate: (2y + 1) dy = x dx.

Integrate: y2 + y = x2/2 + c.

Answer: y2 + y = x2/2 + c.

Q12. (x2 - y x2) dy/dx + y2 + x y2 = 0.

Factor: x2(1 - y) dy/dx + y2(1 + x) = 0.

Separate: (1 - y)/y2 dy = -(1 + x)/x2 dx.

Integrate: -1/y - ln|y| = 1/x - ln|x| + c.

Answer: ln|x/y| = 1/x + 1/y + C.

Q13. sec2 x tan y dx + sec2 y tan x dy = 0.

Separate: tan x sec2 x dx = -tan y sec2 y dy.

Integrate: (1/2) tan2 x = -(1/2) tan2 y + c.

Answer: tan2 x + tan2 y = C.

Q14. (y - x dy/dx) = 2(y2 + dy/dx).

Expand: y - x dy/dx = 2y2 + 2 dy/dx, so (x + 2) dy/dx = y - 2y2.

Separate: dy/(y - 2y2) = dx/(x + 2).

Integrate: ln|y/(1 - 2y)| = ln|x + 2| + c.

Answer: y/(1 - 2y) = C(x + 2).

Q15. 1 + cos x tan y dy/dx = 0.

Separate: tan y dy = -dx/(1 + cos x).

Integrate: -ln|cos y| = -tan(x/2) + c, since Int dx/(1 + cos x) = tan(x/2).

Answer: cos y = C e^(tan(x/2)).

Q16. y - x dy/dx = 3(1 + x dy/dx).

Expand: y - x dy/dx = 3 + 3x dy/dx, so 4x dy/dx = y - 3.

Separate: dy/(y - 3) = dx/(4x).

Integrate: ln|y - 3| = (1/4) ln|x| + c.

Answer: y - 3 = C x^(1/4).

Q17. sec x + tan y dy/dx = 0.

Separate: tan y dy = -sec x dx.

Integrate: -ln|cos y| = -ln|sec x + tan x| + c.

Answer: cos y = C(sec x + tan x).

Q18. (e^x + e^(-x)) dy/dx = e^x - e^(-x). (Print note: the RHS operator is faint in this copy; read as minus, the standard form.)

Separate: dy = (e^x - e^(-x))/(e^x + e^(-x)) dx.

The numerator is the derivative of the denominator, so integrate: y = ln(e^x + e^(-x)) + c.

Answer: y = ln(e^x + e^(-x)) + c.

Exercise 3.8, Q19-Q23 - Differential equations and applications

Q19. General solution of dy/dx - x = x y2; also the particular solution if y = 1 when x = 0.

Rewrite: dy/dx = x(1 + y2).

Separate: dy/(1 + y2) = x dx; integrate: Tan^{-1} y = x2/2 + c.

General solution: y = tan(x2/2 + c).

Particular: y = 1 at x = 0 gives c = pi/4.

Answer: y = tan(x2/2 + pi/4).

Q20. Solve dx/dt = 2x given that x = 4 when t = 0.

Separate: dx/x = 2 dt; integrate: ln|x| = 2t + c, so x = C e^(2t).

Particular: x = 4 at t = 0 gives C = 4.

Answer: x = 4 e^(2t).

Q21. Solve ds/dt + 2s t = 0; also the particular solution if s = 4e when t = 0.

Separate: ds/s = -2t dt; integrate: ln|s| = -t2 + c, so s = C e^(-t2).

Particular: s = 4e at t = 0 gives C = 4e.

Answer: s = 4e^(1 - t2).

Q22. In a culture, bacteria increase at a rate proportional to the number present. If there are 200 initially and they double in 2 hours, find the number present four hours later.

Let N(t) be the number: dN/dt = k N, so N = 200 e^(k t).

Doubling in 2 hours: 400 = 200 e^(2k), so e^(2k) = 2.

After 4 hours: N = 200 e^(4k) = 200 (e^(2k))2 = 200 x 4.

Answer: 800 bacteria.

Q23. A ball is thrown vertically upward with a velocity of 2450 cm/sec. Neglecting air resistance, find (i) velocity at any time t, (ii) distance traveled in any time t, (iii) maximum height attained by the ball. (Take g = 980 cm/sec2.)

(i) dv/dt = -g gives v = 2450 - 980t.

(ii) Since v = dh/dt, h = 2450t - 490t2 (starting from h = 0 at t = 0).

(iii) Maximum height when v = 0, i.e. t = 2450/980 = 2.5 sec; h = 2450(2.5) - 490(2.5)2 = 6125 - 3062.5.

Answer: (i) v = 2450 - 980t cm/sec; (ii) h = 2450t - 490t2 cm; (iii) maximum height 3062.5 cm.

Chapter 4: Introduction to Analytic Geometry

Exercise 4.1 - Distance, midpoint and section formula

Q1. Describe the location in the plane of the point P(x, y) for which (i) x > 0 (ii) x > 0 and y > 0 (iii) x = 0 (iv) y = 0 (v) x < 0 and y >= 0 (vi) x = y (vii) |x| = |y| (viii) |x| >= 3 (ix) x > 2 and y = 2 (x) x and y have opposite signs.

(i) x > 0 means P lies to the right of the y-axis: the right half-plane, y-axis excluded.

(ii) x > 0 and y > 0 means P lies in the interior of the first quadrant.

(iii) x = 0 means P lies on the y-axis; (iv) y = 0 means P lies on the x-axis.

(v) x < 0 and y >= 0 means P lies in the second quadrant together with the negative x-axis.

(vi) x = y means P lies on the line y = x, the bisector of the first and third quadrants.

(vii) |x| = |y| means y = x or y = -x: P lies on either bisector of the quadrants.

(viii) |x| >= 3 means P lies on or outside the vertical strip -3 < x < 3, i.e. on or to the left of x = -3, or on or to the right of x = 3.

(ix) x > 2 and y = 2 means P lies on the ray of the line y = 2 to the right of (2, 2), the point (2, 2) excluded.

Answer: (x) x and y of opposite signs means P lies in the interior of the second or the fourth quadrant.

Q2. Find in each case (i) the distance between the two given points (ii) the midpoint of the line segment joining them: (a) A(3, 1); B(-2, -4) (b) A(-8, 3); B(2, -1) (c) A(-sqrt(5), -1/3); B(-3 sqrt(5), 5).

Distance formula: d = sqrt[(x2 - x1)2 + (y2 - y1)2]; midpoint = ((x1 + x2)/2, (y1 +

y2)/2).

(a) Distance = sqrt[(-5)2 + (-5)2] = sqrt(50) = 5 sqrt(2); midpoint = ((3 - 2)/2, (1 - 4)/2) = (1/2, -3/2).

(b) Distance = sqrt[(10)2 + (-4)2] = sqrt(116) = 2 sqrt(29); midpoint = ((-8 + 2)/2, (3 - 1)/2) = (-3, 1).

(c) Distance = sqrt[(-2 sqrt(5))2 + (16/3)2] = sqrt(20 + 256/9) = sqrt(436/9) = 2 sqrt(109)/3.

Answer: (c) midpoint = ((-sqrt(5) - 3 sqrt(5))/2, (-1/3 + 5)/2) = (-2 sqrt(5), 7/3).

Q3. Which of the following points are at a distance of 15 units from the origin? (a) (sqrt(176), 7) (b) (10, -10) (c) (1, 15) (d) (15/2, 15/2).

A point (x, y) is 15 units from the origin exactly when x2 + y2 = 225.

(a) 176 + 49 = 225, so (sqrt(176), 7) is at distance 15.

(b) 100 + 100 = 200, not 225; (c) 1 + 225 = 226, not 225.

(d) 225/4 + 225/4 = 225/2, not 225.

Answer: only (a) is at a distance of 15 units from the origin.

Q4. Show that (i) A(0, 2), B(sqrt(3), 1) and C(0, -2) are vertices of a right triangle (ii) A(3, 1), B(-2, -3) and C(2, 2) are vertices of an isosceles triangle (iii) A(5, 2), B(-2, 3),

C(-3, -4) and D(4, -5) are vertices of a parallelogram. Is the parallelogram a square?

(i) AB2 = 3 + 1 = 4, BC2 = 3 + 9 = 12, AC2 = 0 + 16 = 16; AB2 + BC2 = AC2, so the triangle is right-angled at B.

(ii) AB2 = 25 + 16 = 41, BC2 = 16 + 25 = 41, AC2 = 1 + 1 = 2; AB = BC, so the triangle is isosceles with vertex B.

(iii) Midpoint of AC = (1, -1) and midpoint of BD = (1, -1); the diagonals bisect each other, so ABCD is a parallelogram.

Each side squared is 50: AB2 = 49 + 1, BC2 = 1 + 49, CD2 = 49 + 1, DA2 = 1 + 49, so all sides are equal; also AC2 = 64 + 36 = 100 and BD2 = 36 + 64 = 100, so the diagonals are equal.

Answer: a rhombus with equal diagonals is a square, so yes, the parallelogram is a square.

Q5. The midpoints of the sides of a triangle are (1, -1), (-4, -3) and (-1, 1). Find the coordinates of the vertices of the triangle.

Let the midpoints of BC, CA, AB be (1, -1), (-4, -3), (-1, 1) respectively.

A vertex is the sum of the two adjacent midpoints minus the opposite midpoint: A = (-4, -3) + (-1, 1) - (1, -1) = (-6, -1).

Similarly B = (-1, 1) + (1, -1) - (-4, -3) = (4, 3), and C = (1, -1) + (-4, -3) - (-1, 1) = (-2, -5).

Check: midpoint of BC = (1, -1), of CA = (-4, -3), of AB = (-1, 1), as given.

Answer: the vertices are (-6, -1), (4, 3) and (-2, -5).

Q6. Find h such that A(sqrt(3), -1), B(0, 2) and C(h, -2) are vertices of a right triangle with right angle at the vertex A.

At A, vectors AB = (-sqrt(3), 3) and AC = (h - sqrt(3), -1) must be perpendicular, so their dot product is 0.

Dot product: -sqrt(3)(h - sqrt(3)) + 3(-1) = -sqrt(3) h + 3 - 3 = -sqrt(3) h.

Setting this to 0 gives h = 0.

Answer: h = 0.

Q7. Find h such that A(-1, h), B(3, 2) and C(7, 3) are collinear.

Collinear points have equal slopes: slope AB = slope BC.

Slope AB = (2 - h)/(3 - (-1)) = (2 - h)/4; slope BC = (3 - 2)/(7 - 3) = 1/4.

Equating: (2 - h)/4 = 1/4, so 2 - h = 1.

Answer: h = 1.

Q8. The points A(-5, -2) and B(5, -4) are ends of a diameter of a circle. Find the centre and radius of the circle.

The centre is the midpoint of the diameter: ((-5 + 5)/2, (-2 - 4)/2) = (0, -3).

AB2 = (10)2 + (-2)2 = 104, so AB = sqrt(104) = 2 sqrt(26); the radius is half the diameter.

Answer: centre (0, -3), radius sqrt(26).

Q9. Find h such that A(h, 1), B(2, 7) and C(-6, -7) are vertices of a right triangle with right angle at the vertex A.

At A, vectors AB = (2 - h, 6) and AC = (-6 - h, -8) must be perpendicular, so their dot product is 0.

Dot product: (2 - h)(-6 - h) + 6(-8) = (h2 + 4h - 12) - 48 = h2 + 4h - 60.

h2 + 4h - 60 = 0 gives (h + 10)(h - 6) = 0.

Answer: h = 6 or h = -10.

Q10. A quadrilateral has vertices A(9, 3), B(-7, 7), C(-3, -7) and D(5, -5). Find the midpoints of its sides. Show that the figure formed by joining the midpoints consecutively is a parallelogram.

Midpoints: AB = (1, 5), BC = (-5, 0), CD = (1, -6), DA = (7, -1).

Joining them consecutively gives a quadrilateral; its diagonals join (1, 5) to (1, -6) and (-5, 0) to (7, -1).

Midpoint of the first diagonal = (1, -1/2); midpoint of the second diagonal = (1, -1/2).

The diagonals of the midpoint figure bisect each other, so it is a parallelogram.

Answer: midpoints (1, 5), (-5, 0), (1, -6), (7, -1); the midpoint figure is a parallelogram.

Q11. Find h such that the quadrilateral with vertices A(-3, 0), B(1, -2), C(5, 0) and D(1, h) is a parallelogram. Is it a square?

For a parallelogram the diagonals bisect each other: midpoint of AC = (1, 0) must equal midpoint of BD = (1, (h - 2)/2).

So (h - 2)/2 = 0, giving h = 2.

With h = 2, each side squared is 20 (AB2 = 16 + 4, and likewise for the others), so it is a rhombus; but diagonal AC = 8 while diagonal BD = 4, so the diagonals are not equal.

Answer: h = 2; it is a rhombus but not a square.

Q12. If two vertices of an equilateral triangle are A(-3, 0) and B(3, 0), find the third vertex. How many of these triangles are possible?

AB = 6, so the third vertex lies on the perpendicular bisector of AB, the line x = 0, at height equal to the altitude of an equilateral triangle of side 6.

Altitude = (sqrt(3)/2) x 6 = 3 sqrt(3).

The third vertex is (0, 3 sqrt(3)) above AB or (0, -3 sqrt(3)) below AB.

Answer: two triangles are possible, with third vertex (0, 3 sqrt(3)) or (0, -3 sqrt(3)).

Q13. Find the points trisecting the join of A(-1, 4) and B(6, 2).

Trisection points divide AB in the ratios 1:2 and 2:1.

First point (ratio 1:2) = ((2(-1) + 6)/3, (2(4) + 2)/3) = (4/3, 10/3).

Second point (ratio 2:1) = ((-1 + 2(6))/3, (4 + 2(2))/3) = (11/3, 8/3).

Answer: (4/3, 10/3) and (11/3, 8/3).

Q14. Find the point three-fifth of the way along the line segment from A(-5, 8) to B(5, 3).

Three-fifth of the way means the point divides AB in the ratio 3:2.

Point = ((2(-5) + 3(5))/5, (2(8) + 3(3))/5) = (5/5, 25/5).

Answer: (1, 5).

Q15. Find the point P on the join of A(1, 4) and B(5, 6) that is twice as far from A as B is from A and lies (i) on the same side of A as B does (ii) on the opposite side of A as B does.

AP = 2 AB, so P = A + 2(B - A) on the same side, and P = A - 2(B - A) on the opposite side; B - A = (4, 2).

(i) P = (1, 4) + 2(4, 2) = (9, 8).

(ii) P = (1, 4) - 2(4, 2) = (-7, 0).

Answer: (i) (9, 8); (ii) (-7, 0).

Q16. Find the point which is equidistant from A(5, 3), B(-2, 2) and C(4, 2). What is the radius of the circumcircle of triangle ABC?

B and C have the same y-coordinate, so the perpendicular bisector of BC is x = (-2 + 4)/2 = 1; let the centre be (1, k).

Equate squared distances to B and A: (1 + 2)2 + (k - 2)2 = (1 - 5)2 + (k - 3)2.

That is 9 + (k - 2)2 = 16 + (k - 3)2, giving 13 - 4k = 25 - 6k, so 2k = 12 and k = 6.

Radius = distance from (1, 6) to B(-2, 2) = sqrt(9 + 16) = 5.

Answer: the equidistant point is (1, 6); circumradius 5.

Q17. The points (4, -2), (-2, 4) and (5, 5) are the vertices of a triangle. Find the in-centre of the triangle.

Side lengths: a = BC = sqrt(49 + 1) = 5 sqrt(2); b = CA = sqrt(1 + 49) = 5 sqrt(2); c = AB = sqrt(36 + 36) = 6 sqrt(2).

In-centre = (aA + bB + cC)/(a + b + c), with a + b + c = 16 sqrt(2).

x-coordinate = (5 sqrt(2)(4) + 5 sqrt(2)(-2) + 6 sqrt(2)(5))/(16 sqrt(2)) = 40/16 = 5/2.

y-coordinate = (5 sqrt(2)(-2) + 5 sqrt(2)(4) + 6 sqrt(2)(5))/(16 sqrt(2)) = 40/16 = 5/2.

Answer: in-centre (5/2, 5/2).

Q18. Find the points that divide the line segment joining A(x1, y1) and B(x2, y2) into four equal parts.

The division points divide AB in the ratios 1:3, 1:1 and 3:1.

Ratio 1:3: ((3x1 + x2)/4, (3y1 + y2)/4).

Ratio 1:1 (midpoint): ((x1 + x2)/2, (y1 + y2)/2).

Ratio 3:1: ((x1 + 3x2)/4, (y1 + 3y2)/4).

Answer: the three points above, in order from A to B.

Exercise 4.2 - Translation and rotation of axes

Q1. The two points P and O' are given in the xy-coordinate system. Find the XY-coordinates of P referred to the translated axes O'X and O'Y: (i) P(3, 2); O'(1, 3) (ii) P(-2, 6); O'(-3, 2) (iii) P(-6, -8); O'(-4, -6) (iv) P(3/2, 5/2); O'(-1/2, 7/2).

For translation through O'(h, k): X = x - h, Y = y - k.

(i) X = 3 - 1 = 2, Y = 2 - 3 = -1, so P = (2, -1).

(ii) X = -2 - (-3) = 1, Y = 6 - 2 = 4, so P = (1, 4).

(iii) X = -6 - (-4) = -2, Y = -8 - (-6) = -2, so P = (-2, -2).

Answer: (iv) X = 3/2 - (-1/2) = 2, Y = 5/2 - 7/2 = -1, so P = (2, -1).

Q2. The xy-coordinate axes are translated through the point whose coordinates are given in the xy-coordinate system. The coordinates of P are given in the XY-coordinate system. Find the coordinates of P in the xy-coordinate system: (i) P(8, 10); O'(3, 4) (ii) P(-5, -3); O'(-2, -6) (iii) P(-3/4, -7/6); O'(1/4, -1/6) (iv) P(4, -3); O'(-2, 3).

For translation through O'(h, k): x = X + h, y = Y + k.

(i) x = 8 + 3 = 11, y = 10 + 4 = 14, so P = (11, 14).

(ii) x = -5 + (-2) = -7, y = -3 + (-6) = -9, so P = (-7, -9).

(iii) x = -3/4 + 1/4 = -1/2, y = -7/6 + (-1/6) = -4/3, so P = (-1/2, -4/3).

Answer: (iv) x = 4 + (-2) = 2, y = -3 + 3 = 0, so P = (2, 0).

Q3. The xy-coordinate axes are rotated about the origin through the indicated angle. The new axes are OX and OY. Find the XY-coordinates of the point P with the given xy-coordinates: (i) P(5, 3); theta = 45 degrees (ii) P(3, -7); theta = 30 degrees (iii) P(11, -15); theta = 60 degrees (iv) P(15, 10); theta = arctan(1/3).

For rotation through theta: X = x cos theta + y sin theta, Y = y cos theta - x sin theta.

(i) theta = 45 degrees: X = (5 + 3)(sqrt(2)/2) = 4 sqrt(2); Y = (3 - 5)(sqrt(2)/2) =

-sqrt(2).

(ii) theta = 30 degrees: X = 3(sqrt(3)/2) - 7(1/2) = (3 sqrt(3) - 7)/2; Y = -7(sqrt(3)/2) - 3(1/2) = (-7 sqrt(3) - 3)/2.

(iii) theta = 60 degrees: X = 11(1/2) - 15(sqrt(3)/2) = (11 - 15 sqrt(3))/2; Y = -15(1/2) - 11(sqrt(3)/2) = (-15 - 11 sqrt(3))/2.

Answer: (iv) theta = arctan(1/3) gives sin theta = 1/sqrt(10), cos theta = 3/sqrt(10): X = 45/sqrt(10) + 10/sqrt(10) = 55/sqrt(10) = 11 sqrt(10)/2; Y = 30/sqrt(10) - 15/sqrt(10) = 15/sqrt(10) = 3 sqrt(10)/2.

Q4. The xy-coordinate axes are rotated about the origin through the indicated angle and the new axes are OX and OY. Find the xy-coordinates of P with the given XY-coordinates: (i) P(-5, 3); theta = 30 degrees (ii) P(-7 sqrt(2), 5 sqrt(2)); theta = 45 degrees.

For the inverse rotation: x = X cos theta - Y sin theta, y = X sin theta + Y cos theta.

(i) theta = 30 degrees: x = -5(sqrt(3)/2) - 3(1/2) = (-5 sqrt(3) - 3)/2; y = -5(1/2) + 3(sqrt(3)/2) = (-5 + 3 sqrt(3))/2.

Answer: (ii) theta = 45 degrees: x = (-7 sqrt(2) - 5 sqrt(2))(sqrt(2)/2) = -12; y = (-7 sqrt(2) + 5 sqrt(2))(sqrt(2)/2) = -2; so P = (-12, -2).

Exercise 4.3 - Equations of straight lines

Q1. Find the slope and inclination of the line joining the points, and sketch each line

in the plane: (i) (-2, 4); (5, 11) (ii) (3, -2); (2, 7) (iii) (4, 6); (4, 8).

Slope m = (y2 - y1)/(x2 - x1); inclination alpha satisfies tan alpha = m, with 0 degrees < alpha < 180 degrees.

(i) m = (11 - 4)/(5 + 2) = 7/7 = 1, so alpha = 45 degrees. Sketch: join (-2, 4) to (5, 11).

(ii) m = (7 + 2)/(2 - 3) = 9/(-1) = -9, so alpha = 180 degrees - arctan 9 = approx. 96.3 degrees. Sketch: join (3, -2) to (2, 7).

Answer: (iii) x2 = x1, so the line is vertical: slope undefined, alpha = 90 degrees. Sketch: the vertical line x = 4 through (4, 6) and (4, 8).

Q2. In the triangle A(8, 6), B(-4, 2), C(-2, -6), find the slope of (i) each side of the triangle (ii) each median of the triangle (iii) each altitude of the triangle.

(i) Sides: AB: (2 - 6)/(-4 - 8) = 1/3; BC: (-6 - 2)/(-2 + 4) = -4; CA: (6 + 6)/(8 + 2) = 6/5.

(ii) Midpoints: BC (-3, -2), CA (3, 0), AB (2, 4). Medians: from A: (-2 - 6)/(-3 - 8) = 8/11; from B: (0 - 2)/(3 + 4) = -2/7; from C: (4 + 6)/(2 + 2) = 5/2.

Answer: (iii) Altitudes are perpendicular to the sides, so their slopes are the negative reciprocals: from A (perpendicular to BC): 1/4; from B (perpendicular to CA): -5/6; from C (perpendicular to AB): -3.

Q3. By means of slopes, show that the following points lie on the same line: (a) (-1, -3); (1, 5); (2, 9) (b) (4, -5); (7, 5); (10, 15) (c) (-4, 6); (3, 8); (10, 10) (d) (a, 2b); (c, a

+ b); (2c - a, 2a).

Three points are collinear when the slope of the first pair equals the slope of the second pair.

(a) Slopes: (5 + 3)/(1 + 1) = 4 and (9 - 5)/(2 - 1) = 4, equal, so collinear.

(b) Slopes: (5 + 5)/(7 - 4) = 10/3 and (15 - 5)/(10 - 7) = 10/3, equal, so collinear.

(c) Slopes: (8 - 6)/(3 + 4) = 2/7 and (10 - 8)/(10 - 3) = 2/7, equal, so collinear.

Answer: (d) Slopes: (a + b - 2b)/(c - a) = (a - b)/(c - a) and (2a - (a + b))/(2c - a - c) = (a - b)/(c - a), equal, so collinear.

Q4. Find k so that the line joining A(7, 3); B(k, -6) and the line joining C(-4, 5); D(-6, 4) are (i) parallel (ii) perpendicular.

Slope of AB = (-6 - 3)/(k - 7) = -9/(k - 7); slope of CD = (4 - 5)/(-6 + 4) = 1/2.

(i) Parallel: slopes equal, so -9/(k - 7) = 1/2, giving k - 7 = -18.

Answer: (i) k = -11. (ii) Perpendicular: product of slopes = -1, so -9/(k - 7) = -2, giving k - 7 = 9/2, so k = 23/2.

Q5. Using slopes, show that the triangle with vertices A(6, 1), B(2, 7) and C(-6, -7) is a right triangle.

Slopes: AB = (7 - 1)/(2 - 6) = -3/2; BC = (-7 - 7)/(-6 - 2) = 7/4; CA = (1 + 7)/(6 + 6) = 2/3.

The product of the slopes of AB and AC is (-3/2)(2/3) = -1, so AB is perpendicular to AC.

Answer: the triangle is right-angled at A, hence a right triangle.

Q6. The three points A(7, -1), B(-2, 2) and C(1, 4) are consecutive vertices of a parallelogram. Find the fourth vertex.

In a parallelogram the diagonals bisect each other, so the fourth vertex D satisfies D = A + C - B.

D = (7 + 1 - (-2), -1 + 4 - 2) = (10, 1).

Answer: the fourth vertex is (10, 1).

Q7. The points A(-1, 2), B(3, -1) and C(6, 3) are consecutive vertices of a rhombus. Find the fourth vertex and show that the diagonals of the rhombus are perpendicular to each other.

Fourth vertex D = A + C - B = (-1 + 6 - 3, 2 + 3 + 1) = (2, 6).

Diagonal AC has direction (7, 1); diagonal BD has direction D - B = (-1, 7).

Their dot product is 7(-1) + 1(7) = 0, so the diagonals are perpendicular.

Answer: fourth vertex (2, 6); diagonals AC and BD are perpendicular.

Q8. Two pairs of points are given. Find whether the two lines determined by these points are (i) parallel (ii) perpendicular (iii) none: (a) (1, -2), (2, 4) and (4, 1), (-8, 2)

(b) (-3, 4), (6, 2) and (4, 5), (-2, -7).

(a) Slopes: (4 + 2)/(2 - 1) = 6 and (2 - 1)/(-8 - 4) = -1/12; product = -1/2, and the slopes are not equal.

Answer: (a) none. (b) Slopes: (2 - 4)/(6 + 3) = -2/9 and (-7 - 5)/(-2 - 4) = 2; product = -4/9, and the slopes are not equal, so (b) none.

Q9. Find an equation of (a) the horizontal line through (7, -9) (b) the vertical line through (-5, 3) (c) the line bisecting the first and third quadrants (d) the line bisecting the second and fourth quadrants.

(a) A horizontal line has equation y = constant: y = -9.

(b) A vertical line has equation x = constant: x = -5.

(c) The bisector of the first and third quadrants is the line y = x.

Answer: (d) The bisector of the second and fourth quadrants is the line y = -x.

Q10. Find an equation of the line (a) through A(-6, 5) having slope 7 (b) through (8, -3) having slope 0 (c) through (-8, 5) having slope undefined (d) through (-5, -3) and (9, -1) (e) y-intercept -7 and slope -5 (f) x-intercept -3 and y-intercept 4 (g) x-intercept -9 and slope -4.

(a) Point-slope: y - 5 = 7(x + 6), i.e. 7x - y + 47 = 0.

(b) Slope 0: y = -3. (c) Slope undefined: x = -8.

(d) Slope = (-1 + 3)/(9 + 5) = 1/7, so y + 3 = (1/7)(x + 5), i.e. x - 7y - 16 = 0.

(e) Slope-intercept: y = -5x - 7. (f) Two-intercept: x/(-3) + y/4 = 1, i.e. 4x - 3y + 12 = 0.

Answer: (g) y = -4(x + 9), i.e. 4x + y + 36 = 0.

Q11. Find an equation of the perpendicular bisector of the segment joining A(3, 5) and B(9, 8).

Midpoint of AB = (6, 13/2); slope of AB = (8 - 5)/(9 - 3) = 1/2, so the perpendicular slope is -2.

Equation: y - 13/2 = -2(x - 6), i.e. y = -2x + 37/2.

Answer: 4x + 2y - 37 = 0.

Q12. Find equations of the sides, altitudes and medians of the triangle whose vertices are A(-3, 2), B(5, 4) and C(3, -8).

Sides: BC: slope 6, y - 4 = 6(x - 5), i.e. 6x - y - 26 = 0; CA: slope -5/3, y + 8 = (-5/3)(x - 3), i.e. 5x + 3y + 9 = 0; AB: slope 1/4, y - 2 = (1/4)(x + 3), i.e. x - 4y + 11 = 0.

Altitudes: from A (perpendicular to BC, slope -1/6): y - 2 = (-1/6)(x + 3), i.e. x + 6y - 9 = 0; from B (perpendicular to CA, slope 3/5): y - 4 = (3/5)(x - 5), i.e. 3x - 5y + 5 = 0; from C (perpendicular to AB, slope -4): y + 8 = -4(x - 3), i.e. 4x + y - 4 = 0.

Midpoints: BC (4, -2), CA (0, -3), AB (1, 3). Medians: from A: y - 2 = (-4/7)(x + 3), i.e. 4x + 7y - 2 = 0; from B: y - 4 = (7/5)(x - 5), i.e. 7x - 5y - 15 = 0.

Answer: median from C: y + 8 = (-11/2)(x - 3), i.e. 11x + 2y - 17 = 0.

Q13. Find an equation of the line through (-4, -6) and perpendicular to a line having slope -3/2.

The perpendicular slope is the negative reciprocal of -3/2, i.e. 2/3.

Equation: y + 6 = (2/3)(x + 4).

Answer: 2x - 3y - 10 = 0.

Q14. Find an equation of the line through (11, -5) and parallel to a line with slope -24.

A parallel line has the same slope, -24.

Equation: y + 5 = -24(x - 11), i.e. y = -24x + 259.

Answer: 24x + y - 259 = 0.

Q15. The points A(-1, 2), B(6, 3) and C(2, -4) are vertices of a triangle. Show that the line joining the midpoint D of AB and the midpoint E of AC is parallel to BC and DE = (1/2) BC.

D = midpoint of AB = (5/2, 5/2); E = midpoint of AC = (1/2, -1).

Slope of DE = (-1 - 5/2)/(1/2 - 5/2) = (-7/2)/(-2) = 7/4; slope of BC = (-4 - 3)/(2 - 6) = 7/4, so DE is parallel to BC.

DE = sqrt[(2)2 + (7/2)2] = sqrt(65)/2; BC = sqrt[(4)2 + (7)2] = sqrt(65).

Answer: DE is parallel to BC and DE = (1/2) BC, as required.

Q16. A milkman can sell 560 litres of milk at Rs. 12.50 per litre and 700 litres of milk at Rs. 12.00 per litre. Assuming the graph of the sale price and the milk sold to be a straight line, find the number of litres of milk that the milkman can sell at Rs. 12.25 per litre.

Take price as x and litres sold as y: the points are (12.50, 560) and (12.00, 700).

Slope = (700 - 560)/(12.00 - 12.50) = 140/(-0.50) = -280 litres per rupee.

At price 12.25, the change from 12.50 is -0.25, so litres = 560 + (-280)(-0.25) = 560 + 70.

Answer: 630 litres.

Q17. The population of Pakistan to the nearest million was 60 million in 1961 and 95 million in 1981. Using t as the number of years after 1961, find an equation of the line that gives the population in terms of t. Use this equation to find the population in (a) 1947 (b) 1997.

Points (t, P): (0, 60) and (20, 95), in millions; slope = (95 - 60)/20 = 7/4.

Equation: P = 60 + (7/4) t million.

(a) 1947 gives t = -14: P = 60 - (7/4)(14) = 60 - 24.5 = 35.5 million.

Answer: (b) 1997 gives t = 36: P = 60 + (7/4)(36) = 60 + 63 = 123 million.

Q18. A house was purchased for Rs. 1 million in 1980. It is worth Rs. 4 million in 1996. Assuming that the value increased by the same amount each year, find an equation that gives the value of the house after t years of the date of purchase. What was its value in 1990?

Points (t, V): (0, 1) and (16, 4), in millions of rupees; slope = (4 - 1)/16 = 3/16.

Equation: V = 1 + (3/16) t million rupees.

In 1990, t = 10: V = 1 + 30/16 = 1 + 1.875 = 2.875 million.

Answer: V = 1 + (3/16) t million rupees; value in 1990 = Rs. 2,875,000.

Q19. Plot the Celsius (C) and Fahrenheit (F) temperature scales on the horizontal axis and the vertical axis respectively. Draw the line joining the freezing point and the boiling point of water. Find an equation giving F temperature in terms of C.

Freezing point: (C, F) = (0, 32); boiling point: (100, 212).

Slope = (212 - 32)/(100 - 0) = 180/100 = 9/5.

With F-intercept 32, the equation is F = 32 + (9/5) C.

Answer: F = (9/5) C + 32.

Q20. The average entry test score of engineering candidates was 592 in the year 1998 while the score was 564 in 2002. Assuming that the relationship between time and score is linear, find the average score for 2006.

Points (year, score): (1998, 592) and (2002, 564); slope = (564 - 592)/4 = -7 per

year.

Let t be years after 1998: score S = 592 - 7t.

For 2006, t = 8: S = 592 - 56.

Answer: average score 536.

Q21. Convert each of the following equations into (i) slope-intercept form (ii) two-intercept form (iii) normal form: (a) 2x - 4y + 11 = 0 (b) 4x + 7y - 2 = 0 (c) 15y - 8x + 13 = 0. Also find the length of the perpendicular from (0, 0) to each line.

(a) Slope-intercept: y = (1/2) x + 11/4. Two-intercept: x/(-11/2) + y/(11/4) = 1. Normal: (-x + 2y)/sqrt(5) = 11/(2 sqrt(5)). Perpendicular from origin: |11|/sqrt(20) = 11 sqrt(5)/10.

(b) Slope-intercept: y = (-4/7) x + 2/7. Two-intercept: x/(1/2) + y/(2/7) = 1. Normal: (4x + 7y)/sqrt(65) = 2/sqrt(65). Perpendicular from origin: 2/sqrt(65).

Answer: (c) Slope-intercept: y = (8/15) x - 13/15. Two-intercept: x/(13/8) + y/(-13/15) = 1. Normal: (8x - 15y)/17 = 13/17. Perpendicular from origin: 13/17.

Q22. In each of the following check whether the two lines are (i) parallel (ii) perpendicular (iii) neither parallel nor perpendicular: (a) 2x + y - 3 = 0; 4x + 2y + 5 = 0 (b) 3y = 2x + 5; 3x + 2y - 8 = 0 (Print note: the operators in the second line of (b) render faintly in this copy; the form 3x + 2y - 8 = 0 is confirmed by the book's text layer.) (c) 4y + 2x - 1 = 0; x - 2y - 7 = 0 (d) 4x - y + 2 = 0; 12x - 3y + 1 = 0 (Print note: the operators in the second line of (d) render faintly in this copy; the

form 12x - 3y + 1 = 0 is confirmed by the book's text layer.) (e) 12x + 35y - 7 = 0; 105x - 36y + 11 = 0.

Slopes: (a) -2 and -2, equal, so parallel. (b) 2/3 and -3/2, product -1, so perpendicular.

(c) -1/2 and 1/2, product -1/4, neither equal nor product -1, so neither.

(d) 4 and 4, equal, so parallel. (e) -12/35 and 105/36 = 35/12, product -1, so perpendicular.

Answer: (a) parallel; (b) perpendicular; (c) neither; (d) parallel; (e) perpendicular.

Q23. Find the distance between the given parallel lines. Sketch the lines. Also find an equation of the parallel line lying midway between them: (a) 3x - 4y + 3 = 0; 3x - 4y + 7 = 0 (b) 12x + 5y - 6 = 0; 12x + 5y + 13 = 0 (c) x + 2y - 5 = 0; 2x + 4y + 1 = 0 (Print note: the operators of the second line of (c) render faintly in this copy; the form 2x + 4y + 1 = 0 is confirmed by the book's text layer.)

(a) Distance = |7 - 3|/sqrt(9 + 16) = 4/5; midway line: 3x - 4y + 5 = 0. Sketch: two parallel lines with the midway line between them.

(b) Distance = |13 - (-6)|/sqrt(144 + 25) = 19/13; midway line: 12x + 5y + 7/2 = 0, i.e. 24x + 10y + 7 = 0.

Answer: (c) Second line divided by 2 is x + 2y + 1/2 = 0; distance = |-5 - 1/2|/sqrt(5) = 11/(2 sqrt(5)) = 11 sqrt(5)/10; midway line: x + 2y - 9/4 = 0, i.e. 4x + 8y - 9 = 0.

Q24. Find an equation of the line through (-4, 7) and parallel to the line 2x - 7y + 4 = 0.

The given line has slope 2/7, so the parallel line has slope 2/7.

Equation: y - 7 = (2/7)(x + 4).

Answer: 2x - 7y + 57 = 0.

Q25. Find an equation of the line through (5, -8) and perpendicular to the join of A(-15, -8), B(10, 7).

Slope of AB = (7 + 8)/(10 + 15) = 15/25 = 3/5, so the perpendicular slope is -5/3.

Equation: y + 8 = (-5/3)(x - 5).

Answer: 5x + 3y - 1 = 0.

Q26. Find equations of two parallel lines perpendicular to 2x - y + 3 = 0 such that the product of the x- and y-intercepts of each is 3.

The given line has slope 2, so the required lines have slope -1/2; write a line as x + 2y + k = 0.

Its intercepts are x = -k and y = -k/2, whose product is k2/2.

Setting k2/2 = 3 gives k2 = 6, so k = sqrt(6) or k = -sqrt(6).

Answer: x + 2y + sqrt(6) = 0 and x + 2y - sqrt(6) = 0.

Q27. One vertex of a parallelogram is (1, 4); the diagonals intersect at (2, 1) and the sides have slopes 1 and -1/7. Find the other three vertices.

The vertex opposite (1, 4) is the reflection in the diagonal centre (2, 1): (3, -2).

Let an adjacent vertex B lie on the slope-1 line through (1, 4): B = (x, x + 3); the other adjacent vertex is its reflection in (2, 1): (4 - x, -x - 1).

This vertex lies on the slope -1/7 line through (1, 4): -x - 1 - 4 = (-1/7)(4 - x - 1), giving 7x + 35 = 3 - x, so x = -4.

Then B = (-4, -1) and the other adjacent vertex is (8, 3).

Answer: the other vertices are (3, -2), (-4, -1) and (8, 3).

Q28. Find whether the given point lies above or below the given line: (a) (5, 8); 2x - 3y + 6 = 0 (b) (-7, 6); 4x + 3y - 9 = 0.

(a) On the line 2x - 3y + 6 = 0, at x = 5, y = (2(5) + 6)/3 = 16/3, which is less than 8, so the point lies above the line.

Answer: (b) On the line 4x + 3y - 9 = 0, at x = -7, y = (9 - 4(-7))/3 = 37/3, which is greater than 6, so the point lies below the line.

Q29. Check whether the given points are on the same or opposite sides of the given line: (a) (0, 0) and (-4, 7); 6x - 7y + 70 = 0 (b) (2, 3) and (-2, 3); 3x - 5y + 8 = 0.

(a) Values of 6x - 7y + 70: at (0, 0) it is 70 (positive); at (-4, 7) it is -24 - 49 + 70 = -3 (negative); opposite signs mean opposite sides.

Answer: (b) Values of 3x - 5y + 8: at (2, 3) it is 6 - 15 + 8 = -1 (negative); at (-2, 3) it is -6 - 15 + 8 = -13 (negative); same sign means the same side.

Q30. Find the distance from the point P(6, -1) to the line 6x - 4y + 9 = 0.

Distance = |6(6) - 4(-1) + 9|/sqrt(36 + 16) = |36 + 4 + 9|/sqrt(52).

That is 49/(2 sqrt(13)) = 49 sqrt(13)/26.

Answer: 49 sqrt(13)/26 units.

Q31. Find the area of the triangular region whose vertices are A(5, 3), B(-2, 2), C(4, 2).

BC is horizontal (y = 2) with length 4 - (-2) = 6; the height from A to BC is 3 - 2 = 1.

Area = (1/2) x base x height = (1/2)(6)(1).

Answer: 3 square units.

Q32. The coordinates of three points are A(2, 3), B(-1, 1) and C(4, -5). By computing the area bounded by ABC check whether the points are collinear.

Area = (1/2)|2(1 + 5) + (-1)(-5 - 3) + 4(3 - 1)| = (1/2)|12 + 8 + 8| = 14.

The area is not zero, so the points do not lie on one line.

Answer: the points are not collinear.

Exercise 4.4 - Angle between lines, concurrency and centres of a triangle

Q1. Find the point of intersection of the lines (Print note: the operators of the second line of each pair render blank in this copy; the forms below are recovered from the book's text layer and cross-checked against the legible first lines.): (i) x - 2y + 1 = 0 and 2x - y + 2 = 0 (ii) 3x + y + 12 = 0 and x + 2y - 1 = 0 (iii) x + 4y - 12 = 0 and x - 3y + 3 = 0.

(i) From the first line x = 2y - 1; putting this in the second: 2(2y - 1) - y + 2 = 0, so 3y = 0 and y = 0, x = -1.

(ii) From the second line x = 1 - 2y; putting this in the first: 3(1 - 2y) + y + 12 = 0, so -5y + 15 = 0, y = 3, x = -5.

Answer: (i) (-1, 0); (ii) (-5, 3); (iii) subtracting the two lines gives 7y - 15 = 0, so y = 15/7 and x = 3y - 3 = 24/7, i.e. (24/7, 15/7).

Q2. Find an equation of the line through (Print note: the operators of the second lines render blank in this copy; the forms below are recovered from the book's text layer.): (i) the point (2, -9) and the intersection of the lines 2x + 5y - 8 = 0 and 3x - 4y - 6 = 0 (ii) the intersection of the lines x - y - 4 = 0 and 7x + y + 20 = 0 and (a) parallel (b) perpendicular to the line 6x + y - 14 = 0 (iii) the intersection of the lines x + 2y + 3 = 0, 3x + 4y + 7 = 0 and making equal intercepts on the axes.

(i) Intersection: 6x + 15y = 24 and 6x - 8y = 12 give 23y = 12, so the point is (62/23, 12/23).

Slope from (2, -9) to (62/23, 12/23) is (12/23 + 9)/(62/23 - 2) = 219/16, so the line is y + 9 = (219/16)(x - 2), i.e. 219x - 16y - 582 = 0.

(ii) Intersection: adding the lines gives 8x + 16 = 0, so the point is (-2, -6). (a)

Parallel to 6x + y - 14 = 0 (slope -6): y + 6 = -6(x + 2), i.e. 6x + y + 18 = 0. (b)

Perpendicular (slope 1/6): y + 6 = (1/6)(x + 2), i.e. x - 6y - 34 = 0.

Answer: (iii) Intersection: doubling the first line and subtracting from the second gives x = -1, y = -1. A line with equal intercepts a has equation x + y = a; through (-1, -1) this gives a = -2, i.e. x + y + 2 = 0.

Q3. Find an equation of the line through the intersection of 16x - 10y - 33 = 0; 12x - 14y - 29 = 0 and the intersection of x - y + 4 = 0; x - 7y + 2 = 0.

First intersection: 7 times the first line minus 5 times the second gives 52x = 86, so x = 43/26; then 10y = 16x - 33 = -170/26, so y = -17/26.

Second intersection: subtracting the lines gives 6y + 2 = 0, so y = -1/3 and x = y - 4 = -13/3.

Slope between the points: (-1/3 + 17/26)/(-13/3 - 43/26) = (25/78)/(-467/78) = -25/467.

Equation through (-13/3, -1/3): y + 1/3 = (-25/467)(x + 13/3).

Answer: 25x + 467y + 264 = 0.

Q4. Find the condition that the lines y = m1 x + c1; y = m2 x + c2 and y = m3 x + c3 are concurrent.

Write the lines as m1 x - y + c1 = 0, m2 x - y + c2 = 0, m3 x - y + c3 = 0.

Three lines are concurrent exactly when the determinant of their coefficients is zero.

Expanding the determinant gives m1(c2 - c3) + m2(c3 - c1) + m3(c1 - c2) = 0.

Answer: the required condition is m1(c2 - c3) + m2(c3 - c1) + m3(c1 - c2) = 0.

Q5. FLAGGED - Determine the value of p such that the lines 2x - 3y - 1 = 0, 3x - y - 5 = 0 and 3x + 4y + 8 = 0 meet at a point. (Print note: as printed in this copy, no line contains p, and the three lines are not concurrent, so no value of p can be determined from the printed question; no value is invented here.)

Intersection of the first two lines: from 3x - y - 5 = 0, y = 3x - 5; putting this in 2x - 3y - 1 = 0 gives 2x - 9x + 15 - 1 = 0, so x = 2 and y = 1.

Check the third line at (2, 1): 3(2) + 4(1) + 8 = 18, not 0, so the third line does not pass through the intersection of the first two.

Answer: FLAGGED - the question as printed has no determinable value of p; the three printed lines do not meet at a point.

Q6. Show that the lines 4x - 3y - 8 = 0, 3x - 4y - 6 = 0 and x - y - 2 = 0 are concurrent and the third line bisects the angle formed by the first two lines.

Intersection of the first two: 4 times the first minus 3 times the second gives 7x = 14, so x = 2 and y = 0; the third line gives 2 - 0 - 2 = 0, so all three pass through (2, 0) and are concurrent.

Angle bisectors of the first two lines: (4x - 3y - 8)/5 + (3x - 4y - 6)/5 = 0 gives (7x - 7y - 14)/5 = 0, i.e. x - y - 2 = 0.

Answer: the third line is one of the angle bisectors of the first two lines, as required.

Q7. The vertices of a triangle are A(-2, 3), B(-4, 1) and C(3, 5). Find coordinates of the (i) centroid (ii) orthocentre (iii) circumcentre of the triangle. Are these three points collinear?

(i) Centroid = ((-2 - 4 + 3)/3, (3 + 1 + 5)/3) = (-1, 3).

(ii) Altitude from A (perpendicular to BC, slope 4/7): 7x + 4y + 2 = 0. Altitude from B (perpendicular to AC, slope 2/5): 5x + 2y + 18 = 0. Solving gives x = -34/3, y = 58/3; this also satisfies the third altitude x + y - 8 = 0.

(iii) Perpendicular bisector of AB: x + y + 1 = 0. Perpendicular bisector of AC: 5x + 2y - 21/2 = 0. Solving gives x = 25/6, y = -31/6; this point is equidistant from A, B and C.

Answer: centroid (-1, 3), orthocentre (-34/3, 58/3), circumcentre (25/6, -31/6). Slope from orthocentre to centroid is -49/31, and slope from centroid to circumcentre is also -49/31, so the three points are collinear (the Euler line).

Q8. Check whether the lines 4x - 3y - 8 = 0; 3x - 4y - 6 = 0; x - y - 2 = 0 are concurrent. If so, find the point where they meet.

Intersection of the first two lines: 4 times the first minus 3 times the second gives 7x = 14, so x = 2; then 8 - 3y - 8 = 0 gives y = 0.

Check the third line at (2, 0): 2 - 0 - 2 = 0, satisfied.

Answer: yes, concurrent at (2, 0).

Q9. Find the coordinates of the vertices of the triangle formed by the lines x - 2y - 6 = 0; 3x - y + 3 = 0; 2x + y - 4 = 0. Also find measures of the angles of the triangle.

Vertices: first and second lines meet where x - 2y = 6 and 3x - y = -3, giving (-12/5, -21/5). Second and third meet where 3x - y = -3 and 2x + y = 4, giving (1/5, 18/5). First and third meet where x - 2y = 6 and 2x + y = 4, giving (14/5, -8/5).

Slopes: first line 1/2, second line 3, third line -2. Angle between first and second: tan = (3 - 1/2)/(1 + 3/2) = 1, so 45 degrees. Angle between second and third: tan = (-2 - 3)/(1 - 6) = 1, so 45 degrees.

Answer: vertices (-12/5, -21/5), (1/5, 18/5), (14/5, -8/5); angles 45 degrees, 45 degrees and 90 degrees (the third angle, between slopes -2 and 1/2, has denominator 1 + (-2)(1/2) = 0).

Q10. Find the angle measured from the line l1 to the line l2 where (a) l1 joining (2, 7) and (7, 10); l2 joining (1, 1) and (-5, 3) (b) l1 joining (3, -1) and (5, 7); l2 joining (2, 4) and (-8, 2) (c) l1 joining (1, -7) and (6, -4); l2 joining (-1, 2) and (-6, -1) (d) l1 joining (-9, -1) and (3, -5); l2 joining (2, 7) and (-6, -7). Also find the acute angle in each case.

Use tan theta = (m2 - m1)/(1 + m1 m2). (a) m1 = 3/5, m2 = -1/3: tan theta = -7/6, so theta = 180 degrees - 49.4 degrees = 130.6 degrees; acute angle 49.4 degrees.

(b) m1 = 4, m2 = 1/5: tan theta = -19/9, so theta = 180 degrees - 64.7 degrees = 115.3 degrees; acute angle 64.7 degrees.

(c) m1 = 3/5, m2 = 3/5: the slopes are equal, so the lines are parallel; angle 0 degrees, acute angle 0 degrees.

Answer: (d) m1 = -1/3, m2 = 7/4: tan theta = 5, so theta = 78.7 degrees, already acute.

Q11. Find the interior angles of the triangle whose vertices are (a) A(-2, 11), B(-6, -3), C(4, -9) (b) A(6, 1), B(2, 7), C(-6, -7) (c) A(2, -5), B(-4, -3), C(-1, 5) (d) A(2, 8), B(-5, 4), C(4, -9).

Use cos of the angle between the two side vectors at each vertex. (a) At A: cos = 64/sqrt(5777), angle approx. 32.6 degrees; at B: cos = -44/sqrt(28832), angle 105.0 degrees; at C: angle approx. 42.3 degrees.

(b) At A the side slopes are -3/2 and 2/3, whose product is -1, so angle A = 90 degrees; at B: cos approx. 0.447, angle 63.4 degrees; at C: angle 26.6 degrees.

(c) Angles: at A approx. 54.9 degrees; at B approx. 87.9 degrees; at C approx. 37.3 degrees.

Answer: (d) Angles: at A approx. 67.0 degrees; at B approx. 85.0 degrees; at C approx. 28.0 degrees. Each set sums to 180 degrees.

Q12. Find the interior angles of the quadrilateral whose vertices are A(5, 2), B(-2, 3), C(-3, -4) and D(4, -5).

At A, vectors AB = (-7, 1) and AD = (-1, -7) have dot product 7 - 7 = 0, so angle A = 90 degrees.

At B, vectors BA = (7, -1) and BC = (-1, -7) have dot product -7 + 7 = 0, so angle B = 90 degrees.

At C, vectors CB = (1, 7) and CD = (7, -1) have dot product 7 - 7 = 0, so angle C = 90 degrees; similarly angle D = 90 degrees.

Answer: all interior angles are 90 degrees; in fact all sides equal sqrt(50), so the quadrilateral is a square.

Q13. Show that the points A(0, 0), B(2, 1), C(3, 3), D(1, 2) are the vertices of a rhombus. Find its interior angles.

Side lengths: AB = sqrt(5), BC = sqrt(5), CD = sqrt(5), DA = sqrt(5); all sides equal, so ABCD is a rhombus.

At A, vectors AB = (2, 1) and AD = (1, 2) give cos A = 4/5, so angle A = approx. 36.9 degrees.

At B, vectors BA = (-2, -1) and BC = (1, 2) give cos B = -4/5, so angle B = approx. 143.1 degrees.

Answer: interior angles 36.9 degrees, 143.1 degrees, 36.9 degrees, 143.1 degrees.

Q14. Find the area of the region bounded by the triangle whose sides are 7x - y - 10 = 0; 10x + y - 14 = 0; 3x + 2y + 3 = 0.

Vertices: first and second lines meet at (24/17, 2/17); second and third at (31/17, -72/17); first and third at (1, -3).

Area = (1/2)|24/17(-72/17 + 3) + 31/17(-3 - 2/17) + 1(2/17 + 72/17)| = (1/2)(889/289).

Answer: 889/578 square units (approx. 1.54).

Q15. The vertices of a triangle are A(-2, 3), B(-4, 1) and C(3, 5). Find the centre of the circumcircle of the triangle.

This is the triangle of Q7. Its perpendicular bisector of AB is x + y + 1 = 0 and of AC is 5x + 2y - 21/2 = 0.

Solving gives x = 25/6 and y = -31/6.

Answer: circumcentre (25/6, -31/6).

Q16. Express the given system of equations in matrix form. Find in each case whether the lines are concurrent: (a) x + 3y - 2 = 0; 2x - y + 4 = 0; x - 11y + 14 = 0 (b) 2x + 3y + 4 = 0; x - 2y - 3 = 0; 3x + y - 8 = 0 (c) 3x - 4y - 2 = 0; x + 2y - 4 = 0; 3x - 2y + 5 = 0.

Concurrency holds exactly when the determinant of the coefficient matrix (columns x, y, constant) is zero.

(a) Matrix [[1, 3, -2], [2, -1, 4], [1, -11, 14]] times [x, y, 1] = 0; determinant = 30 - 72 + 42 = 0, so concurrent.

(b) Matrix [[2, 3, 4], [1, -2, -3], [3, 1, -8]]; determinant = 38 - 3 + 28 = 63, not zero, so not concurrent.

Answer: (c) Matrix [[3, -4, -2], [1, 2, -4], [3, -2, 5]]; determinant = 6 + 68 + 16 = 90, not zero, so not concurrent.

Q17. Find a system of linear equations corresponding to the given matrix form.

Check whether the lines represented by the system are concurrent: (a) [[1, 0, -1], [2, 0, 1], [0, -1, 6]] [x, y, 1] = 0 (b) [[1, 1, 2], [2, 4, -3], [3, 6, -5]] [x, y, 1] = 0.

(a) Equations: x - 1 = 0; 2x + 1 = 0; -y + 6 = 0. The first two give x = 1 and x = -1/2, which cannot both hold, so the lines are not concurrent (determinant -1, not zero).

Answer: (b) Equations: x + y + 2 = 0; 2x + 4y - 3 = 0; 3x + 6y - 5 = 0. The last two are parallel (2x + 4y = 3 and 3x + 6y = 5), so the lines are not concurrent (determinant -1, not zero).

Exercise 4.5 - Homogeneous equation of the second degree (pair of lines through the origin)

Q1. Find the lines represented by 10x2 - 23xy - 5y2 = 0 and also find the measure of the angle between them. (Problems 1-6.)

Factor: 10x2 - 23xy - 5y2 = (5x + y)(2x - 5y) = 0.

Lines: 5x + y = 0 (slope -5) and 2x - 5y = 0 (slope 2/5).

Angle: with a = 10, h = -23/2, b = -5: tan theta = 2 sqrt(h2 - ab)/(a + b) = 27/5, so the acute angle is arctan(27/5) = approx. 79.5 degrees.

Answer: lines y = -5x and y = (2/5)x; acute angle approx. 79.5 degrees.

Q2. Find the lines represented by 3x2 + 7xy + 2y2 = 0 and also find the measure of the angle between them.

Factor: 3x2 + 7xy + 2y2 = (3x + y)(x + 2y) = 0.

Lines: 3x + y = 0 and x + 2y = 0.

Angle: with a = 3, h = 7/2, b = 2: tan theta = 2 sqrt(h2 - ab)/(a + b) = 5/5 = 1.

Answer: lines y = -3x and y = -x/2; angle 45 degrees.

Q3. Find the lines represented by 9x2 + 24xy + 16y2 = 0 and also find the measure of the angle between them.

Factor: 9x2 + 24xy + 16y2 = (3x + 4y)2 = 0.

The two lines coincide: 3x + 4y = 0 (a double line).

Answer: coincident lines 3x + 4y = 0; angle between them 0 degrees.

Q4. Find the lines represented by 2x2 + 3xy - 5y2 = 0 and also find the measure of the angle between them.

Factor: 2x2 + 3xy - 5y2 = (2x + 5y)(x - y) = 0.

Lines: 2x + 5y = 0 (slope -2/5) and x - y = 0 (slope 1).

Angle: with a = 2, h = 3/2, b = -5: tan theta = 2 sqrt(h2 - ab)/(a + b) = 7/(-3), so the acute angle is arctan(7/3) = approx. 66.8 degrees.

Answer: lines y = -(2/5)x and y = x; acute angle approx. 66.8 degrees.

Q5. Find the lines represented by 6x2 - 19xy + 15y2 = 0 and also find the measure of the angle between them.

Factor: 6x2 - 19xy + 15y2 = (3x - 5y)(2x - 3y) = 0.

Lines: 3x - 5y = 0 (slope 3/5) and 2x - 3y = 0 (slope 2/3).

Angle: with a = 6, h = -19/2, b = 15: tan theta = 2 sqrt(h2 - ab)/(a + b) = 1/21.

Answer: lines y = (3/5)x and y = (2/3)x; angle approx. 2.7 degrees.

Q6. Find the lines represented by x2 + 2xy sec(alpha) + y2 = 0 and also find the measure of the angle between them.

Here a = 1, h = sec(alpha), b = 1. Dividing by y2 and solving for x/y gives x/y = -sec(alpha) + tan(alpha) or x/y = -sec(alpha) - tan(alpha).

Lines: x = (-sec(alpha) + tan(alpha)) y and x = (-sec(alpha) - tan(alpha)) y.

Angle: tan theta = 2 sqrt(h2 - ab)/(a + b) = 2 tan(alpha)/2 = tan(alpha).

Answer: angle theta = alpha (for alpha acute).

Q7. Find a joint equation of the lines through the origin and perpendicular to the lines: x2 - 2xy tan(alpha) - y2 = 0.

For the given pair, the slopes satisfy -m2 - 2 tan(alpha) m + 1 = 0, so the product of the slopes is -1: the two given lines are already perpendicular to each other.

Hence the lines through the origin perpendicular to them are the same pair of lines.

Answer: the joint equation is unchanged: x2 - 2xy tan(alpha) - y2 = 0.

Q8. Find a joint equation of the lines through the origin and perpendicular to the lines: ax2 + 2hxy + by2 = 0.

Let the given lines be y = m1 x and y = m2 x, with m1 + m2 = -2h/b and m1 m2 = a/b.

The perpendiculars through the origin are x + m1 y = 0 and x + m2 y = 0.

Their joint equation is (x + m1 y)(x + m2 y) = x2 + (m1 + m2) xy + m1 m2 y2 = x2 - (2h/b) xy + (a/b) y2 = 0.

Answer: bx2 - 2hxy + ay2 = 0.

Q9. Find the area of the region bounded by: 10x2 - xy - 21y2 = 0 and x + y + 1 = 0.

Factor: 10x2 - xy - 21y2 = (5x + 7y)(2x - 3y) = 0, two lines through the origin.

Intersection with x + y + 1 = 0: with 5x + 7y = 0, the point is (-7/2, 5/2); with 2x - 3y = 0, the point is (-3/5, -2/5).

The region is the triangle with vertices (0, 0), (-7/2, 5/2), (-3/5, -2/5).

Area = (1/2)|(-7/2)(-2/5) - (-3/5)(5/2)| = (1/2)(7/5 + 3/2) = 29/20.

Answer: 29/20 square units.

Chapter 5: Linear Inequalities and Linear Programming

Exercise 5.1 - Graphs of linear inequalities

Q1. Graph the solution set of each of the following linear inequalities in the xy-plane: (i) 2x + y <= 6 (ii) 3x + 7y >= 21 (iii) 3x - 2y >= 6 (iv) 5x - 4y <= 20 (v) 2x + 1 >= 0 (vi) 3y - 4 <= 0.

Draw the boundary line and test the origin (0, 0); shade the side that satisfies the inequality.

(i) Line 2x + y = 6 through (3, 0) and (0, 6); origin satisfies 0 <= 6, so shade the origin side (below the line).

(ii) Line 3x + 7y = 21 through (7, 0) and (0, 3); origin fails 0 >= 21, so shade the side away from the origin (above the line).

(iii) Line 3x - 2y = 6 through (2, 0) and (0, -3); origin fails 0 >= 6, so shade the side away from the origin.

(iv) Line 5x - 4y = 20 through (4, 0) and (0, -5); origin satisfies 0 <= 20, so shade the origin side.

Answer: (v) 2x + 1 >= 0 gives x >= -1/2: shade the vertical half-plane to the right of the line x = -1/2. (vi) 3y - 4 <= 0 gives y <= 4/3: shade the horizontal half-plane below the line y = 4/3.

Q2. Indicate the solution set of the following systems of linear inequalities by shading: (i) 2x - 3y <= 6; 2x + 3y <= 12 (ii) x + y >= 5; x - y <= 1 (iii) 3x + 7y >=

21; x - y <= 2 (iv) 4x - 3y <= 12 (v) 3x + 7y >= 21; x >= -3/2; y <= 4.

Shade the intersection of the half-planes. (i) The two lines meet at (9/2, 1); both inequalities hold at the origin, so the region is the wedge on the origin side of both lines, with vertex (9/2, 1), unbounded to the left.

(ii) The lines x + y = 5 and x - y = 1 meet at (3, 2); the region is above both lines (y >= 5 - x and y >= x - 1), with vertex (3, 2), unbounded upward.

(iii) The lines meet at (7/2, 3/2); the region is above 3x + 7y = 21 and above x - y = 2 in the sense y >= x - 2, with vertex (7/2, 3/2), unbounded upward.

(iv) A single half-plane: the line 4x - 3y = 12 through (3, 0) and (0, -4); the origin satisfies 0 <= 12, so shade the origin side.

Answer: (v) The region is bounded on the left by x = -3/2 and above by y = 4, and lies above 3x + 7y = 21; its vertices are (-3/2, 51/14) and (-3/2, 4), and it extends unbounded to the right.

Q3. Indicate the solution region of the following systems of linear inequalities by shading: (i) 2x - 3y <= 6; 2x + 3y <= 12; y >= 0 (ii) x + y <= 5; y - 2x <= 2; x >= 0 (iii) x + y >= 5; x - y >= 1; y >= 0 (iv) 3x + 7y <= 21; x - y <= 2; x >= 0 (v) 3x + 7y <= 21; x - y <= 2; y >= 0 (vi) 3x + 7y <= 21; 2x - y >= -3; x >= 0.

(i) Between the lines, above y = 0: vertices (3, 0) and (9/2, 1); the region extends unbounded to the left along y = 0.

(ii) Below both lines, x >= 0: vertices (0, 2) and (1, 4); the region extends unbounded downward.

(iii) Between y = 5 - x (below) and y = x - 1 (above), y >= 0: vertices (3, 2) and (5, 0); the region extends unbounded to the right.

(iv) Triangle with vertices (0, -2), (0, 3) and (7/2, 3/2), bounded.

(v) Vertices (2, 0) and (7/2, 3/2); the region extends unbounded to the left along y = 0.

Answer: (vi) The three lines are concurrent at (0, 3); the region is x >= 0, below both lines, with the single corner (0, 3), extending unbounded downward.

Q4. Graph the solution region of the following system of linear inequalities and find the corner points in each case: (i) 2x - 3y <= 6; 2x + 3y <= 12; x >= 0 (ii) x + y <= 5; -2x + y <= 2; y >= 0 (iii) 3x + 7y <= 21; 2x - y <= -3; y >= 0 (iv) 3x + 2y >= 6; x + 3y <= 6; y >= 0 (v) 5x + 7y <= 35; -x + 3y <= 3; x >= 0 (vi) 5x + 7y <= 35; x - 2y <= 2; x >= 0.

(i) Triangle with corner points (0, -2), (0, 4) and (9/2, 1).

(ii) Triangle with corner points (-1, 0), (5, 0) and (1, 4).

(iii) Corner points (0, 3) and (-3/2, 0); the region is unbounded (the wedge between y = 2x + 3 and 3x + 7y = 21, y >= 0, opening to the left).

(iv) Triangle with corner points (2, 0), (6, 0) and (6/7, 12/7).

(v) Corner points (0, 1) and (42/11, 25/11); the region is unbounded downward (no y >= 0 constraint).

Answer: (vi) Triangle with corner points (0, -1), (0, 5) and (84/17, 25/17).

Q5. Graph the solution region of the following system of linear inequalities by shading: (i) 3x - 4y <= 12; 3x + 2y >= 3; x + 2y <= 9 (ii) 3x - 4y <= 12; x + 2y <= 6; x + y >= 1 (iii) 2x + y <= 4; 2x - 3y >= 12; x + 2y <= 6 (iv) 2x + y <= 10; x + y <= 7; -2x + y <= 4 (v) 2x + 3y <= 18; 2x + y <= 10; -2x + y <= 2 (vi) 3x - 2y >= 3; x + 4y <= 12; 3x + y <= 12.

(i) Triangle with vertices (-3, 6), (2, -3/2) and (6, 3/2).

(ii) Triangle with vertices (-4, 5), (16/7, -9/7) and (24/5, 3/5).

(iii) The lines 2x + y = 4 and 2x - 3y = 12 meet at the single corner (3, -2); the region lies below all three lines and extends unbounded downward.

(iv) Corner points (1, 6) and (3, 4); the region lies below all three lines and extends unbounded downward.

(v) Corner points (3/2, 5) and (3, 4); the region lies below all three lines and extends unbounded downward.

Answer: (vi) Corner points (18/7, 33/14) and (36/11, 24/11); the region lies below x + 4y = 12 and 3x + y = 12 and above 3x - 2y = 3, extending unbounded downward.

Exercise 5.2 - Feasible regions and corner points

Q1. Graph the feasible region of the following system of linear inequalities and find

the corner points in each case: (i) 2x - 3y <= 6; 2x + 3y <= 12; x >= 0, y >= 0 (ii) x + y <= 5; -2x + y <= 2; x >= 0, y >= 0 (iii) x + y <= 5; -2x + y >= 2; x >= 0 (iv) 3x + 7y <= 21; x - y <= 3; x >= 0, y >= 0 (v) 3x + 2y > 6; x + y <= 4; x >= 0, y >= 0 (vi) 5x + 7y <= 35; x - 2y <= 4; x >= 0, y >= 0.

(i) Corner points (0, 0), (3, 0), (9/2, 1), (0, 4).

(ii) Corner points (0, 0), (5, 0), (1, 4), (0, 2).

(iii) Triangle with corner points (0, 2), (1, 4), (0, 5).

(iv) Corner points (0, 0), (3, 0), (21/5, 6/5), (0, 3).

(v) Corner points (4, 0), (0, 4), (0, 3), (2, 0); the boundary 3x + 2y = 6 is open (strict inequality) and is not included.

Answer: (vi) Corner points (0, 0), (4, 0), (98/17, 15/17), (0, 5).

Q2. Graph the feasible region of the following system of linear inequalities and find the corner points in each case: (i) 2x + y <= 10; x + 4y <= 12; x + 2y <= 10; x >= 0, y >= 0 (ii) 2x + 3y <= 18; 2x + y <= 10; x + 4y <= 12; x >= 0, y >= 0 (iii) 2x + 3y <= 18; x + 4y <= 12; 3x + y <= 12; x >= 0, y >= 0 (iv) x + 2y <= 14; 3x + 4y <= 36; 2x + y <= 10; x >= 0, y >= 0 (v) x + 3y <= 15; 2x + y <= 12; 4x + 3y <= 24; x >= 0, y >= 0 (vi) 2x + y <= 20; 8x + 15y <= 120; x + y <= 11; x >= 0, y >= 0.

(i) Corner points (0, 0), (5, 0), (4, 2), (0, 3).

(ii) Corner points (0, 0), (5, 0), (4, 2), (0, 3).

(iii) Corner points (0, 0), (4, 0), (36/11, 24/11), (0, 3).

(iv) Corner points (0, 0), (5, 0), (2, 6), (0, 7).

(v) Corner points (0, 0), (6, 0), (3, 4), (0, 5).

Answer: (vi) Corner points (0, 0), (10, 0), (9, 2), (45/7, 32/7), (0, 8).

Exercise 5.3 - Linear programming problems

Q1. Maximize f(x, y) = 2x + 5y subject to the constraints 2y - x <= 8; x - y <= 4; x >= 0; y >= 0.

Feasible corner points: (0, 0), (0, 4), (16, 12), (4, 0).

Values of f: f(0, 0) = 0; f(0, 4) = 20; f(16, 12) = 32 + 60 = 92; f(4, 0) = 8.

Answer: maximum value 92 at (16, 12).

Q2. Maximize f(x, y) = x + 3y subject to the constraints 2x + 5y <= 30; 5x + 4y <= 20; x >= 0; y >= 0.

The two lines meet outside the first quadrant (at negative x), so in the first quadrant the second constraint dominates: the feasible region is the triangle (0, 0), (4, 0), (0, 5).

Values of f: f(0, 0) = 0; f(4, 0) = 4; f(0, 5) = 15.

Answer: maximum value 15 at (0, 5).

Q3. Maximize z = 2x + 3y subject to the constraints 3x + 4y <= 12; 2x + y <= 4; 4x - y <= 4; x >= 0; y >= 0.

Feasible corner points: (0, 0), (0, 3), (4/5, 12/5), (4/3, 4/3), (1, 0).

Values of z: z(0, 3) = 9; z(4/5, 12/5) = 44/5 = 8.8; z(4/3, 4/3) = 20/3; z(1, 0) = 2.

Answer: maximum value 9 at (0, 3).

Q4. Minimize z = 2x + y subject to the constraints x + y >= 3; 7x + 5y <= 35; x >= 0; y >= 0.

Feasible corner points: (0, 3), (0, 7), (5, 0), (3, 0).

Values of z: z(0, 3) = 3; z(0, 7) = 7; z(5, 0) = 10; z(3, 0) = 6.

Answer: minimum value 3 at (0, 3).

Q5. Maximize the function defined as f(x, y) = 2x + 3y subject to the constraints 2x + y <= 8; x + 2y <= 14; x >= 0; y >= 0.

Feasible corner points: (0, 0), (4, 0), (2/3, 20/3), (0, 7).

Values of f: f(4, 0) = 8; f(2/3, 20/3) = 4/3 + 20 = 64/3; f(0, 7) = 21.

Answer: maximum value 64/3 at (2/3, 20/3).

Q6. Minimize z = 3x + y subject to the constraints 3x + 5y >= 15; x + 6y >= 9; x >= 0; y >= 0.

Feasible corner points: (0, 3), (45/13, 12/13), (9, 0); the region is unbounded, but z

has positive coefficients so its minimum occurs at a corner.

Values of z: z(0, 3) = 3; z(45/13, 12/13) = 147/13; z(9, 0) = 27.

Answer: minimum value 3 at (0, 3).

Q7. Each unit of food X costs Rs. 25 and contains 2 units of protein and 4 units of iron while each unit of food Y costs Rs. 30 and contains 3 units of protein and 2 units of iron. Each animal must receive at least 12 units of protein and 16 units of iron each day. How many units of each food should be fed to each animal at the smallest possible cost?

Let x units of X and y units of Y be fed. Minimize cost C = 25x + 30y subject to 2x + 3y >= 12 (protein), 4x + 2y >= 16 (iron), x >= 0, y >= 0.

Feasible corner points: (6, 0) (satisfies both: protein 12, iron 24), (3, 2) (intersection: protein 12, iron 16), (0, 8) (protein 24, iron 16).

Costs: C(6, 0) = 150; C(3, 2) = 75 + 60 = 135; C(0, 8) = 240.

Answer: smallest cost Rs. 135 with 3 units of food X and 2 units of food Y per animal per day.

Q8. A dealer wishes to purchase a number of fans and sewing machines. He has only Rs. 5760 to invest and has space at most for 20 items. A fan costs him Rs. 360 and a sewing machine costs Rs. 240. His expectation is that he can sell a fan at a profit of Rs. 22 and a sewing machine at a profit of Rs. 18. Assuming that he can sell all the items that he can buy, how should he invest his money in order to maximize

his profit?

Let x fans and y sewing machines be bought. Maximize profit P = 22x + 18y subject to 360x + 240y <= 5760 (money), x + y <= 20 (space), x >= 0, y >= 0.

Feasible corner points: (0, 0), (16, 0), (8, 12) (intersection of 3x + 2y = 48 and x + y = 20), (0, 20).

Profits: P(16, 0) = 352; P(8, 12) = 176 + 216 = 392; P(0, 20) = 360.

Answer: maximum profit Rs. 392 by buying 8 fans and 12 sewing machines.

Q9. A machine can produce product A by using 2 units of chemical and 1 unit of a compound or can produce product B by using 1 unit of chemical and 2 units of the compound. Only 800 units of chemical and 1000 units of the compound are available. The profits per unit of A and B are Rs. 30 and Rs. 20 respectively, maximize the profit function.

Let x units of A and y units of B be produced. Maximize profit P = 30x + 20y subject to 2x + y <= 800 (chemical), x + 2y <= 1000 (compound), x >= 0, y >= 0.

Feasible corner points: (0, 0), (400, 0), (200, 400) (intersection), (0, 500).

Profits: P(400, 0) = 12000; P(200, 400) = 6000 + 8000 = 14000; P(0, 500) = 10000.

Answer: maximum profit Rs. 14000 by producing 200 units of A and 400 units of B.

Exercise 6.1 - The circle

Q1. In each of the following, find an equation of the circle with (a) centre at (5, -2) and radius 4 (b) centre at (sqrt(2), -3 sqrt(3)) and radius 2 sqrt(2) (c) ends of a diameter at (-3, 2) and (5, -6).

(a) (x - 5)2 + (y + 2)2 = 16, i.e. x2 + y2 - 10x + 4y + 13 = 0.

(b) (x - sqrt(2))2 + (y + 3 sqrt(3))2 = 8, i.e. x2 + y2 - 2 sqrt(2) x + 6 sqrt(3) y + 21 = 0.

Answer: (c) Centre is the midpoint (1, -2); radius is half the distance between the ends, sqrt(128)/2 = 4 sqrt(2). Equation: (x - 1)2 + (y + 2)2 = 32, i.e. x2 + y2 - 2x + 4y - 27 = 0.

Q2. Find the centre and radius of the circle with the given equation (a) x2 + y2 + 12x - 10y = 0 (b) 5x2 + 5y2 + 14x + 12y - 10 = 0 (c) x2 + y2 - 6x + 4y + 13 = 0 (d) 4x2 + 4y2 - 8x + 12y - 25 = 0.

Complete the squares. (a) (x + 6)2 + (y - 5)2 = 61: centre (-6, 5), radius sqrt(61).

(b) Dividing by 5: x2 + y2 + (14/5)x + (12/5)y - 2 = 0, so (x + 7/5)2 + (y + 6/5)2 = 27/5: centre (-7/5, -6/5), radius 3 sqrt(15)/5.

(c) (x - 3)2 + (y + 2)2 = 0: centre (3, -2), radius 0, a degenerate point circle.

Answer: (d) Dividing by 4: x2 + y2 - 2x + 3y - 25/4 = 0, so (x - 1)2 + (y + 3/2)2 = 19/2: centre (1, -3/2), radius sqrt(38)/2.

Q3. Write an equation of the circle that passes through the given points (a) A(4, 5), B(-4, -3), C(8, -3) (b) A(-7, 7), B(5, -1), C(10, 0) (c) A(a, 0), B(0, b), C(0, 0) (d) A(5, 6), B(-3, 2), C(3, -4).

Use x2 + y2 + 2gx + 2fy + c = 0 and solve for g, f, c. (a) Subtracting the equations in pairs gives g = -2, f = 1, c = -35: x2 + y2 - 4x + 2y - 35 = 0.

(b) Solving gives g = -5, f = -12, c = 0: x2 + y2 - 10x - 24y = 0.

(c) Through the origin c = 0; through (a, 0) gives g = -a/2; through (0, b) gives f = -b/2: x2 + y2 - ax - by = 0.

Answer: (d) Solving gives g = -7/3, f = -4/3, c = -65/3: 3x2 + 3y2 - 14x - 8y - 65 = 0.

Q4. In each of the following, find an equation of the circle passing through (a) A(3, -1), B(0, 1) and having centre at 4x - 3y - 3 = 0 (b) A(-3, 1) with radius 2 and centre at 2x - 3y + 3 = 0 (c) A(5, 1) and tangent to the line 2x - y - 10 = 0 at B(3, -4) (d) A(1, 4), B(-1, 8) and tangent to the line x + 3y - 3 = 0.

(a) Equidistance from A and B gives 3h - 2k = 9/2; with 4h - 3k = 3 this gives centre (15/2, 9) and radius squared 481/4: x2 + y2 - 15x - 18y + 17 = 0.

(b) Centre (h, k) on 2x - 3y + 3 = 0 at distance 2 from (-3, 1) gives 13k2 + 10k - 3 = 0, so k = 3/13 or k = -1: circles (x + 3)2 + (y + 1)2 = 4 and (x + 15/13)2 + (y - 3/13)2 = 4.

(c) The centre lies on the perpendicular to 2x - y - 10 = 0 at B, i.e. x + 2y + 5 = 0, and is equidistant from A and B, giving centre (-26, 21/2): x2 + y2 + 52x - 21y - 265 = 0.

Answer: (d) Equidistance from A and B gives h = 2k - 12; equating distance to the line with distance to A gives k2 - 18k + 65 = 0, so k = 5 or 13: circles x2 + y2 + 4x - 10y + 19 = 0 (centre (-2, 5)) and x2 + y2 - 28x - 26y + 115 = 0 (centre (14, 13)).

Q5. Find an equation of a circle of radius a and lying in the second quadrant such that it is tangent to both the axes.

A circle in the second quadrant tangent to both axes has centre (-a, a) and radius a.

Answer: (x + a)2 + (y - a)2 = a2, i.e. x2 + y2 + 2ax - 2ay + a2 = 0.

Q6. Show that the lines 3x - 2y = 0 and 2x + 3y - 13 = 0 are tangents to the circle x2 + y2 + 6x - 4y = 0.

The circle has centre (-3, 2) and radius sqrt(9 + 4) = sqrt(13).

Distance from (-3, 2) to 3x - 2y = 0 is |-9 - 4|/sqrt(13) = sqrt(13), equal to the radius.

Answer: distance from (-3, 2) to 2x + 3y - 13 = 0 is |-6 + 6 - 13|/sqrt(13) = sqrt(13), equal to the radius; so both lines are tangents.

Q7. Show that the circles x2 + y2 + 2x - 2y - 7 = 0 and x2 + y2 - 6x + 4y + 9 = 0 touch externally.

First circle: centre (-1, 1), radius sqrt(1 + 1 + 7) = 3. Second circle: centre (3, -2), radius sqrt(9 + 4 - 9) = 2.

Distance between centres = sqrt(16 + 9) = 5 = 3 + 2, the sum of the radii.

Answer: the circles touch externally.

Q8. Show that the circles x2 + y2 + 2x - 8 = 0 and x2 + y2 - 6x + 6y - 46 = 0 touch internally.

First circle: centre (-1, 0), radius sqrt(1 + 8) = 3. Second circle: centre (3, -3), radius sqrt(9 + 9 + 46) = 8.

Distance between centres = sqrt(16 + 9) = 5 = 8 - 3, the difference of the radii.

Answer: the circles touch internally.

Q9. Find equations of the circles of radius 2 and tangent to the line x - y - 4 = 0 at A(1, -3).

The centre lies on the perpendicular to x - y - 4 = 0 through A(1, -3), i.e. x + y + 2 = 0, at distance 2 from A.

This gives (h - 1)2 + (k + 3)2 = 4 with k = -h - 2, i.e. 2(h - 1)2 = 4, so h = 1 + sqrt(2) or h = 1 - sqrt(2).

Answer: centres (1 + sqrt(2), -3 - sqrt(2)) and (1 - sqrt(2), -3 + sqrt(2)); equations x2 + y2 - 2(1 + sqrt(2))x + 2(3 + sqrt(2))y + 10 + 8 sqrt(2) = 0 and x2 + y2 - 2(1 - sqrt(2))x + 2(3 - sqrt(2))y + 10 - 8 sqrt(2) = 0.

Exercise 6.2 - Tangents and normals to a circle

Q1. Write down equations of the tangent and normal to the circle (i) x2 + y2 = 25 at (4, 3) and at (5 cos(theta), 5 sin(theta)) (ii) 3x2 + 3y2 + 5x - 13y + 2 = 0 at (1,

10/3).

(i) At (4, 3): tangent 4x + 3y = 25; normal (through the origin and the point) 3x - 4y = 0.

At (5 cos(theta), 5 sin(theta)): tangent x cos(theta) + y sin(theta) = 5; normal x sin(theta) - y cos(theta) = 0.

Answer: (ii) In standard form x2 + y2 + (5/3)x - (13/3)y + 2/3 = 0, so g = 5/6, f = -13/6. Tangent T = 0: 33x + 21y - 103 = 0. Normal: (y - 10/3)(11/6) = (x - 1)(7/6), i.e. 21x - 33y + 89 = 0.

Q2. Write down equations of the tangent and normal to the circle 4x2 + 4y2 - 16x + 24y - 117 = 0 at the points on the circle whose abscissa is -4.

In standard form x2 + y2 - 4x + 6y - 117/4 = 0. Putting x = -4 gives y2 + 6y + 11/4 = 0, so y = -1/2 or y = -11/2.

At (-4, -1/2): tangent -24x + 10y - 91 = 0; normal 5x + 12y + 26 = 0.

Answer: at (-4, -11/2): tangent 24x + 10y + 151 = 0; normal 5x - 12y - 46 = 0.

Q3. Check the position of the point (5, 6) with respect to the circle (i) x2 + y2 = 81 (ii) 2x2 + 2y2 + 12x - 8y + 1 = 0.

(i) Substituting: 25 + 36 = 61 < 81, so the point lies inside the circle.

Answer: (ii) Dividing by 2: x2 + y2 + 6x - 4y + 1/2 = 0; substituting gives 25 + 36 + 30 - 24 + 1/2 = 135/2 > 0, so the point lies outside the circle.

Q4. Find the length of the tangent drawn from the point (-5, 4) to the circle 5x2 + 5y2 - 10x + 15y - 131 = 0.

In standard form x2 + y2 - 2x + 3y - 131/5 = 0.

Substituting (-5, 4): 25 + 16 + 10 + 12 - 131/5 = 184/5.

Answer: length of the tangent = sqrt(184/5) = 2 sqrt(230)/5.

Q5. Find the length of the chord cut off from the line 2x + 3y = 13 by the circle x2 + y2 = 26.

Centre (0, 0), radius sqrt(26). Distance from the centre to the line = 13/sqrt(13) = sqrt(13).

Half the chord = sqrt(26 - 13) = sqrt(13).

Answer: length of the chord = 2 sqrt(13).

Q6. Find the coordinates of the points of intersection of the line x + 2y = 6 with the circle x2 + y2 - 2x - 2y - 39 = 0.

Putting x = 6 - 2y in the circle gives 5y2 - 22y - 15 = 0, so y = 5 or y = -3/5.

Answer: points of intersection (-4, 5) and (36/5, -3/5).

Q7. Find equations of the tangents to the circle x2 + y2 = 2 (i) parallel to the line x - 2y + 1 = 0 (ii) perpendicular to the line 3x + 2y = 6.

(i) Tangents parallel to x - 2y + 1 = 0 have the form x - 2y + c = 0; distance from the origin equal to the radius sqrt(2) gives |c|/sqrt(5) = sqrt(2), so c = + sqrt(10) or

- sqrt(10).

Answer: (i) x - 2y + sqrt(10) = 0 and x - 2y - sqrt(10) = 0. (ii) Tangents perpendicular to 3x + 2y = 6 have slope 2/3, i.e. form 2x - 3y + c = 0; |c|/sqrt(13) = sqrt(2) gives c = + sqrt(26) or - sqrt(26): 2x - 3y + sqrt(26) = 0 and 2x - 3y - sqrt(26) = 0.

Q8. Find equations of the tangents drawn from (i) (0, 5) to x2 + y2 = 16 (ii) (-1, 2) to x2 + y2 + 4x + 2y = 0 (iii) (-7, -2) to (x + 1)2 + (y - 2)2 = 26. Also find the points of contact.

(i) Chord of contact: 5y = 16, i.e. y = 16/5; with the circle this gives x = + 12/5 or - 12/5. Tangents: 3x + 4y = 20 and -3x + 4y = 20; points of contact (12/5, 16/5) and (-12/5, 16/5).

(ii) Chord of contact: x + 3y = 0; with the circle this gives (0, 0) and (-3, 1). Tangents: 2x + y = 0 and x - 2y + 5 = 0.

Answer: (iii) Chord of contact: 3x + 2y + 12 = 0; with the circle this gives (-2, -3) and (-6, 3). Tangents: x + 5y + 17 = 0 and 5x - y + 33 = 0.

Q9. Find an equation of the chord of contact of the tangents drawn from (4, 5) to the circle 2x2 + 2y2 - 8x + 12y + 21 = 0.

In standard form x2 + y2 - 4x + 6y + 21/2 = 0.

Chord of contact T = 0: 4x + 5y - 2(x + 4) + 3(y + 5) + 21/2 = 0.

Answer: 4x + 16y + 35 = 0.

Exercise 6.3 - Analytic proofs of properties of a circle

Q1. Prove that normal lines of a circle pass through the centre of the circle.

For the circle x2 + y2 + 2gx + 2fy + c = 0, the normal at P(x1, y1) is (y - y1)(x1 + g) = (x - x1)(y1 + f).

The centre is (-g, -f). Putting x = -g and y = -f gives left side (-f - y1)(x1 + g) = -(y1 + f)(x1 + g) and right side (-g - x1)(y1 + f) = -(x1 + g)(y1 + f).

Answer: both sides are equal, so the centre lies on the normal; hence every normal passes through the centre.

Q2. Prove that the straight line drawn from the centre of a circle perpendicular to a tangent passes through the point of tangency.

Let P(x1, y1) be the point of tangency on x2 + y2 + 2gx + 2fy + c = 0, whose centre is C(-g, -f).

The tangent at P has slope -(x1 + g)/(y1 + f), so the line through C perpendicular to it has slope (y1 + f)/(x1 + g) and equation y + f = ((y1 + f)/(x1 + g))(x + g).

Answer: putting x = x1 gives y + f = y1 + f, i.e. y = y1, so the line passes through P(x1, y1), the point of tangency.

Q3. Prove that the mid point of the hypotenuse of a right triangle is the circumcentre of the triangle.

Place the right angle at the origin: let the triangle have vertices C(0, 0), A(a, 0) and B(0, b), with hypotenuse AB.

The midpoint of AB is M(a/2, b/2). Then MA = MB = sqrt(a2 + b2)/2.

Answer: also MC = sqrt((a/2)2 + (b/2)2) = sqrt(a2 + b2)/2, so M is equidistant from A, B and C; hence M is the circumcentre.

Q4. Prove that the perpendicular dropped from a point of a circle on a diameter is a mean proportional between the segments into which it divides the diameter.

Take the diameter AB on the x-axis with A(-a, 0), B(a, 0), so the circle is x2 + y2 = a2. Let P(x1, y1) be a point on the circle and M(x1, 0) the foot of the perpendicular from P to AB.

Then PM = y1, AM = x1 + a and MB = a - x1, so AM times MB = (a + x1)(a - x1) = a2 - x12.

Answer: since P lies on the circle, y12 = a2 - x12, so PM2 = AM times MB, i.e. PM is the mean proportional between AM and MB.

Exercise 6.4 - Parabola

Q1. Find the focus, vertex and directrix of the parabola. Sketch its graph. (i) y2 = 8x (ii) x2 = -16y (iii) x2 = 5y (iv) y2 = -12x (v) x2 = 4(y - 1) (vi) y2 = -8(x - 3) (vii) (x - 1)2 = 8(y + 2) (viii) y = 6x2 - 1 (ix) x + 8 - y2 + 2y = 0 (x) x2 - 4x - 8y + 4 = 0.

(i) 4a = 8, a = 2: vertex (0, 0), focus (2, 0), directrix x = -2, opening right. (ii) 4a = 16, a = 4: vertex (0, 0), focus (0, -4), directrix y = 4, opening downward.

(iii) 4a = 5, a = 5/4: vertex (0, 0), focus (0, 5/4), directrix y = -5/4, opening upward.

(iv) 4a = 12, a = 3: vertex (0, 0), focus (-3, 0), directrix x = 3, opening left.

(v) Vertex (0, 1), a = 1: focus (0, 2), directrix y = 0. (vi) Vertex (3, 0), a = 2: focus (1, 0), directrix x = 5.

(vii) Vertex (1, -2), a = 2: focus (1, 0), directrix y = -4. (viii) x2 = (1/6)(y + 1), a = 1/24: vertex (0, -1), focus (0, -23/24), directrix y = -25/24.

Answer: (ix) (y - 1)2 = x + 9, a = 1/4: vertex (-9, 1), focus (-35/4, 1), directrix x = -37/4. (x) (x - 2)2 = 8y, a = 2: vertex (2, 0), focus (2, 2), directrix y = -2.

Q2. Write an equation of the parabola with given elements. (i) Focus (-3, 1); directrix x = 3 (ii) Focus (2, 5); directrix y = 1 (iii) Focus (-3, 1); directrix x - 2y - 3 = 0 (iv) Focus (1, 2); vertex (3, 2) (v) Focus (-1, 0); vertex (-1, 2) (vi) Directrix x = -2; Focus (2, 2) (vii) Directrix y = 3; vertex (2, 2) (viii) Directrix y = 1, length of latus rectum is 8. Opens downward. (ix) Axis y = 0, through (2, 1) and (11, -2) (x) Axis parallel to y-axis, the points (0, 3), (3, 4) and (4, 11) lie on the graph.

(i) Vertex is the midpoint (0, 1) of focus and foot of perpendicular, a = 3, opening left: (y - 1)2 = -12x. (ii) Vertex (2, 3), a = 2, opening upward: (x - 2)2 = 8(y - 3).

(iii) By definition (x + 3)2 + (y - 1)2 = (x - 2y - 3)2/5, i.e. 4x2 + y2 + 4xy + 36x - 22y + 41 = 0.

(iv) Vertex (3, 2), a = 2, focus to the left of the vertex, opening left: (y - 2)2 = -8(x - 3). (v) Vertex (-1, 2), a = 2, focus below the vertex, opening downward: (x + 1)2 = -8(y - 2).

(vi) Vertex (0, 2), a = 2, opening right: (y - 2)2 = 8x. (vii) Focus (2, 1), a = 1, opening downward: (x - 2)2 = -4(y - 2).

(viii) Note: the printed data do not fix the position of the axis. With latus rectum 8, a = 2, and the vertex 2 units below the directrix y = 1, the general equation is (x - h)2 = -8(y + 1); with the axis taken as the y-axis (h = 0) this is x2 = -8(y + 1).

(ix) With axis y = 0, y2 = 4a(x - h): from the two points, 1 = 8a - 4ah and 4 = 44a - 4ah, giving a = 1/12 and h = -1: 3y2 = x + 1.

Answer: (x) y = Ax2 + Bx + C through the three points gives C = 3, A = 5/3, B = -14/3: 5x2 - 14x - 3y + 9 = 0.

Q3. Find an equation of the parabola having its focus at the origin and directrix, parallel to the (i) x-axis (ii) y-axis.

(i) Let the directrix parallel to the x-axis be y = k. By definition, distance from P(x, y) to (0, 0) equals distance to the line: x2 + y2 = (y - k)2, i.e. x2 = -2ky + k2.

Answer: (ii) Let the directrix parallel to the y-axis be x = k. Then x2 + y2 = (x - k)2, i.e. y2 = -2kx + k2. (In particular, with the directrix at distance 2a from the focus, these are x2 = 4ay and y2 = 4ax up to the sign of the side on which the directrix lies.)

Q4. Show that an equation of the parabola with focus at (a cos(alpha), a sin(alpha)) and directrix x cos(alpha) + y sin(alpha) + a = 0 is (x sin(alpha) - y cos(alpha))2 = 4a(x cos(alpha) + y sin(alpha)).

By definition, PF2 = PM2: (x - a cos(alpha))2 + (y - a sin(alpha))2 = (x cos(alpha) + y sin(alpha) + a)2.

Expanding the left side gives x2 + y2 - 2a(x cos(alpha) + y sin(alpha)) + a2;

expanding the right side gives x2 cos2(alpha) + y2 sin2(alpha) + 2xy sin(alpha) cos(alpha) + 2a(x cos(alpha) + y sin(alpha)) + a2.

Answer: subtracting gives x2 sin2(alpha) + y2 cos2(alpha) - 2xy sin(alpha) cos(alpha) = 4a(x cos(alpha) + y sin(alpha)), i.e. (x sin(alpha) - y cos(alpha))2 = 4a(x cos(alpha) + y sin(alpha)).

Q5. Show that the ordinate at any point P of the parabola is a mean proportional between the length of the latus rectum and the abscissa of P.

Let P = (at2, 2at) be a point on y2 = 4ax. Its ordinate is 2at and its abscissa is at2; the length of the latus rectum is 4a.

Answer: the mean proportional between 4a and at2 is sqrt(4a times at2) = 2at, which is the ordinate of P.

Q6. A comet has a parabolic orbit with the earth at the focus. When the comet is 150,000 km from the earth, the line joining the comet and the earth makes an angle of 30 degrees with the axis of the parabola. How close will the comet come to the earth?

For the parabola y2 = 4ax with focus (a, 0), the distance from the focus to a point P is r = x + a, and the angle condition gives y = r sin(30 degrees) = 75,000 km.

Then y2 = 4a(r - a) gives 75,0002 = 4a(150,000 - a), i.e. a2 - 150,000a + 1,406,250,000 = 0, so a = 10,048 km or a = 139,952 km.

Answer: the perihelion (closest approach) is a = approx. 10,048 km; the second root corresponds to the supplementary angle on the other side of the focus.

Q7. Find an equation of the parabola formed by the cables of a suspension bridge whose span is a m and the vertical height of the supporting towers is b m.

Take the vertex at the centre of the bridge on the road surface, with the axis vertical: x2 = 4Ay. The towers stand at x = + a/2 and x = - a/2 with height b, so (a/2)2 = 4Ab, giving 4A = a2/(4b).

Answer: x2 = (a2/(4b)) y, with y measured upward from the road surface at the centre of the span.

Q8. A parabolic arch has a 100 m base and height 25 m. Find the height of the arch at the point 30 m from the centre of the base.

Take the vertex at the top (0, 25) with the axis vertical, opening downward: x2 = -4a(y - 25). The base ends (+-50, 0) lie on the arch, so 2500 = 100a, giving a = 25.

At x = 30: 900 = -100(y - 25), so y - 25 = -9 and y = 16.

Answer: height of the arch 30 m from the centre is 16 m.

Q9. Show that tangent at any point P of a parabola makes equal angles with the line PF and the line through P parallel to the axis of the parabola, F being focus. (These angles are called respectively angle of incidence and angle of reflection).

Let P = (at2, 2at) on y2 = 4ax. The tangent at P is ty = x + at2, which meets the axis at T(-at2, 0).

Then TF = a + at2, and PF = at2 + a (distance from focus equals distance from directrix), so triangle TPF is isosceles with TF = PF.

Answer: hence the angle at T (equal, by parallel lines, to the angle between the tangent and the line through P parallel to the axis) equals the angle at P between PF and the tangent.

Exercise 6.5 - Ellipse

Q1. Find an equation of the ellipse with given data and sketch its graph: (i) Foci (+-3, 0) and minor axis of length 10 (ii) Foci (0, -1) and (0, -5) and major axis of length 6 (iii) Foci (+-3 sqrt(3), 0) and vertices (+-6, 0) (iv) Vertices (-1, 1), (5, 1); foci (4, 1) and (0, 1) (v) Foci (+-sqrt(5), 0) and passing through the point (3/2, sqrt(3)) (vi) Vertices (0, +-5), eccentricity 3/5 (vii) Centre (0, 0), focus (0, -3), vertex (0, 4) (viii) Centre (2, 2), major axis parallel to y-axis and of length 8 units, minor axis parallel to x-axis and of length 6 units (ix) Centre (0, 0), symmetric with respect to both the axes and passing through the points (2, 3) and (6, 1) (x) Centre (0, 0), major axis horizontal, the points (3, 1), (4, 0) lie on the graph.

(i) c = 3, b = 5, a2 = 25 + 9 = 34: x2/34 + y2/25 = 1. (ii) Centre (0, -3), c = 2, a = 3, b2 = 5, major axis vertical: x2/5 + (y + 3)2/9 = 1.

(iii) a = 6, c = 3 sqrt(3), b2 = 36 - 27 = 9: x2/36 + y2/9 = 1. (iv) Centre (2, 1), a = 3, c = 2, b2 = 5: (x - 2)2/9 + (y - 1)2/5 = 1.

(v) c = sqrt(5); putting (3/2, sqrt(3)) in x2/a2 + y2/(a2 - 5) = 1 gives a2 = 9 (the other root 5/4 is rejected since a2 > c2): x2/9 + y2/4 = 1.

(vi) a = 5, e = 3/5, c = 3, b2 = 16, major axis vertical: x2/16 + y2/25 = 1. (vii) a = 4, c = 3, b2 = 7, major axis vertical: x2/7 + y2/16 = 1.

(viii) Centre (2, 2), a = 4, b = 3: (x - 2)2/9 + (y - 2)2/16 = 1. (ix) With x2/A + y2/B = 1 through (2, 3) and (6, 1): A = 40, B = 10: x2/40 + y2/10 = 1.

Answer: (x) (4, 0) is a vertex, so a = 4; (3, 1) gives 9/16 + 1/b2 = 1, so b2 = 16/7: x2 + 7y2 = 16.

Q2. Find the centre, foci, eccentricity, vertices and directrices of the ellipse, whose equation is given: (i) x2 + 4y2 = 16 (ii) 9x2 + y2 = 18 (iii) 25x2 + 9y2 = 225 (iv) (2x - 1)2/4 + (y + 2)2/16 = 1 (v) x2 + 16x + 4y2 - 16y + 76 = 0 (vi) 25x2 + 4y2 - 250x - 16y + 541 = 0.

(i) x2/16 + y2/4 = 1: centre (0, 0), a = 4, b = 2, c = 2 sqrt(3), e = sqrt(3)/2, vertices (+-4, 0), foci (+-2 sqrt(3), 0), directrices x = +-8 sqrt(3)/3.

(ii) x2/2 + y2/18 = 1: centre (0, 0), a = 3 sqrt(2), b = sqrt(2), c = 4, e = 2 sqrt(2)/3, vertices (0, +-3 sqrt(2)), foci (0, +-4), directrices y = +-9/2.

(iii) x2/9 + y2/25 = 1: centre (0, 0), a = 5, b = 3, c = 4, e = 4/5, vertices (0, +-5), foci (0, +-4), directrices y = +-25/4.

(iv) (x - 1/2)2 + (y + 2)2/4 = 1: centre (1/2, -2), a = 2, b = 1, c = sqrt(3), e = sqrt(3)/2, vertices (1/2, 0), (1/2, -4), foci (1/2, -2 +- sqrt(3)), directrices y = -2 +- 4/sqrt(3).

(v) (x + 8)2/4 + (y - 2)2 = 1: centre (-8, 2), a = 2, b = 1, c = sqrt(3), e = sqrt(3)/2, vertices (-6, 2), (-10, 2), foci (-8 +- sqrt(3), 2), directrices x = -8 +- 4/sqrt(3).

Answer: (vi) (x - 5)2/4 + (y - 2)2/25 = 1: centre (5, 2), a = 5, b = 2, c = sqrt(21), e =

sqrt(21)/5, vertices (5, 7), (5, -3), foci (5, 2 +- sqrt(21)), directrices y = 2 +- 25/sqrt(21).

Q3. Let a be a positive number and 0 < c < a. Let F(-c, 0) and F'(c, 0) be two given points. Prove that the locus of points P(x, y) such that |PF| + |PF'| = 2a, is an ellipse.
The condition is sqrt((x + c)2 + y2) + sqrt((x - c)2 + y2) = 2a, i.e. sqrt((x + c)2 + y2) = 2a - sqrt((x - c)2 + y2).

Squaring and simplifying gives (a2 - c2)x2 + a2y2 = a2(a2 - c2).

Answer: since 0 < c < a, b2 = a2 - c2 > 0, and dividing gives x2/a2 + y2/b2 = 1, an ellipse with foci (+-c, 0).

Q4. Use problem 3 to find equation of the ellipse as locus of points P(x, y) such that the sum of the distances from P to the points (0, 0) and (1, 1) is 2.
The foci are (0, 0) and (1, 1), so the centre is (1/2, 1/2), 2c = sqrt(2) (c = 1/sqrt(2)) and 2a = 2 (a = 1); hence b2 = 1 - 1/2 = 1/2.

With u = (x + y - 1)/sqrt(2) along the major axis and v = (x - y)/sqrt(2) perpendicular to it, the equation is u2 + v2/(1/2) = 1.

Answer: (x + y - 1)2 + 2(x - y)2 = 2, i.e. 3x2 - 2xy + 3y2 - 2x - 2y - 1 = 0.

Q5. Prove that the latus rectum of the ellipse x2/a2 + y2/b2 = 1 is 2b2/a.
The latus rectum is the chord through a focus perpendicular to the major axis; take the focus (ae, 0), so the chord is x = ae.

Putting x = ae in the ellipse gives y2/b2 = 1 - e2 = b2/a2, so y = +-b2/a.

Answer: the length of the chord is 2b2/a.

Q6. The major axis of an ellipse in standard form lies along the x-axis and has length 4 sqrt(2). The distance between the foci equals the length of the minor axis. Write an equation of the ellipse.

Major axis 2a = 4 sqrt(2) gives a = 2 sqrt(2). Distance between foci 2c equals minor axis 2b, so c = b.

Then a2 = b2 + c2 = 2b2 gives b2 = a2/2 = 4.

Answer: x2/8 + y2/4 = 1.

Q7. An asteroid has elliptic orbit with the sun at one focus. Its distance from the sun ranges from 17 million miles to 183 million miles. Write an equation of the orbit of the asteroid.

Perihelion a - c = 17 and aphelion a + c = 183 (million miles), so a = 100 and c = 83.

Then b2 = a2 - c2 = 10,000 - 6,889 = 3,111.

Answer: with the centre at the origin, x2/10,000 + y2/3,111 = 1 (distances in million miles).

Q8. An arch in the shape of a semi-ellipse is 90m wide at the base and 30m high at the centre. At what distance from the centre is the arch 20 sqrt(2) m high?

Take the semi-ellipse x2/452 + y2/302 = 1 with y >= 0 (a = 45, b = 30).

At height y = 20 sqrt(2): x2/2025 = 1 - 800/900 = 1/9, so x2 = 225 and x = 15.

Answer: the arch is 20 sqrt(2) m high at a distance of 15 m from the centre.

Q9. The moon orbits the earth in an elliptic path with earth at one focus. The major and minor axes of the orbit are 768,806 km and 767,746 km respectively. Find the greatest and least distances (in Astronomy called the apogee and perigee respectively) of the moon from the earth.

Semi-axes: a = 384,403 km, b = 383,873 km. Then c = sqrt(a2 - b2) = sqrt(530 times 768,276) = approx. 20,179 km.

Greatest distance (apogee) = a + c = approx. 404,582 km; least distance (perigee) = a - c = approx. 364,224 km.

Answer: apogee approx. 404,582 km, perigee approx. 364,224 km.

Exercise 6.6 - Hyperbola

Q1. Find an equation of the hyperbola with the given data. Sketch the graph of each. (i) Centre (0, 0), focus (6, 0), vertex (4, 0) (ii) Foci (+-5, 0), vertex (3, 0) (iii) Foci (2 +- 5 sqrt(2), -7), length of the transverse axis 10 (iv) Foci (0, +-6), e = 2 (v) Foci (0, +-9), directrices y = +-4 (vi) Centre (2, 2), horizontal transverse axis of length 6 and eccentricity e = 2 (vii) Vertices (2, +-3), (0, 5) lies on the curve (viii) Foci (5, -2), (5, 4) and one vertex (5, 3).

(i) a = 4, c = 6, b2 = 36 - 16 = 20: x2/16 - y2/20 = 1. (ii) a = 3, c = 5, b2 = 16: x2/9

- y2/16 = 1.

(iii) Centre (2, -7), c = 5 sqrt(2), a = 5, b2 = 50 - 25 = 25: (x - 2)2/25 - (y + 7)2/25 = 1.

(iv) c = 6, a = c/e = 3, b2 = 36 - 9 = 27, transverse axis vertical: y2/9 - x2/27 = 1.

(v) c = 9 and a/e = 4 with ae = 9 give a2 = 36, so a = 6, e = 3/2, b2 = 45: y2/36 - x2/45 = 1.

(vi) Centre (2, 2), a = 3, c = ae = 6, b2 = 27: (x - 2)2/9 - (y - 2)2/27 = 1.

(vii) Centre (2, 0), a = 3, transverse axis vertical; (0, 5) on y2/9 - (x - 2)2/b2 = 1 gives 25/9 - 4/b2 = 1, so b2 = 9/4: y2 - 4(x - 2)2 = 9.

Answer: (viii) Centre (5, 1), c = 3, a = 2, b2 = 5, transverse axis vertical: (y - 1)2/4 - (x - 5)2/5 = 1.

Q2. Find the centre, foci, eccentricity, vertices and equations of directrices of each of the following: (i) x2 - y2 = 9 (ii) x2/4 - y2/9 = 1 (iii) y2/16 - x2/9 = 1 (iv) y2/4 - x2 = 1 (v) (x - 1)2/2 - (y - 1)2/9 = 1 (vi) (y + 2)2/9 - (x - 2)2/16 = 1 (vii) 9x2 - 12x - y2 - 2y + 2 = 0 (viii) 4y2 + 12y - x2 + 4x + 1 = 0 (ix) x2 - y2 + 8x - 2y - 10 = 0 (x) 9x2 - y2 - 36x - 6y + 18 = 0.

(i) x2/9 - y2/9 = 1: centre (0, 0), a = 3, b = 3, c = 3 sqrt(2), e = sqrt(2), vertices (+-3, 0), foci (+-3 sqrt(2), 0), directrices x = +-3/sqrt(2).

(ii) a = 2, b = 3, c = sqrt(13), e = sqrt(13)/2, vertices (+-2, 0), foci (+-sqrt(13), 0),

directrices x = +-4/sqrt(13).

(iii) a = 4, b = 3, c = 5, e = 5/4, vertices (0, +-4), foci (0, +-5), directrices y = +-16/5.

(iv) a = 2, b = 1, c = sqrt(5), e = sqrt(5)/2, vertices (0, +-2), foci (0, +-sqrt(5)), directrices y = +-8/sqrt(5).

(v) Centre (1, 1), a = sqrt(2), b = 3, c = sqrt(11), e = sqrt(22)/2, vertices (1 +- sqrt(2), 1), foci (1 +- sqrt(11), 1), directrices x = 1 +- 2/sqrt(11).

(vi) Centre (2, -2), a = 3, b = 4, c = 5, e = 5/3, vertices (2, 1), (2, -5), foci (2, 3), (2, -7), directrices y = -1/5, y = -19/5.

(vii) (x - 2/3)2/(1/9) - (y + 1)2 = 1: centre (2/3, -1), a = 1/3, b = 1, c = sqrt(10)/3, e = sqrt(10), vertices (1, -1), (1/3, -1), foci (2/3 +- sqrt(10)/3, -1), directrices x = 2/3 +- 1/(3 sqrt(10)).

(viii) (y + 3/2)2 - (x - 2)2/4 = 1: centre (2, -3/2), a = 1, b = 2, c = sqrt(5), e = sqrt(5), vertices (2, -1/2), (2, -5/2), foci (2, -3/2 +- sqrt(5)), directrices y = -3/2 +- 1/sqrt(5).

(ix) (x + 4)2/25 - (y + 1)2/25 = 1: centre (-4, -1), a = 5, b = 5, c = 5 sqrt(2), e = sqrt(2), vertices (1, -1), (-9, -1), foci (-4 +- 5 sqrt(2), -1), directrices x = -4 +- 5/sqrt(2).

Answer: (x) (x - 2)2 - (y + 3)2/9 = 1: centre (2, -3), a = 1, b = 3, c = sqrt(10), e =

sqrt(10), vertices (3, -3), (1, -3), foci (2 +- sqrt(10), -3), directrices x = 2 +- 1/sqrt(10).

Q3. Let 0 < a < c and F'(-c, 0), F(c, 0) be two fixed points. Show that the set of points P(x, y) such that |PF| - |PF'| = 2a, is the hyperbola x2/a2 - y2/(c2 - a2) = 1. (F, F' are foci of the hyperbola.)

The condition is |sqrt((x - c)2 + y2) - sqrt((x + c)2 + y2)| = 2a; take the positive branch sqrt((x - c)2 + y2) = 2a + sqrt((x + c)2 + y2).

Squaring and simplifying twice gives (c2 - a2)x2 - a2y2 = a2(c2 - a2).

Answer: since c > a, b2 = c2 - a2 > 0, and dividing gives x2/a2 - y2/b2 = 1, a hyperbola with foci (+-c, 0).

Q4. Using Problem 3, find an equation of the hyperbola with foci (-5, -5) and (5, 5), vertices (-3 sqrt(2), -3 sqrt(2)) and (3 sqrt(2), 3 sqrt(2)).

Centre (0, 0), transverse axis along y = x; c = distance to a focus = 5 sqrt(2), a = distance to a vertex = 6, so b2 = 50 - 36 = 14.

With u = (x + y)/sqrt(2) along the axis and v = (x - y)/sqrt(2) perpendicular, u2/36 - v2/14 = 1.

Answer: (x + y)2/72 - (x - y)2/28 = 1, i.e. 11x2 - 50xy + 11y2 + 504 = 0.

Q5. For any point on a hyperbola the difference of its distances from the points (2, 2) and (10, 2) is 6. Find an equation of the hyperbola.

The points are the foci: centre (6, 2), 2c = 8 so c = 4, and 2a = 6 so a = 3; hence b2

= 16 - 9 = 7.

Answer: (x - 6)2/9 - (y - 2)2/7 = 1.

Q6. Two listening posts hear the sound of an enemy gun. The difference in time is one second. If the listening posts are 1400 feet apart, write an equation of the hyperbola passing through the position of the enemy gun. (Sound travels at 1080 ft/sec).

The difference of distances from the two posts is 1080 times 1 = 1080 feet, so 2a = 1080 and a = 540; the posts are the foci, so 2c = 1400 and c = 700.

Then b2 = 7002 - 5402 = 198,400.

Answer: with the posts at (+-700, 0), x2/291,600 - y2/198,400 = 1 (distances in feet).

Exercise 6.7 - Tangents and normals to conics

Q1. Find equations of the tangent and normal to each of the following at the indicated point: (i) y2 = 4ax at (a t2, 2a t) (ii) x2/a2 + y2/b2 = 1 at (a cos(theta), b sin(theta)) (iii) x2/a2 - y2/b2 = 1 at (a sec(theta), b tan(theta)).

(i) Tangent: y(2at) = 2a(x + at2), i.e. ty = x + at2. Normal (slope -t): y - 2at = -t(x - at2).

(ii) Tangent: x cos(theta)/a + y sin(theta)/b = 1. Normal: y - b sin(theta) = (a sin(theta)/(b cos(theta)))(x - a cos(theta)), i.e. a x sin(theta) - b y cos(theta) = (a2 - b2) sin(theta) cos(theta).

Answer: (iii) Tangent: x sec(theta)/a - y tan(theta)/b = 1. Normal: y - b tan(theta) = (-a sin(theta)/b)(x - a sec(theta)).

Q2. Write equation of the tangent to the given conic at the indicated point (i) 3x2 = -16y at the points whose ordinate is -3 (ii) 3x2 - 7y2 = 20 at the points where y = -1 (iii) 3x2 - 7y2 + 2x - y - 48 = 0 at the point where x = 4.

(i) Ordinate -3 gives 3x2 = 48, so x = +-4. Tangents: at (4, -3): x + 2y + 6 = 0; at (-4, -3): x - 2y - 6 = 0.

(ii) y = -1 gives 3x2 = 27, so x = +-3. Tangents: at (3, -1): 9x + 7y = 20; at (-3, -1): -9x + 7y = 20.

Answer: (iii) x = 4 gives 7y2 + y - 8 = 0, so y = 1 or y = -8/7. Tangents: at (4, 1): 26x - 15y - 89 = 0; at (4, -8/7): 182x + 105y - 608 = 0.

Q3. Find equations of the tangents to each of the following through the given point: (i) x2 + y2 = 25 through (7, -1) (ii) y2 = 12x through (1, 4) (iii) x2 - 2y2 = 2 through (1, -2).

(i) Chord of contact: 7x - y = 25; with the circle this gives points of contact (3, -4) and (4, 3). Tangents: 3x - 4y = 25 and 4x + 3y = 25.

(ii) Chord of contact: 2y = 3x + 3; with the parabola this gives points of contact (3, 6) and (1/3, 2). Tangents: y = x + 3 and y = 3x + 1.

Answer: (iii) Chord of contact: x + 4y = 2; with the hyperbola this gives points of contact (-2, 1) and (10/7, 1/7). Tangents: x + y + 1 = 0 and 5x - y - 7 = 0.

Q4. Find equations of the normals to the parabola y2 = 8x which are parallel to the line 2x + 3y = 10.

Here a = 2. The normal at (2t2, 4t) has slope -t; parallel to 2x + 3y = 10 (slope -2/3) requires t = 2/3.

The point is (8/9, 8/3), and the normal is y - 8/3 = -(2/3)(x - 8/9).

Answer: 18x + 27y - 88 = 0.

Q5. Find equations of the tangents to the ellipse x2/4 + y2 = 1 which are parallel to the line 2x - 4y + 5 = 0.

The line has slope 1/2. A tangent of slope m to x2/4 + y2 = 1 is y = mx +- sqrt(4m2 + 1).

With m = 1/2: y = (1/2)x +- sqrt(2).

Answer: x - 2y + 2 sqrt(2) = 0 and x - 2y - 2 sqrt(2) = 0.

Q6. Find equations of the tangents to the conic 9x2 - 4y2 = 36 parallel to 5x - 2y + 7 = 0.

The conic is x2/4 - y2/9 = 1; the line has slope 5/2. A tangent of slope m is y = mx +- sqrt(4m2 - 9).

With m = 5/2: y = (5/2)x +- 4.

Answer: 5x - 2y + 8 = 0 and 5x - 2y - 8 = 0.

Q7. Find equations of the common tangents to the given conics (i) x2 = 80y and x2 + y2 = 81 (ii) y2 = 16x and x2 = 2y.

(i) A tangent of slope m to x2 = 80y is y = mx - 20m2; its distance from the origin must equal 9, giving 400m4 = 81(1 + m2), so m2 = 9/16 and m = +-3/4.

Tangents: y = (3/4)x - 45/4 and y = -(3/4)x - 45/4.

Answer: (ii) A tangent of slope m to y2 = 16x is y = mx + 4/m, and to x2 = 2y is y = mx - (1/2)m2; equating gives m3 = -8, so m = -2 and the tangent is y = -2x - 2. Also y = 0 is tangent to both conics at the origin.

Q8. Find the points of intersection of the given conics (i) x2/18 + y2/8 = 1 and x2/3 - y2/3 = 1 (ii) x2 + y2 = 8 and x2 - y2 = 1 (iii) 3x2 - 4y2 = 12 and 3y2 - 2x2 = 7 (iv) 3x2 + 5y2 = 60 and 9x2 + y2 = 124 (v) 4x2 + y2 = 16 and x2 + y2 + y + 8 = 0.

(i) From x2 - y2 = 3 and 4x2 + 9y2 = 72: 13x2 = 99, so the four points are (+-3 sqrt(11/13), +-2 sqrt(15/13)).

(ii) Adding and subtracting: 2x2 = 9 and 2y2 = 7, so the four points are (+-3/sqrt(2), +-sqrt(7/2)).

(iii) Solving gives x2 = 64 and y2 = 45, so the four points are (+-8, +-3 sqrt(5)).

(iv) Solving gives y2 = 4 and x2 = 40/3, so the four points are (+-2 sqrt(10/3), +-2).

Answer: (v) Subtracting gives y = 3x2 - 24; substituting gives 9x4 - 140x2 + 560 = 0, whose discriminant is negative, so the conics have no real points of intersection.

Exercise 6.8 - Translation and rotation of axes

Q1. Find an equation of each of the following with respect to new parallel axes obtained by shifting the origin to the indicated point: (i) x2 + 16y - 16 = 0, O'(0, 1) (ii) 4x2 + y2 + 16x - 10y + 37 = 0, O'(2, 5) (iii) 9x2 + 4x2 + 18x - 16y - 11 = 0, O'(-1, 2) (iv) x2 - y2 + 4x + 8y - 11 = 0, O'(-2, 4) (v) 9x2 - 4y2 + 36x + 8y - 4 = 0, O'(2, 1). (Print note for (iii): as printed in this copy the equation reads 9x2 + 4x2, i.e. 13x2; solved as printed below. If the second term were the misprint for 4y2, the shifted equation would be the clean 9X2 + 4Y2 = 36.)

(i) x = X, y = Y + 1: X2 + 16Y = 0. (ii) x = X + 2, y = Y + 5: 4X2 + Y2 + 32X + 60 = 0.

(iii) As printed: x = X - 1, y = Y + 2 in 13x2 + 18x - 16y - 11 = 0 gives 13X2 - 8X - 16Y - 48 = 0.

(iv) x = X - 2, y = Y + 4: X2 - Y2 + 1 = 0.

Answer: (v) x = X + 2, y = Y + 1: 9X2 - 4Y2 + 72X + 108 = 0.

Q2. Find coordinates of the new origin (axes remaining parallel) so that first degree terms are removed from the transformed equation of each of the following. Also find the transformed equation: (i) 3x2 - 2y2 + 24x + 12y + 24 = 0 (ii) 25x2 + 9y2 + 50x - 36y - 164 = 0 (iii) x2 - y2 - 6x + 2y + 7 = 0.

(i) 3(x + 4)2 - 2(y - 3)2 = 6: new origin (-4, 3), transformed equation 3X2 - 2Y2 = 6.

(ii) 25(x + 1)2 + 9(y - 2)2 = 225: new origin (-1, 2), transformed equation 25X2 + 9Y2 = 225.

Answer: (iii) (x - 3)2 - (y - 1)2 = 1: new origin (3, 1), transformed equation X2 - Y2 = 1.

Q3. In each of the following, find an equation referred to the new axes obtained by rotation of axes about the origin through the given angle: (i) xy = 1, theta = 45 degrees (ii) 7x2 - 8xy + y2 - 9 = 0, theta = arctan 2 (iii) 9x2 + 12xy + 4y2 - x - y = 0, theta = arctan(2/3) (iv) x2 - 2xy + y2 - 2 sqrt(2) x - 2 sqrt(2) y + 2 = 0, theta = 45 degrees.

(i) With x = (X - Y)/sqrt(2), y = (X + Y)/sqrt(2): (X2 - Y2)/2 = 1, i.e. X2 - Y2 = 2.

(ii) With cos(theta) = 1/sqrt(5), sin(theta) = 2/sqrt(5): the XY term cancels and -X2 + 9Y2 - 9 = 0.

(iii) With cos(theta) = 3/sqrt(13), sin(theta) = 2/sqrt(13): 3x + 2y = sqrt(13) X, so 13X2 - (5X + Y)/sqrt(13) = 0, i.e. 13 sqrt(13) X2 - 5X - Y = 0.

Answer: (iv) x - y = -sqrt(2) Y and x + y = sqrt(2) X, so 2Y2 - 4X + 2 = 0, i.e. Y2 - 2X + 1 = 0.

Q4. Find measure of the angle through which the axes be rotated so that the product term XY is removed from the transformed equation. Also find the transformed equation: (i) 2x2 + 6xy + 10y2 - 11 = 0 (ii) xy + 4x - 3y - 10 = 0 (iii) 5x2 - 6xy + 5y2 - 8 = 0.

(i) tan(2 theta) = 6/(2 - 10) = -3/4, so 2 theta = 143.13 degrees and theta = 71.57 degrees; transformed coefficients 37/5 and 23/5 give 37X2 + 23Y2 = 55.

(ii) tan(2 theta) is infinite, so theta = 45 degrees; transformed equation X2 - Y2 +

sqrt(2) X - 7 sqrt(2) Y - 20 = 0.

Answer: (iii) tan(2 theta) is infinite, so theta = 45 degrees; transformed equation 2X2 + 8Y2 - 8 = 0, i.e. X2 + 4Y2 = 4.

Exercise 6.9 - The general equation of second degree

Q1. By a rotation of axes, eliminate the xy-term in each of the following equations. Identify the conic and find its elements: (i) 4x2 - 4xy + y2 - 6 = 0 (ii) x2 - 2xy + y2 - 8x - 8y = 0 (iii) x2 + 2xy + y2 + 2 sqrt(2) x - 2 sqrt(2) y + 2 = 0 (iv) x2 + xy + y2 - 4 = 0 (v) 7x2 - 6 sqrt(3) xy + 13y2 - 16 = 0 (vi) 4x2 - 4xy + 7y2 + 12x + 6y - 9 = 0 (vii) xy - 4x - 2y = 0 (viii) x2 + 4xy - 2y2 - 6 = 0 (ix) x2 - 4xy - 2y2 + 10x + 4y = 0.

(i) Eigenvalues 5 and 0: rotated form 5X2 = 6, i.e. a pair of parallel lines 2x - y = +-sqrt(6) (a degenerate conic).

(ii) Rotation 45 degrees: Y2 = 4 sqrt(2) X, a parabola with vertex at the origin in XY-coordinates, a = sqrt(2), focus (sqrt(2), 0), directrix X = -sqrt(2).

(iii) Rotation 45 degrees: X2 = 2(Y - 1/2), a parabola with vertex (0, 1/2) in XY-coordinates, a = 1/2, focus (0, 1), directrix Y = 0.

(iv) Rotation 45 degrees: (3/2)X2 + (1/2)Y2 = 4, i.e. X2/(8/3) + Y2/8 = 1, an ellipse with semi-major axis 2 sqrt(2) along Y, semi-minor axis 2 sqrt(2/3), eccentricity sqrt(2/3).

(v) Rotation 30 degrees: 16X2 + 4Y2 = 16, i.e. X2 + Y2/4 = 1, an ellipse with semi-major axis 2 along Y, semi-minor axis 1, eccentricity sqrt(3)/2.

(vi) Rotation with tan(2 theta) = 4/3: 8X2 + 3Y2 + 6 sqrt(5) X - 9 = 0, i.e. an ellipse with centre (-3 sqrt(5)/8, 0) in XY-coordinates, semi-major axis sqrt(39/8) along Y, eccentricity sqrt(10)/4.

(vii) Rotation 45 degrees: (X - 3 sqrt(2))2/16 - (Y - sqrt(2))2/16 = 1, a rectangular hyperbola with centre (3 sqrt(2), sqrt(2)) in XY-coordinates, a = b = 4, eccentricity sqrt(2).

(viii) Rotation with tan(2 theta) = 4/3: 2X2 - 3Y2 = 6, i.e. X2/3 - Y2/2 = 1, a hyperbola with a = sqrt(3), b = sqrt(2), eccentricity sqrt(5/3).

Answer: (ix) Rotation with tan(2 theta) = -4/3: a hyperbola with centre (-9/(2 sqrt(5)), -8/(3 sqrt(5))) in XY-coordinates, a2 = 23/12, b2 = 23/18, eccentricity sqrt(5/3).

Q2. Show that (i) 10xy + 8x - 15y - 12 = 0 and (ii) 6x2 + xy - y2 - 21x - 8y + 9 = 0 each represents a pair of straight lines and find an equation of each line.

(i) Factoring: 10xy + 8x - 15y - 12 = (5y + 4)(2x - 3), so the lines are y = -4/5 and x = 3/2.

Answer: (ii) The homogeneous part factors as (3x - y)(2x + y); writing (3x - y + a)(2x + y + b) and matching coefficients gives a = -9, b = -1 (and ab = 9 checks), so the lines are 3x - y - 9 = 0 and 2x + y - 1 = 0.

Q3. Find an equation of the tangent to each of the given conics at the indicated point. (i) 3x2 - 7y2 + 2x - y - 48 = 0 at (4, 1) (ii) x2 + 5xy - 4y2 + 4 = 0 at y = -1 (iii) x2 + 4xy - 3y2 - 5x - 9y + 6 = 0 at x = 3.

(i) The point lies on the conic (48 - 7 + 8 - 1 - 48 = 0). Tangent T = 0: 26x - 15y - 89 = 0.

(ii) y = -1 gives x2 - 5x = 0, so the points are (0, -1) and (5, -1). Tangents: at (0, -1): 5x - 8y - 8 = 0; at (5, -1): 5x + 33y + 8 = 0.

Answer: (iii) x = 3 gives -3y2 + 3y = 0, so the points are (3, 0) and (3, 1). Tangents: at (3, 0): x + 3y - 3 = 0; at (3, 1): 5x - 3y - 12 = 0.

Chapter 7: Vectors

Exercise 7.1 - Vectors in the plane

Q1. Write the vector PQ in the form x i + y j. (i) P(2, 3), Q(6, -2) (ii) P(0, 5), Q(-1, -6).

(i) PQ = Q - P = (6 - 2) i + (-2 - 3) j = 4 i - 5 j.

Answer: (ii) PQ = (-1 - 0) i + (-6 - 5) j = -i - 11 j.

Q2. Find the magnitude of the vector u: (i) u = 2 i - 7 j (ii) u = i + j (iii) u = [3, -4].

(i) |u| = sqrt(4 + 49) = sqrt(53). (ii) |u| = sqrt(1 + 1) = sqrt(2).

Answer: (iii) |u| = sqrt(9 + 16) = 5.

Q3. If u = 2 i - 7 j, v = i - 6 j and w = -i + j. Find the following vectors: (i) u + v - w (ii) 2 u - 3 v + 4 w (iii) (1/2) u + (1/2) v + (1/2) w.

(i) u + v - w = (2 + 1 + 1) i + (-7 - 6 - 1) j = 4 i - 14 j.

(ii) 2 u - 3 v + 4 w = (4 - 3 - 4) i + (-14 + 18 + 4) j = -3 i + 8 j.

Answer: (iii) (1/2)(u + v + w) = (1/2)(1 i - 12 j) = i - 6 j.

Q4. Find the sum of the vectors AB and CD, given the four points A(1, -1), B(2, 0), C(-1, 3) and D(-2, 2).

AB = (2 - 1, 0 + 1) = (1, 1) and CD = (-2 + 1, 2 - 3) = (-1, -1).

Answer: AB + CD = (0, 0), the zero vector.

Q5. Find the vector from the point A to the origin where AB = 4 i - 2 j and B is the point (-2, 5).

A = B - AB = (-2 - 4, 5 + 2) = (-6, 7).

Answer: the vector from A to the origin is O - A = 6 i - 7 j.

Q6. Find a unit vector in the direction of the vector given below: (i) v = 2 i - j (ii) v = (1/2) i + (sqrt(3)/2) j (iii) v = (sqrt(3)/2) i - (1/2) j.

(i) |v| = sqrt(5), so the unit vector is (2/sqrt(5)) i - (1/sqrt(5)) j.

(ii) |v| = sqrt(1/4 + 3/4) = 1, so v is already a unit vector.

Answer: (iii) |v| = sqrt(3/4 + 1/4) = 1, so v is already a unit vector.

Q7. If A, B and C are respectively the points (2, -4), (4, 0) and (1, 6). Use vector method to find the coordinates of the point D if: (i) ABCD is a parallelogram (ii) ADBC is a parallelogram.

(i) In ABCD, D = A + C - B = (2 + 1 - 4, -4 + 6 - 0) = (-1, 2).

Answer: (ii) In ADBC, D = A + B - C = (2 + 4 - 1, -4 + 0 - 6) = (5, -10).

Q8. If B, C and D are respectively (4, 1), (-2, 3) and (-8, 0). Use vector method to find the coordinates of the point: (i) A if ABCD is a parallelogram. (ii) E if AEBD is a parallelogram.

(i) A = B + D - C = (4 - 8 + 2, 1 + 0 - 3) = (-2, -2).

Answer: (ii) With A(-2, -2), E = A + B - D = (-2 + 4 + 8, -2 + 1 - 0) = (10, -1).

Q9. If O is the origin and OP = AB, find the point P when A and B are (-3, 7) and (1, 0) respectively.

AB = (1 + 3, 0 - 7) = (4, -7), and OP = AB with O the origin.

Answer: P = (4, -7).

Q10. Use vectors, to show that ABCD is a parallelogram, when the points A, B, C and D are respectively (0, 0), (a, 0), (b, c) and (b - a, c).

AB = (a, 0) and DC = (b - (b - a), c - 0) = (a, 0), so AB = DC.

Also BC = (b - a, c) and AD = (b - a, c), so BC = AD.

Answer: opposite sides are equal and parallel, so ABCD is a parallelogram.

Q11. If AB = CD, find the coordinates of the point A when points B, C, D are (1, 2), (-2, 5), (4, 11) respectively.

CD = (4 + 2, 11 - 5) = (6, 6), so AB = (6, 6).

Answer: A = B - AB = (1 - 6, 2 - 6) = (-5, -4).

Q12. Find the position vectors of the point of division of the line segments joining the following pair of points, in the given ratio: (i) Point C with position vector 2 i - 3 j and point D with position vector 3 i + 2 j in the ratio 4 : 3 (ii) Point E with position vector 5 j and point F with position vector 4 i + j in ratio 2 : 5.

(i) The dividing point is (3 C + 4 D)/7 = (18/7) i - (1/7) j.

Answer: (ii) The dividing point is (5 E + 2 F)/7 = (8/7) i + (27/7) j.

Q13. Prove that the line segment joining the mid points of two sides of a triangle is parallel to the third side and half as long.

Let the vertices have position vectors a, b, c. The mid points of AB and AC are (a + b)/2 and (a + c)/2.

Their difference is (c - b)/2 = (1/2) BC.

Answer: the segment is parallel to BC and half its length.

Q14. Prove that the line segments joining the mid points of the sides of a quadrilateral taken in order form a parallelogram.

Let the mid points of AB, BC, CD, DA be P, Q, R, S. By the mid point theorem in triangle ABC, PQ = (1/2) AC; in triangle CDA, RS = (1/2) AC, so PQ is parallel and equal to RS.

Answer: similarly QR = SP = (1/2) BD, so PQRS is a parallelogram.

Exercise 7.2 - Vectors in space, direction cosines

Q1. Let A = (2, 5), B = (-1, 1) and C = (2, -6). Find (i) AB (ii) 2 AB - CB (iii) 2 CB - 2 CA.

(i) AB = (-1 - 2, 1 - 5) = (-3, -4). CB = (-1 - 2, 1 + 6) = (-3, 7) and CA = (0, 11).

(ii) 2 AB - CB = (-6, -8) - (-3, 7) = (-3, -15).

Answer: (iii) 2 CB - 2 CA = (-6, 14) - (0, 22) = (-6, -8).

Q2. Let u = i + 2 j - k, v = 3 i - 2 j + 2 k, w = 5 i - j + 3 k. Find the indicated vector or number. (i) u + 2 v + w (ii) u - 3 w (iii) |3 v + w|.

(i) u + 2 v + w = (1 + 6 + 5, 2 - 4 - 1, -1 + 4 + 3) = (12, -3, 6).

(ii) u - 3 w = (1 - 15, 2 + 3, -1 - 9) = (-14, 5, -10).

Answer: (iii) 3 v + w = (14, -7, 9), so |3 v + w| = sqrt(196 + 49 + 81) = sqrt(326).

Q3. Find the magnitude of the vector v and write the direction cosines of v. (i) v = 2 i + 3 j + 4 k (ii) v = i - j - k (iii) v = 4 i - 5 j.

(i) |v| = sqrt(29); direction cosines (2/sqrt(29), 3/sqrt(29), 4/sqrt(29)).

(ii) |v| = sqrt(3); direction cosines (1/sqrt(3), -1/sqrt(3), -1/sqrt(3)).

Answer: (iii) |v| = sqrt(41); direction cosines (4/sqrt(41), -5/sqrt(41), 0).

Q4. Find a, so that |a i + (a + 1) j + 2 k| = 3.

Squaring: a2 + (a + 1)2 + 4 = 9, i.e. 2a2 + 2a - 4 = 0, i.e. a2 + a - 2 = 0.

Answer: a = 1 or a = -2.

Q5. Find a unit vector in the direction of v = i + 2 j - k.

|v| = sqrt(6).

Answer: the unit vector is (1/sqrt(6)) i + (2/sqrt(6)) j - (1/sqrt(6)) k.

Q6. If a = 3 i - j - 4 k, b = -2 i - 4 j - 3 k and c = i + 2 j - k. Find a unit vector parallel to 3 a - 2 b + 4 c.

3 a - 2 b + 4 c = (9 + 4 + 4, -3 + 8 + 8, -12 + 6 - 4) = (17, 13, -10).

Its magnitude is sqrt(289 + 169 + 100) = sqrt(558) = 3 sqrt(62).

Answer: a unit vector parallel to it is (17 i + 13 j - 10 k)/sqrt(558) (the opposite sign is also parallel).

Q7. Find a vector whose (i) magnitude is 4 and is parallel to 2 i - 3 j + 6 k (ii) magnitude is 2 and is parallel to -i + j + k.

(i) |2 i - 3 j + 6 k| = 7, so the vector is 4(2 i - 3 j + 6 k)/7 = (8/7) i - (12/7) j + (24/7) k (or its negative).

Answer: (ii) |-i + j + k| = sqrt(3), so the vector is (2/sqrt(3))(-i + j + k) (or its negative).

Q8. If u = 2 i + 3 j + 4 k, v = -i + 3 j - k and w = i + 6 j + z k represent the sides of a triangle. Find the value of z.

For the three vectors to close a triangle, one must be the sum of the other two. Here u + v = (1, 6, 3).

Answer: w = u + v requires z = 3.

Q9. The position vectors of the points A, B, C and D are 2 i - j + k, 3 i + j, 2 i + 4 j - 2 k and -i - 2 j + k respectively. Show that AB is parallel to CD.

AB = (1, 2, -1) and CD = (-3, -6, 3) = -3 AB.

Answer: CD is a scalar multiple of AB, so AB is parallel to CD.

Q10. Two vectors v and w in space are parallel if there is a scalar c such that v = c w. (a) Find two vectors of length 2 parallel to the vector v = 2 i - 4 j + 4 k. (b) Find the constant a so that the vectors v = i - 3 j + 4 k and w = a i + 9 j - 12 k are parallel. (c) Find a vector of length 5 in the direction opposite that of v = i - 2 j + 3 k. (d) Find a and b so that the vectors 3 i - j + 4 k and a i + b j - 2 k are parallel.

(a) |v| = 6; the two vectors are +-(2/3) i -+(4/3) j -+(4/3) k, i.e. (2/3, -4/3, 4/3) and (-2/3, 4/3, -4/3).

(b) w = -3 v, so a = -3.

(c) |v| = sqrt(14); the vector is -5 v/sqrt(14) = -(5/sqrt(14)) i + (10/sqrt(14)) j - (15/sqrt(14)) k.

Answer: (d) the common ratio is -2/4 = -1/2, so a = -3/2 and b = 1/2.

Q11. Find the direction cosines for the given vector: (i) v = 3 i - j + 2 k (ii) 6 i - 2 j + k (iii) PQ, where P = (-2, 1, 5) and Q = (1, 3, 1).

(i) |v| = sqrt(14): direction cosines (3/sqrt(14), -1/sqrt(14), 2/sqrt(14)).

(ii) |v| = sqrt(41): direction cosines (6/sqrt(41), -2/sqrt(41), 1/sqrt(41)).

Answer: (iii) PQ = (3, 2, -4), |PQ| = sqrt(29): direction cosines (3/sqrt(29), 2/sqrt(29), -4/sqrt(29)).

Q12. Which of the following triples can be the direction angles of a single vector: (i) 45 degrees, 45 degrees, 60 degrees (ii) 30 degrees, 45 degrees, 60 degrees (iii) 45 degrees, 60 degrees, 60 degrees.

A triple is possible exactly when cos2(a) + cos2(b) + cos2(g) = 1.

(i) 1/2 + 1/2 + 1/4 = 5/4: not possible. (ii) 3/4 + 1/2 + 1/4 = 3/2: not possible.

Answer: (iii) 1/2 + 1/4 + 1/4 = 1: possible.

Exercise 7.3 - The scalar product of two vectors

Q1. Find the cosine of the angle theta between u and v: (i) u = 3 i + j - k, v = 2 i - j + k (ii) u = i - 3 j + 4 k, v = 4 i - j + 3 k (iii) u = [3, 5], v = [6, -2] (iv) u = [2, 3, 1], v = [2, 4, 1].

(i) u . v = 6 - 1 - 1 = 4; |u| = sqrt(11), |v| = sqrt(6): cos(theta) = 4/sqrt(66).

(ii) u . v = 4 + 3 + 12 = 19; |u| = |v| = sqrt(26): cos(theta) = 19/26.

(iii) u . v = 18 - 10 = 8; |u| = sqrt(34), |v| = sqrt(40): cos(theta) = 2/sqrt(85).

Answer: (iv) u . v = 4 + 12 + 1 = 17; |u| = sqrt(14), |v| = sqrt(21): cos(theta) = 17/sqrt(294) = 17/(7 sqrt(6)).

Q2. Calculate the projection of a along b and projection of b along a when: (i) a = -i + k, b = j + k (ii) a = 3 i + j - k, b = -2 i - j + k.

(i) a . b = 1, |a| = |b| = sqrt(2). Scalar projection of a along b = 1/sqrt(2), vector projection (1/2)(j + k); scalar projection of b along a = 1/sqrt(2), vector projection (1/2)(-i + k).

Answer: (ii) a . b = -8, |a| = sqrt(11), |b| = sqrt(6). Scalar projection of a along b = -8/sqrt(6), vector projection (8/3) i + (4/3) j - (4/3) k; scalar projection of b along a = -8/sqrt(11), vector projection -(24/11) i - (8/11) j + (8/11) k.

Q3. Find a real number a so that the vectors u and v are perpendicular. (i) u = 2 a i + j - k, v = i + a j + 4 k (ii) u = a i + 2 a j + 3 k, v = i + a j + 3 k.

(i) u . v = 2a + a - 4 = 3a - 4 = 0, so a = 4/3.

Answer: (ii) u . v = a + 2a2 + 9 = 0 has discriminant 1 - 72 < 0, so no real number a makes them perpendicular.

Q4. Find the number z so that the triangle with vertices A(1, -1, 0), B(-2, 2, 1) and C(0, 2, z) is a right triangle with right angle at C.

CA = (1, -3, -z) and CB = (-2, 0, 1 - z); the right angle at C requires CA . CB = 0.

-2 - z(1 - z) = 0, i.e. z2 - z - 2 = 0.

Answer: z = 2 or z = -1.

Q5. If v is a vector for which v . i = 0, v . j = 0, v . k = 0, find v.

Writing v = (x, y, z): v . i = x = 0, v . j = y = 0, v . k = z = 0.

Answer: v = 0, the zero vector.

Q6. (i) Show that the vectors 3 i - 2 j + k, i - 3 j + 5 k and 2 i + j - 4 k form a right angle. (ii) Show that the set of points P = (1, 3, 2), Q = (4, 1, 4) and R = (6, 5, 5) form a right triangle. (In (ii) the book prints the third point as P again; it is read as R.)

(i) Call the vectors v1, v2, v3. Then v2 = v1 - v3, and v1 . v3 = 6 - 2 - 4 = 0, so v1 is perpendicular to v3; also |v2|2 = 35 = 14 + 21 = |v1|2 + |v3|2, so they form a right-angled triangle with the right angle between v1 and v3.

Answer: (ii) PQ = (3, -2, 2) and QR = (2, 4, 1) with PQ . QR = 6 - 8 + 2 = 0, so triangle PQR is right-angled at Q.

Q7. Show that mid point of hypotenuse a right triangle is equidistant from its vertices.

Put the right angle at the origin: let the other vertices have position vectors a and b with a . b = 0. The mid point of the hypotenuse is m = (a + b)/2.

Then |m|2 = (|a|2 + |b|2)/4 and |m - a|2 = |b - a|2/4 = (|a|2 + |b|2)/4; the same holds for the third vertex.

Answer: the mid point is equidistant from all three vertices.

Q8. Prove that perpendicular bisectors of the sides of a triangle are concurrent.

Let a point X lie on the bisectors of AB and AC: then (X - (A + B)/2) . (B - A) = 0 and (X - (A + C)/2) . (C - A) = 0, i.e. X . (B - A) = (|B|2 - |A|2)/2 and X . (C - A) = (|C|2 - |A|2)/2.

Subtracting gives X . (C - B) = (|C|2 - |B|2)/2, i.e. X also lies on the bisector of BC.

Answer: the three bisectors meet at X, so they are concurrent.

Q9. Prove that the altitudes of a triangle are concurrent.

Let H satisfy (H - A) . (B - C) = 0 and (H - B) . (C - A) = 0, i.e. AH is perpendicular to BC and BH is perpendicular to CA.

Adding the two expanded equations gives (H - C) . (B - A) = 0.

Answer: CH is perpendicular to AB, so the three altitudes meet at H.

Q10. Prove that the angle in a semi circle is a right angle.

Let O be the centre, A = -a, B = a and P = p on the circle, so |p| = |a|.

Then PA . PB = (-a - p) . (a - p) = |p|2 - |a|2 = 0.

Answer: angle APB is a right angle.

Q11. Prove that cos(a + b) = cos a cos b - sin a sin b.

Take unit vectors u = (cos a, sin a) and w = (cos b, -sin b); the angle between them is a + b.

Then u . w = cos a cos b - sin a sin b, and also u . w = |u| |w| cos(a + b) = cos(a + b).

Answer: cos(a + b) = cos a cos b - sin a sin b.

Q12. Prove that in any triangle ABC. (i) b = c cos A + a cos C (ii) c = a cos B + b cos A (iii) b2 = c2 + a2 - 2 c a cos B (iv) c2 = a2 + b2 - 2 a b cos C.

Place A at the origin and AC on the x-axis: C = (b, 0), B = (c cos A, c sin A). Then a2 = (b - c cos A)2 + (c sin A)2, and cos C = (b - c cos A)/a, so a cos C = b - c cos A, proving (i); (ii) follows the same way.

For (iii), put B at the origin with BA and BC as vectors of lengths c and a at angle B: then b2 = |BA - BC|2 = c2 + a2 - 2 c a cos B; (iv) is the same argument at C.

Answer: all four formulas hold.

Exercise 7.4 - The cross product of two vectors

Q1. Compute the cross product a x b and b x a. Check your answer by showing that each a and b is perpendicular to a x b and b x a. (i) a = 2 i + j - k, b = i - j + k (ii) a = i - j, b = i + j (Print note: in this copy the k-components of (ii) are blank/missing in the print, confirmed at 300 DPI; the part is solved below for the legible plane vectors a = i - j, b = i + j, and is flagged, not taken as the book's full data.) (iii) a = 3 i - 2 j + k, b = i + j (iv) a = -4 i + j - 2 k, b = 2 i + j + k.

(i) a x b = -3 j - 3 k and b x a = 3 j + 3 k; a . (a x b) = -3 + 3 = 0 and b . (a x b) = 3 - 3 = 0, and the same for b x a.

(ii) Flagged (see head): for a = i - j, b = i + j, a x b = 2 k and b x a = -2 k; a . (2 k) = 0 and b . (2 k) = 0.

(iii) a x b = -i + j + 5 k and b x a = i - j - 5 k; a . (a x b) = -3 - 2 + 5 = 0 and b . (a x b) = -1 + 1 = 0.

Answer: (iv) a x b = 3 i - 6 k and b x a = -3 i + 6 k; a . (a x b) = -12 + 12 = 0 and b . (a x b) = 6 - 6 = 0.

Q2. Find a unit vector perpendicular to the plane containing a and b. Also find sine of the angle between them. (i) a = 2 i - 6 j - 3 k, b = 4 i + 3 j - k (ii) a = -i - j - k, b = 2 i - 3 j + 4 k (iii) a = 2 i - 2 j + 4 k, b = -i + j - 2 k (iv) a = i - j, b = i + j (Print note: as in Q1(ii), the k-components of (iv) are blank/missing in the print; solved below for the legible plane vectors, and flagged.)

(i) a x b = 15 i - 10 j + 30 k, |a x b| = 35, |a| = 7, |b| = sqrt(26): unit vector +-(3 i - 2 j + 6 k)/7, sin(theta) = 5/sqrt(26).

(ii) a x b = -7 i + 2 j + 5 k, |a x b| = sqrt(78), |a| = sqrt(3), |b| = sqrt(29): unit vector +-(-7 i + 2 j + 5 k)/sqrt(78), sin(theta) = sqrt(26/29).

(iii) b = -a/2, so a and b are parallel: a x b = 0 and sin(theta) = 0, and there is no unique unit vector perpendicular to their plane.

Answer: (iv) Flagged (see head): for a = i - j, b = i + j, a x b = 2 k, |a| = |b| = sqrt(2): unit vector +-k, sin(theta) = 1.

Q3. Find the area of the triangle, determined by the point P, Q and R. (i) P(0, 0, 0);

Q(2, 3, 2); R(-1, 1, 4) (ii) P(1, -1, -1); Q(2, 0, -1); R(0, 2, 1).

(i) PQ = (2, 3, 2), PR = (-1, 1, 4); PQ x PR = 10 i - 10 j + 5 k with magnitude 15, so the area is (1/2)(15) = 15/2 square units.

Answer: (ii) PQ = (1, 1, 0), PR = (-1, 3, 2); PQ x PR = 2 i - 2 j + 4 k with magnitude 2 sqrt(6), so the area is sqrt(6) square units.

Q4. Find the area of parallelogram, whose vertices are: (i) A(0, 0, 0); B(1, 2, 3); C(2, -1, 1); D(3, 1, 4) (ii) A(1, 2, -1); B(4, 2, -3); C(6, -5, 2); D(9, -5, 0) (iii) A(-1, 1, 1); B(-1, 2, 2); C(-3, 4, -5); D(-3, 5, -4).

(i) AB = (1, 2, 3), AD = (3, 1, 4); AB x AD = 5 i + 5 j - 5 k with magnitude 5 sqrt(3), so the area is 5 sqrt(3) square units.

(ii) AB = (3, 0, -2), AD = (8, -7, 1); AB x AD = -14 i - 19 j - 21 k with magnitude sqrt(998), so the area is sqrt(998) square units.

Answer: (iii) AB = (0, 1, 1), AD = (-2, 4, -5); AB x AD = -9 i - 2 j + 2 k with magnitude sqrt(89), so the area is sqrt(89) square units.

Q5. Which vectors, if any, are perpendicular or parallel (i) u = 5 i - j + k; v = j - 5 k; w = -15 i + 3 j - 3 k (ii) u = i + 2 j - k; v = -i + j + k; w = -(pi/2) i - pi j + (pi/2) k.

(i) w = -3 u, so w is parallel to u. The dot products are u . v = -6, u . w = -81 and v . w = 18, none zero, so no pair is perpendicular.

Answer: (ii) w = -(pi/2) u, so w is parallel to u; u . v = -1 + 2 - 1 = 0, so u is perpendicular to v, and hence v is also perpendicular to w.

Q6. Prove that: a x (b + c) + b x (c + a) + c x (a + b) = 0.

Expanding by distributivity: a x b + a x c + b x c + b x a + c x a + c x b.

Since b x a = -a x b, c x b = -b x c and c x a = -a x c, the six terms cancel in pairs.

Answer: the sum is 0.

Q7. If a + b + c = 0, then prove that a x b = b x c = c x a.

From c = -(a + b): b x c = b x (-(a + b)) = -b x a = a x b.

Also c x a = (-(a + b)) x a = -(b x a) = a x b.

Answer: a x b = b x c = c x a.

Q8. Prove that: sin(a - b) = sin a cos b - cos a sin b. (Print note: as printed in this copy the formula is given with a + sign before the second term, which is the formula for sin(a + b), not sin(a - b); the correct sin(a - b) formula is proved below and the misprint is flagged, not silently corrected.)

Take unit vectors u = (cos a, sin a, 0) and v = (cos b, sin b, 0). The k-component of u x v is cos a sin b - sin a cos b, and it equals sin(b - a) = -sin(a - b).

Answer: hence sin(a - b) = sin a cos b - cos a sin b, while sin a cos b + cos a sin b is sin(a + b).

Q9. If a x b = 0 and a . b = 0, what conclusion can be drawn about a or b?

a x b = 0 means a and b are parallel (or one is zero); a . b = 0 means they are perpendicular (or one is zero).

Answer: a non-zero pair cannot be both parallel and perpendicular, so at least one of a, b must be the zero vector.

Exercise 7.5 - Scalar triple product of vectors

Q1. Find the volume of the parallelepiped for which the given vectors are three edges. (i) u = 3 i + 2 k; v = i + 2 j + k; w = -j + 4 k (ii) u = i - 4 j - k; v = i - j - 2 k; w = 2 i - 3 j + k (iii) u = i - 2 j - 3 k; v = 2 i - j - k; w = j + k.

(i) [u v w] = 3(8 + 1) - 2 = 25, so the volume is 25 cubic units.

(ii) [u v w] = -7 + 20 + 1 = 14, so the volume is 14 cubic units.

Answer: (iii) [u v w] = 4 - 6 = -2, so the volume is 2 cubic units.

Q2. Verify that a . b x c = b . c x a = c . a x b if a = 3 i - j + 5 k, b = 4 i + 3 j - 2 k, and c = 2 i + 5 j + k.

b x c = 13 i - 8 j + 14 k, so a . (b x c) = 39 + 8 + 70 = 117.

c x a = 26 i - 7 j - 17 k, so b . (c x a) = 104 - 21 + 34 = 117.

Answer: a x b = -13 i + 26 j + 13 k, so c . (a x b) = -26 + 130 + 13 = 117. All three are equal.

Q3. Prove that the vectors i - 2 j + 3 k, -2 i + 3 j - 4 k and i - 3 j + 5 k are coplanar.

Their scalar triple product is the determinant with rows (1, -2, 3), (-2, 3, -4), (1, -3, 5).

Expanding: (15 - 12) + 2(-10 + 4) + 3(6 - 3) = 3 - 12 + 9 = 0.

Answer: the triple product is zero, so the vectors are coplanar.

Q4. Find the constant a such that the vectors are coplanar. (i) i - j + k, i - 2 j - 3 k and 3 i - a j + 5 k. (ii) i - 2 a j - k, i - j + 2 k and a i - j + k.

(i) The determinant is (-10 - 3a) + 14 + (-a + 6) = 10 - 4a; setting it to zero gives a = 5/2.

Answer: (ii) The determinant is 1 + 2a - 4a2 + 1 - a = 2 + a - 4a2; setting it to zero gives a = (1 +- sqrt(33))/8.

Q5. (a) Find the value of: (i) 2 i x 2 j . k (ii) 3 j . k x i (iii) [k i j] (iv) [i i k] (b) Prove that u . (v x w) + v . (w x u) + w . (u x v) = 3 u . (v x w).

(a) (i) (2 i x 2 j) . k = 4 k . k = 4. (ii) 3 j . (k x i) = 3 j . j = 3. (iii) [k i j] = k . (i x j) = k . k = 1. (iv) [i i k] = 0, since two vectors are repeated.

Answer: (b) Each term equals the same triple product [u v w] by cyclic permutation, so the sum is 3 [u v w] = 3 u . (v x w).

Q6. Find volume of the Tetrahedron with the vertices (i) (0, 1, 2), (3, 2, 1), (1, 2, 1) and (5, 5, 6) (ii) (2, 1, 8), (3, 2, 9), (2, 1, 4) and (3, 3, 10).

(i) Edge vectors from (0, 1, 2) are (3, 1, -1), (1, 1, -1), (5, 4, 4); their determinant is 24 - 9 + 1 = 16, so the volume is 16/6 = 8/3 cubic units.

Answer: (ii) Edge vectors from (2, 1, 8) are (1, 1, 1), (0, 0, -4), (1, 2, 2); their determinant is 8 - 4 = 4, so the volume is 4/6 = 2/3 cubic units.

Q7. Find the work done, if the point at which the constant force F = 4 i + 3 j + 5 k is applied to an object, moves from P1(3, 1, -2) to P2(2, 4, 6).

Displacement d = P2 - P1 = (-1, 3, 8).

Work = F . d = -4 + 9 + 40 = 45 units.

Q8. A particle, acted by constant forces 4 i + j - 3 k and 3 i - j - k, is displaced from A(1, 2, 3) to B(5, 4, 1). Find the work done.

Resultant force = (7, 0, -4); displacement = (4, 2, -2).

Answer: work = 28 + 0 + 8 = 36 units.

Q9. A particle is displaced from the point A(5, -5, -7) to the point B(6, 2, -2) under the action of constant forces defined by 10 i - j + 11 k, 4 i + 5 j + 9 k and -2 i + j - 9 k. Show that the total work done by the forces is 102 units.

Resultant force = (12, 5, 11); displacement = (1, 7, 5).

Answer: work = 12 + 35 + 55 = 102 units, as required.

Q10. A force of magnitude 6 units acting parallel to 2 i - 2 j + k displaces, the point of application from (1, 2, 3) to (5, 3, 7). Find the work done.

The direction has magnitude 3, so F = 6(2 i - 2 j + k)/3 = 4 i - 4 j + 2 k; displacement = (4, 1, 4).

Answer: work = 16 - 4 + 8 = 20 units.

Q11. A force F = 3 i + 2 j - 4 k is applied at the point (1, -1, 2). Find the moment of

the force about the point (2, -1, 3).

r = (1, -1, 2) - (2, -1, 3) = (-1, 0, -1).

Answer: moment = r x F = 2 i - 7 j - 2 k.

Q12. A force F = 4 i - 3 k, passes through the point A(2, -2, 5). Find the moment of F about the point B(1, -3, 1).

r = A - B = (1, 1, 4).

Answer: moment = r x F = -3 i + 19 j - 4 k.

Q13. Give a force F = 2 i + j - 3 k acting at a point A(1, -2, 1). Find the moment of F about the point B(2, 0, -2).

r = A - B = (-1, -2, 3).

Answer: moment = r x F = 3 i + 3 j + 3 k.

Q14. Find the moment about A(1, 1, 1) of each of the concurrent forces i - 2 j, 3 i + 2 j - k, 5 j + 2 k, where P(2, 0, 1) is their point of concurrency.

r = P - A = (1, -1, 0).

Moments: r x (i - 2 j) = -k; r x (3 i + 2 j - k) = i + j + 5 k; r x (5 j + 2 k) = -2 i - 2 j + 5 k.

Answer: the three moments are -k, i + j + 5 k and -2 i - 2 j + 5 k.

Q15. A force F = 7 i + 4 j - 3 k is applied at P(1, -2, 3). Find its moment about the

point Q(2, 1, 1).

r = P - Q = (-1, -3, 2).

Answer: moment = r x F = i + 11 j + 17 k.